

HR / POWER

Report di identificazione

Run 0bd77479254d4d2bb827b24f15aa7709

Dataset: G1_Ambigui

Creazione: 2026-09-28T18:35:19.826447+00:00 | Stato: completed

Struttura: p1d_full | $\tau dx/dt + x = K (P(t-L)-P_0)$; $HR_{hat} = B+x$

Inizializzazione: estimated_equilibrium | $P_0 = \text{mean}(P_{pre})$; $HR_0 = HR(0)$; $x(0) = HR_0 - B$; B estimated; $P(t < 0) = P_0$

Dataset: 82 segmenti; train richiesti: 81; risultati: 81; validi: 81; falliti: 0.

Ambito: fit in-sample sui soli segmenti train. Val/test non utilizzati. Aggregati non pesati per durata, calcolati solo sui record fitted; n indica i valori numerici disponibili. n.d. = non disponibile o non applicabile. Nessuna conclusione fisiologica automatica.

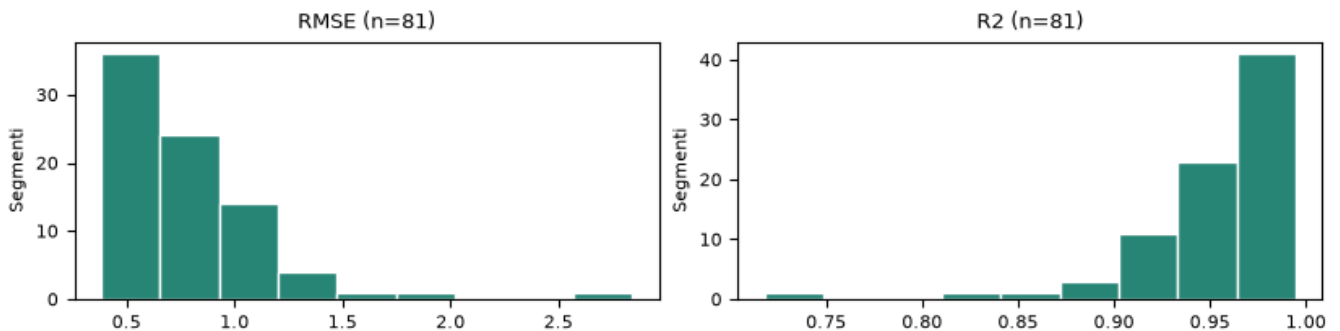
Pre-window richiesta: si; durata: 10 s. P0: pre_window_mean; HR0: first_segment_sample

Pre-window disponibili: 81 / 81 segmenti train inventariati.

Qualita della traiettoria

RMSE, MAE e bias in bpm; R2 e amplitude_ratio adimensionali. Bias positivo: HR osservata maggiore della prevista.

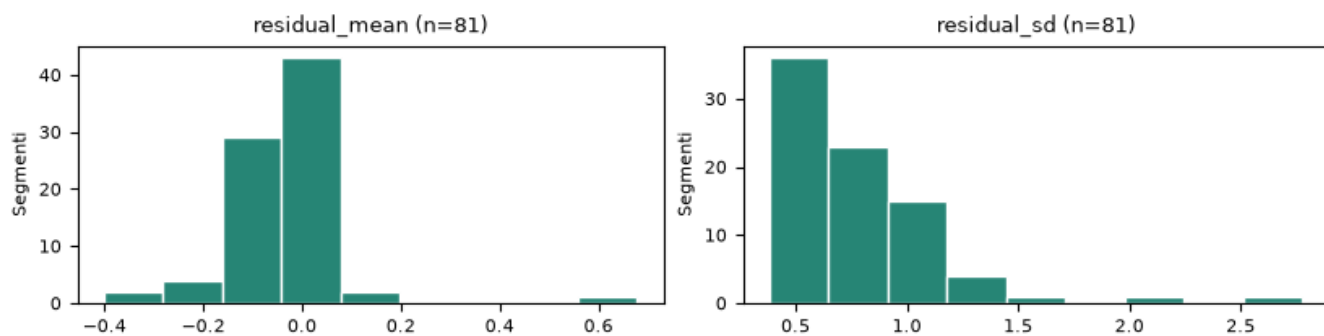
Metrica	n	Media	Mediana	SD	Min	Max
RMSE	81	0.78977	0.70282	0.37641	0.38122	2.8418
MAE	81	0.64205	0.5745	0.29462	0.29506	2.0654
bias	81	-0.028858	-0.032918	0.11607	-0.40092	0.6783
R2	81	0.95349	0.96461	0.041403	0.71787	0.99471
amplitude_ratio	81	1.0387	1.0353	0.073711	0.73058	1.2213



Residui

e = HR osservata - HR prevista. SD campionaria (ddof=1); ACF centrata, lag in campioni. Le segnalazioni sono conservate e non impongono nuove esclusioni.

Metrica	n	Media	Mediana	SD	Min	Max
residual_mean	81	-0.028858	-0.032918	0.11607	-0.40092	0.6783
residual_sd	81	0.79269	0.70996	0.37089	0.38273	2.7782
residual_autocorrelation_lag1	81	0.66982	0.69866	0.16177	0.24108	0.91275
residual_autocorrelation_lag5	81	-0.043713	-0.054275	0.20253	-0.43436	0.5096



Segnalazione	Segmenti
at_bounds	47
autocorrelated_residuals	70
equivalent_predictions_different_parameters	8
irregular_sampling	1
multistart_disagreement	77
near_bounds	3
recording_gaps	1

Multistart

Ricerca: profile_multistart. I tentativi comprendono gli eventuali raffinamenti da griglia; n_multistart indica solo gli start originali.

Start registrati: 1296; convergenti: 1287. Accordo di tutti gli start: 4 / 81 fit validi.

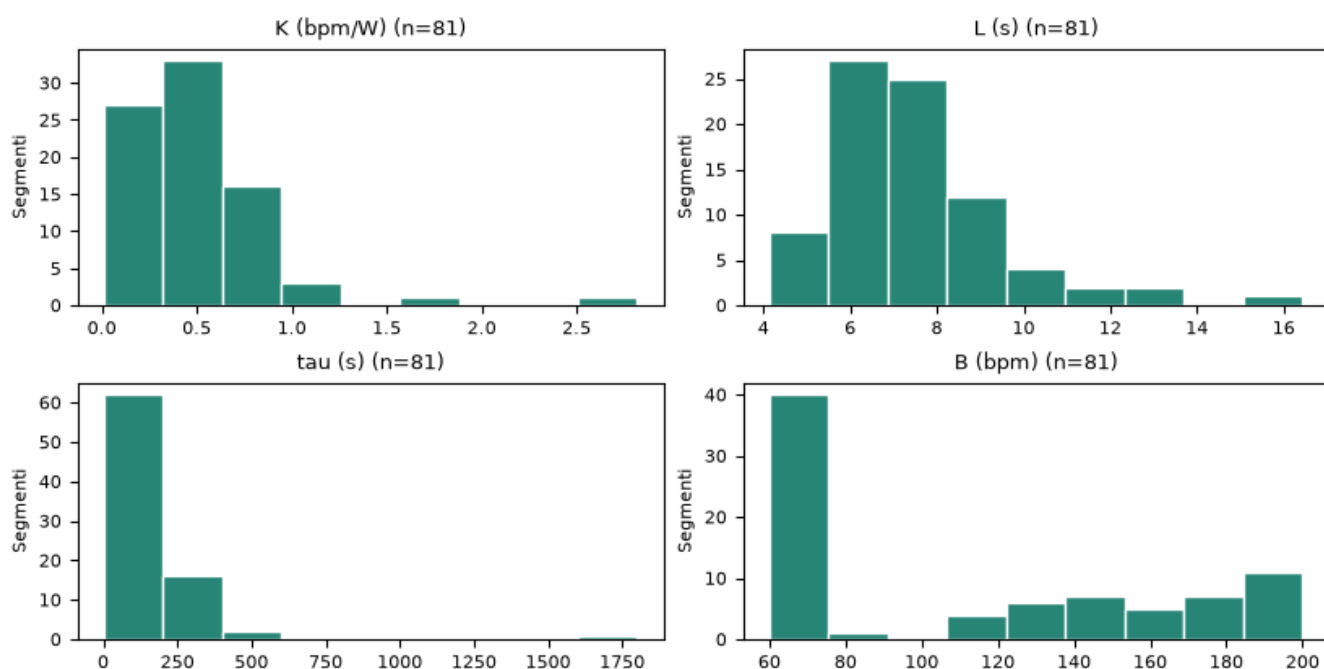
Diagnostica	n	Media	Mediana	SD	Min	Max
SSE	1296	96.733	26.703	448.15	7.121	10764
nfev	1296	17.692	13	44.489	3	500

Distribuzione e precisione dei parametri

SE e CI95 locali di Wald, condizionati ai riferimenti iniziali. RSE in percentuale. Bounds, autocorrelazione e rango ridotto possono limitare la precisione. Un parametro fissato non ha SE/CI stimabili.

Parametro	n	Media	Mediana	SD	Min	Max
K (bpm/W)	81	0.49417	0.48474	0.38594	0.010947	2.8176
L (s)	81	7.4881	7.3063	1.9663	4.139	16.436
tau (s)	81	162.54	143.76	207.89	2.3825	1800
B (bpm)	81	110.48	80.212	54.944	60	200

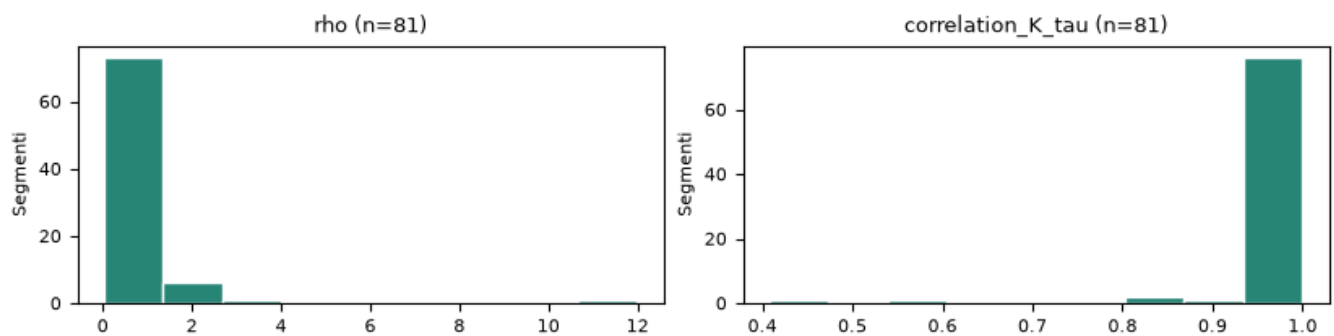
Parametro	SE mediana	RSE mediana %	AI bound	Vicino al bound
K	0.50202	86.465	0	0
L	0.37826	5.2743	0	0
tau	135.72	87.173	1	1
B	71.153	74.039	46	49



Identificabilita

- $\rho = (T-L)/\tau$: tempo disponibile dopo il ritardo in unita di τ . Valori piccoli descrivono una frazione breve del transitorio; nessuna soglia automatica.
- Correlazioni parametriche vicine a -1 o +1 segnalano compensazione locale tra stime; non misurano la qualita della traiettoria.
- Rango Jacobiano: confrontare con p , il numero di parametri liberi. Rango $< p$ indica direzioni non identificabili localmente.
- Multistart: SSE simili con parametri differenti indicano ambiguita. CI e correlazioni restano approssimazioni locali.

Diagnostica	n	Media	Mediana	SD	Min	Max
ρ	81	0.67582	0.28416	1.408	0.027075	11.989
correlation_K_tau	81	0.97734	0.99924	0.082748	0.40787	0.99996
Jacobian rank	81	4	4	0	4	4
Jacobian condition	81	56923	42155	57862	989.47	3.3326e+05



Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_082-083.csv

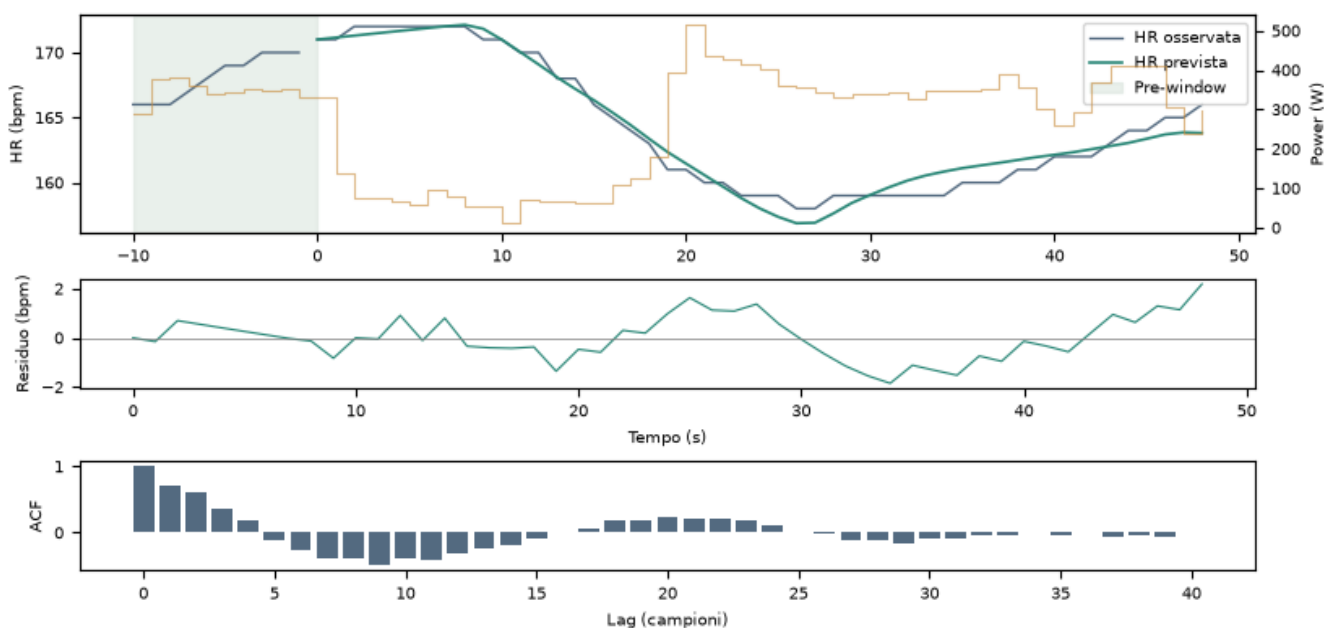
ID: 0046978383f7350729609ddf | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=347.4 W; HR0=171 bpm.

HR(0) osservata fissata; B=186.02 bpm stimato; $x(0)=-15.018$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.88597	0.70501	0.006524	0.9682	1.0899
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.006524	0.89513	0.71967	-0.10145	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.403	0.23424	-0.056103 / 0.86211	58.123	no	no
L	7.5202	0.402	6.7322 / 8.3081	5.3456	no	no
tau	99.262	55.72	-9.9487 / 208.47	56.134	no	no
B	186.02	13.043	160.45 / 211.58	7.0119	no	no

rho=0.40781; Jacobian rank=4/4; cond=12590; accordo multistart=no.

Correlazioni: {"K_L": -0.8416954227963402, "K_tau": 0.9976470596145021, "K_B": 0.9961312706920267, "L_tau": -0.8492209257303313, "L_B": -0.8323562232277962, "tau_B": 0.9885657730440092}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 38.462 / 470.66

Griglia: 42441 coppie L/tau; SSE multistart originale: 38.462; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_019-020.csv

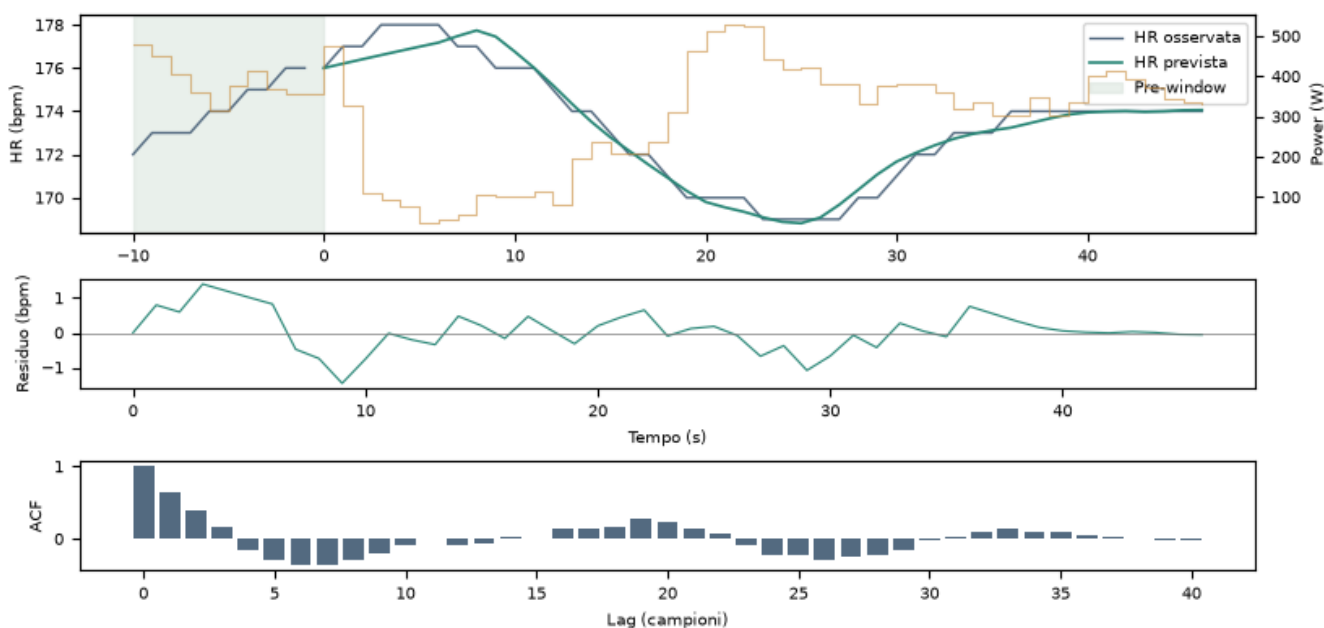
ID: 03a88eec0d8d28fba5080aee | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=388.3 W; HR0=176 bpm.

HR(0) osservata fissata; B=190.42 bpm stimato; x(0)=-14.425 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.5572	0.4052	0.05802	0.95911	0.99154
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.05802	0.56016	0.63609	-0.29791	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21945	0.13218	-0.039619 / 0.47852	60.232	no	no
L	6.5826	0.36061	5.8758 / 7.2894	5.4783	no	no
tau	70.634	41.592	-10.886 / 152.15	58.884	no	no
B	190.42	10.498	169.85 / 211	5.5127	no	no

rho=0.55805; Jacobian rank=4/4; cond=17591; accordo multistart=no.

Correlazioni: {"K_L": -0.8598831286212045, "K_tau": 0.998171134760951, "K_B": 0.9993454254872761, "L_tau": -0.8642515377793382, "L_B": -0.8598603893385399, "tau_B": 0.9961040997391005}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 14.592 / 291.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 14.592; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_064-065.csv

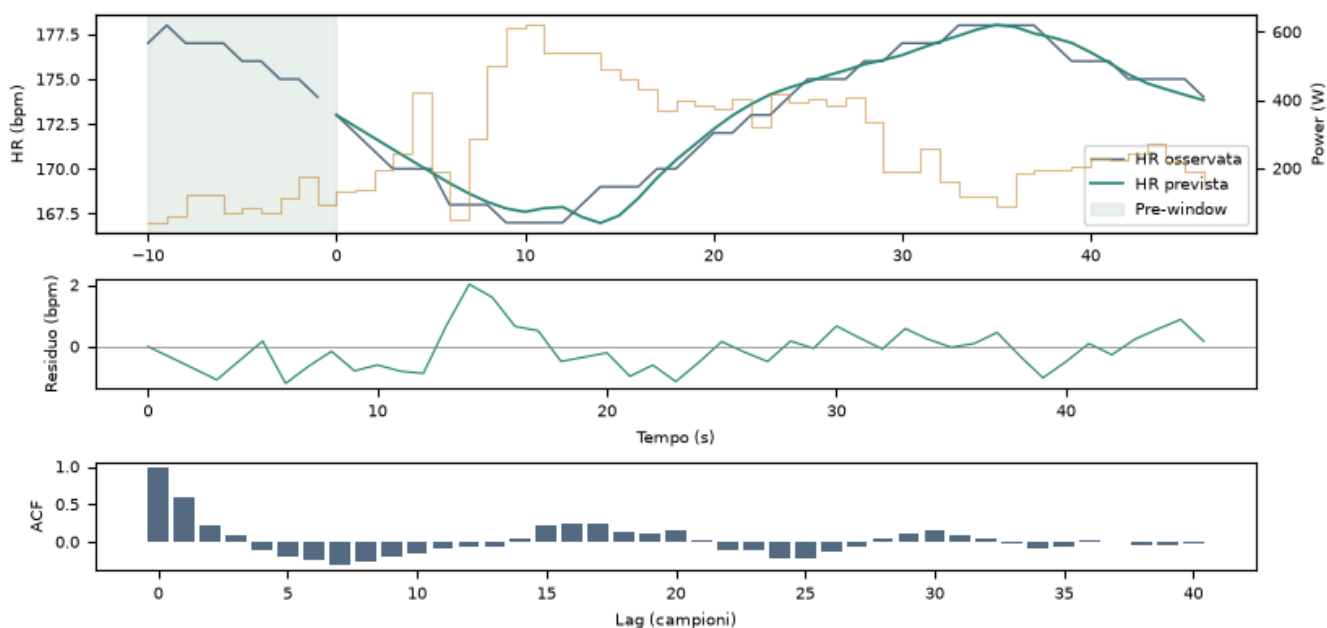
ID: 046d12fab1baeeb84bc07fe1 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=94.6 W; HR0=173 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=113 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.67603	0.53419	-0.098671	0.96486	1.0042
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.098671	0.67602	0.5845		-0.18688



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.59111	0.75433	-0.88737 / 2.0696	127.61	no	no
L	6.4721	0.37364	5.7397 / 7.2044	5.7732	no	no
tau	175.37	225.02	-265.67 / 616.4	128.31	no	no
B	60	143.91	-222.06 / 342.06	239.85	si	si

rho=0.2254; Jacobian rank=4/4; cond=75495; accordo multistart=no.

Correlazioni: {"K_L": -0.8752141404696814, "K_tau": 0.9996473961410781, "K_B": -0.9999690164603428, "L_tau": -0.8704767763843027, "L_B": 0.8778227780436747, "tau_B": -0.999508618284183}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 21.48 / 54.839

Griglia: 42441 coppie L/tau; SSE multistart originale: 21.48; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_066-067.csv

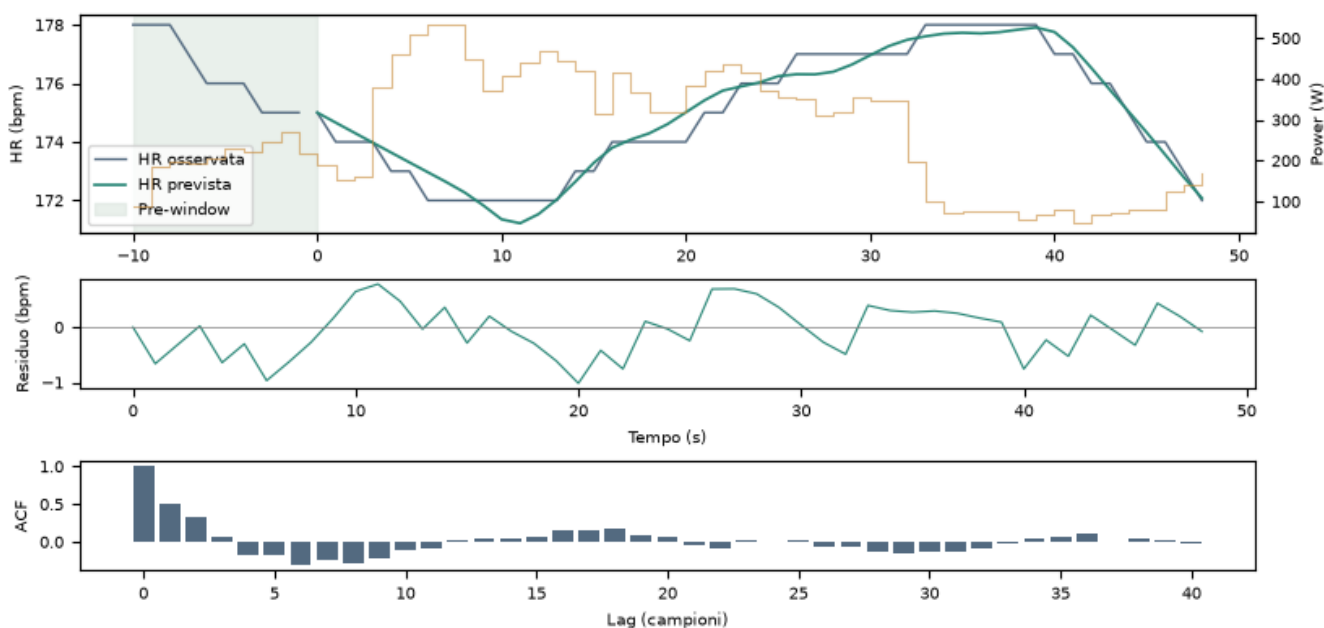
ID: 083b93d8299323434bc95990 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=205.7 W; HR0=175 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=115 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.44705	0.3672	-0.05071	0.95441	1.1146
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.05071	0.44877	0.50353		-0.17443



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.005	2.811	-4.5045 / 6.5145	279.69	no	no
L	7.497	0.32884	6.8524 / 8.1415	4.3863	no	no
tau	335.77	944.19	-1514.8 / 2186.4	281.2	no	no
B	60	321.03	-569.22 / 689.22	535.06	si	si

rho=0.12063; Jacobian rank=4/4; cond=1.4582e+05; accordo multistart=si.

Correlazioni: {"K_L": -0.8971227670376702, "K_tau": 0.9999269707662394, "K_B": -0.9999958235137529, "L_tau": -0.8972174867837505, "L_B": 0.8975097092139283, "tau_B": -0.9999137267644962}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 9.7928 / 9.7928

Griglia: 42441 coppie L/tau; SSE multistart originale: 9.7928; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_012-013.csv

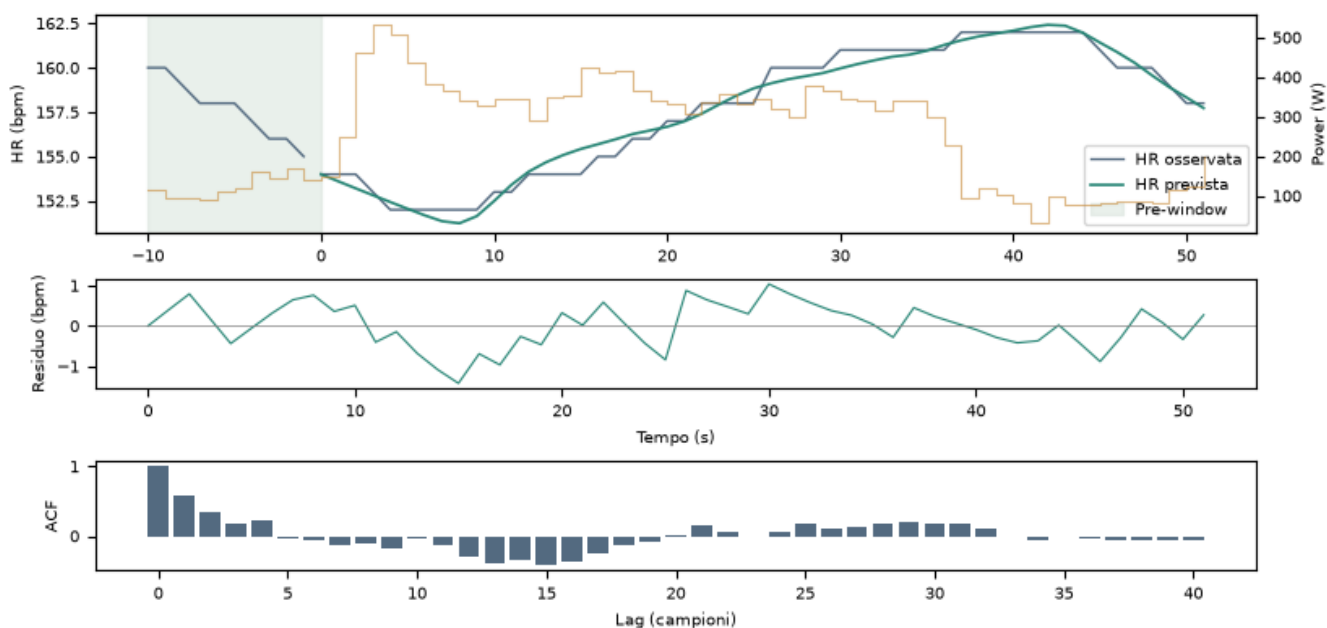
ID: 092301ef17b6a3cbc1f57202 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=124.5 W; HR0=154 bpm.

HR(0) osservata fissata; B=123.54 bpm stimato; x(0)=30.457 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.542	0.44529	0.013764	0.97658	1.118
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.013764	0.54711	0.58299	-0.024963	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.24986	0.086007	0.081286 / 0.41844	34.422	no	no
L	6.4908	0.33391	5.8363 / 7.1452	5.1443	no	no
tau	75.815	26.645	23.591 / 128.04	35.145	no	no
B	123.54	11.6	100.81 / 146.28	9.3896	no	no

rho=0.58707; Jacobian rank=4/4; cond=20990; accordo multistart=no.

Correlazioni: {"K_L": -0.8851283175140984, "K_tau": 0.9926338161194102, "K_B": -0.9986853571636144, "L_tau": -0.8825449427028821, "L_B": 0.8874384090730999, "tau_B": -0.9858669624360841}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 15.276 / 198.43

Griglia: 42441 coppie L/tau; SSE multistart originale: 15.276; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_035-036.csv

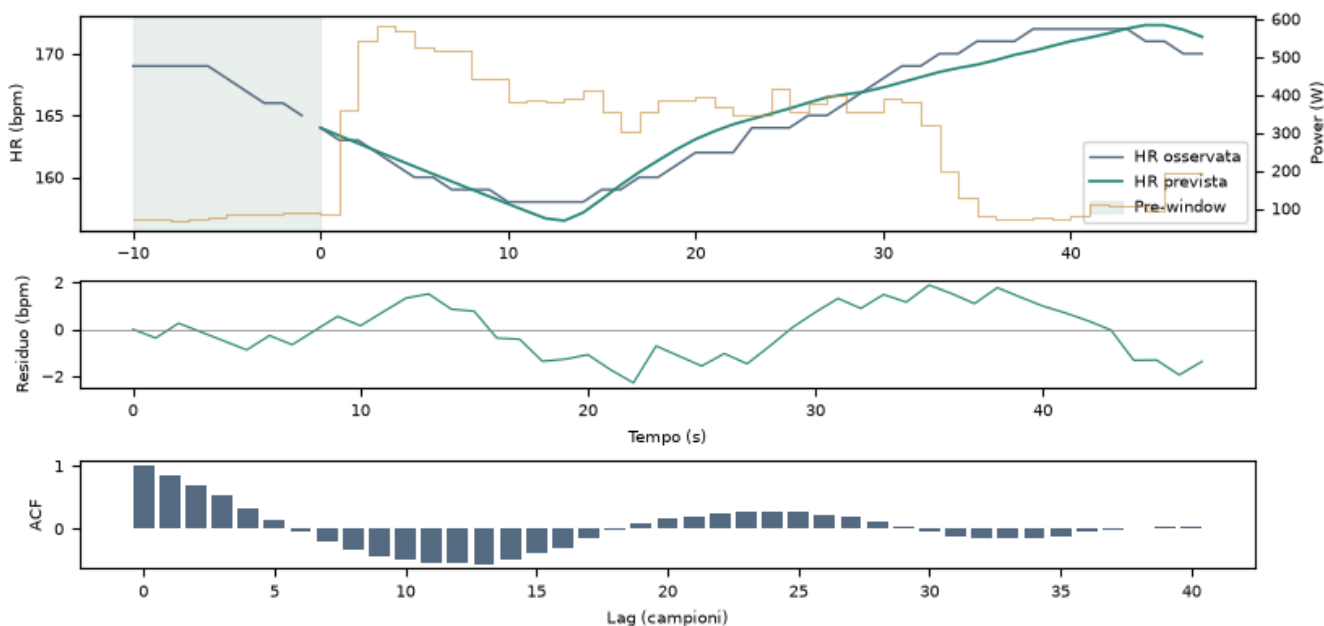
ID: 0c9802b7ccbe588d6a3df30 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=79.3 W; HR0=164 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=104 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1041	0.94735	-0.04779	0.9522	1.1306
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.04779	1.1147	0.8416		0.13231



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.57406	0.7388	-0.87399 / 2.0221	128.7	no	no
L	11.6	0.50995	10.6 / 12.599	4.3962	no	no
tau	163.95	215.55	-258.53 / 586.42	131.48	no	no
B	60	133.74	-202.13 / 322.13	222.9	si	si

rho=0.21593; Jacobian rank=4/4; cond=50157; accordo multistart=no.

Correlazioni: {"K_L": -0.7386460902055484, "K_tau": 0.9993072008470644, "K_B": -0.9998263051307166, "L_tau": -0.7257059843259347, "L_B": 0.7488905946160133, "tau_B": -0.9986417852904379}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 58.511 / 68.594

Griglia: 42441 coppie L/tau; SSE multistart originale: 58.511; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_012-013.csv

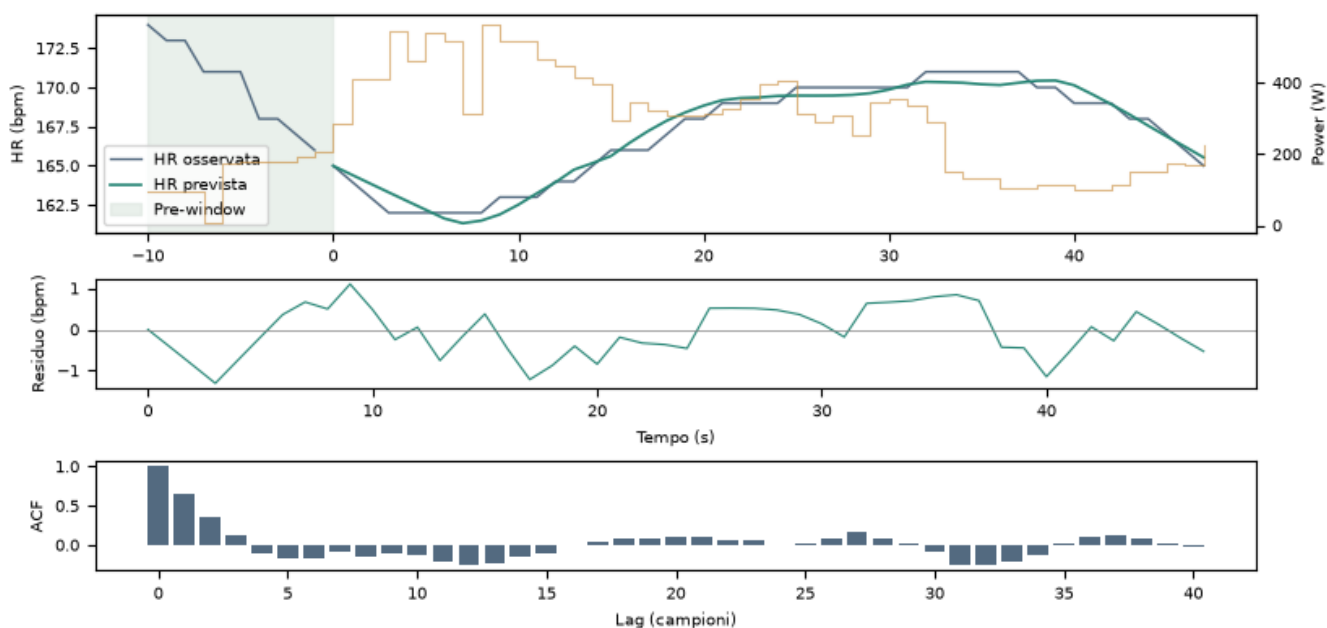
ID: 0e4eaf7bc9e70345c1512fb1 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=140.6 W; HR0=165 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=105 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.59701	0.51357	-0.053862	0.96326	1.0127
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.053862	0.60087	0.64626	-0.18063	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.64996	0.86124	-1.0381 / 2.338	132.51	no	no
L	6.5268	0.34883	5.8431 / 7.2105	5.3446	no	no
tau	184.37	244.1	-294.05 / 662.8	132.39	no	no
B	60	141.94	-218.21 / 338.21	236.57	si	si

rho=0.21952; Jacobian rank=4/4; cond=83220; accordo multistart=no.

Correlazioni: {"K_L": -0.9068993802935756, "K_tau": 0.9995275607121639, "K_B": -0.9999681299085469, "L_tau": -0.9066870525043066, "L_B": 0.9079782119136978, "tau_B": -0.99931616423889}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 17.108 / 252.16

Griglia: 42441 coppie L/tau; SSE multistart originale: 17.108; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_026-027.csv

ID: 0e92d097302f2cd47b9b0c86 | Stato: fitted

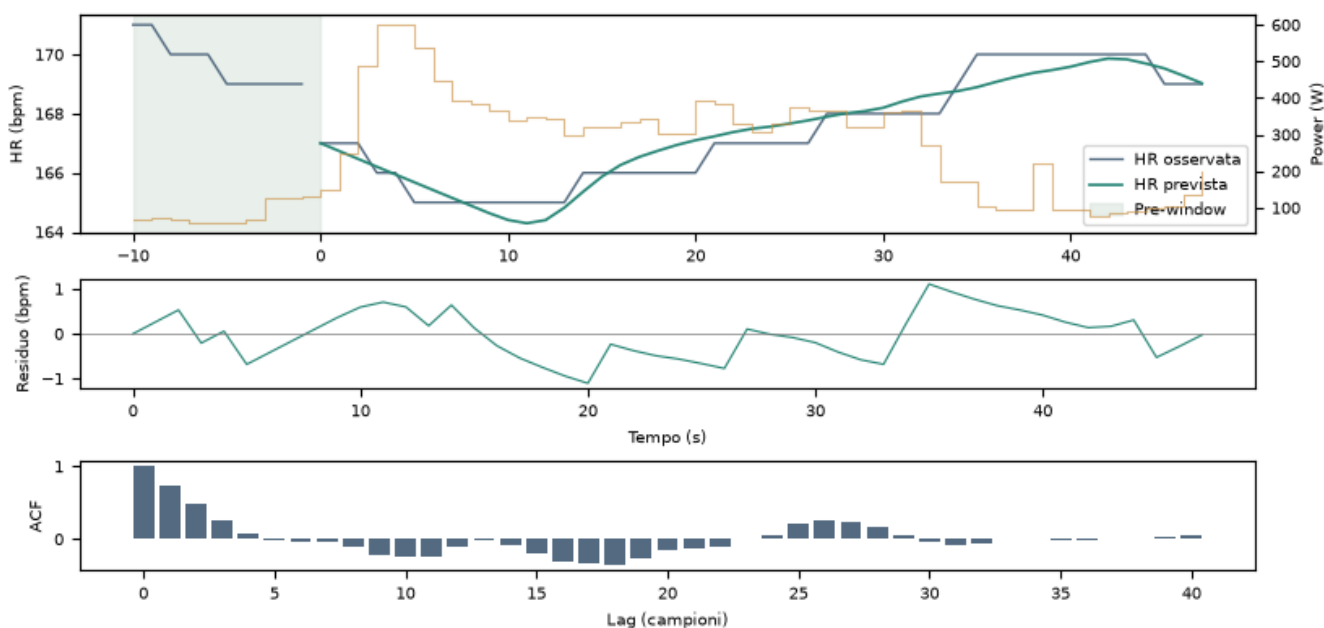
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=84.3 W; HR0=167 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=107 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.51686	0.43078	-0.024327	0.91522	1.1134

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.024327	0.52175	0.73042	-0.015475



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.62444	2.49	-4.256 / 5.5048	398.76	no	no
L	9.7063	0.56631	8.5963 / 10.816	5.8344	no	no
tau	402.68	1616.3	-2765.2 / 3570.6	401.38	no	no
B	60	426.95	-776.82 / 896.82	711.58	no	si

rho=0.092613; Jacobian rank=4/4; cond=2.7995e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.7554648540549632, "K_tau": 0.9998511431075096, "K_B": -0.9999805852554728, "L_tau": -0.749405695889704, "L_B": 0.7587455023686257, "tau_B": -0.9997513224280455}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 12.823 / 12.989

Griglia: 42441 coppie L/tau; SSE multistart originale: 12.823; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_064-065.csv

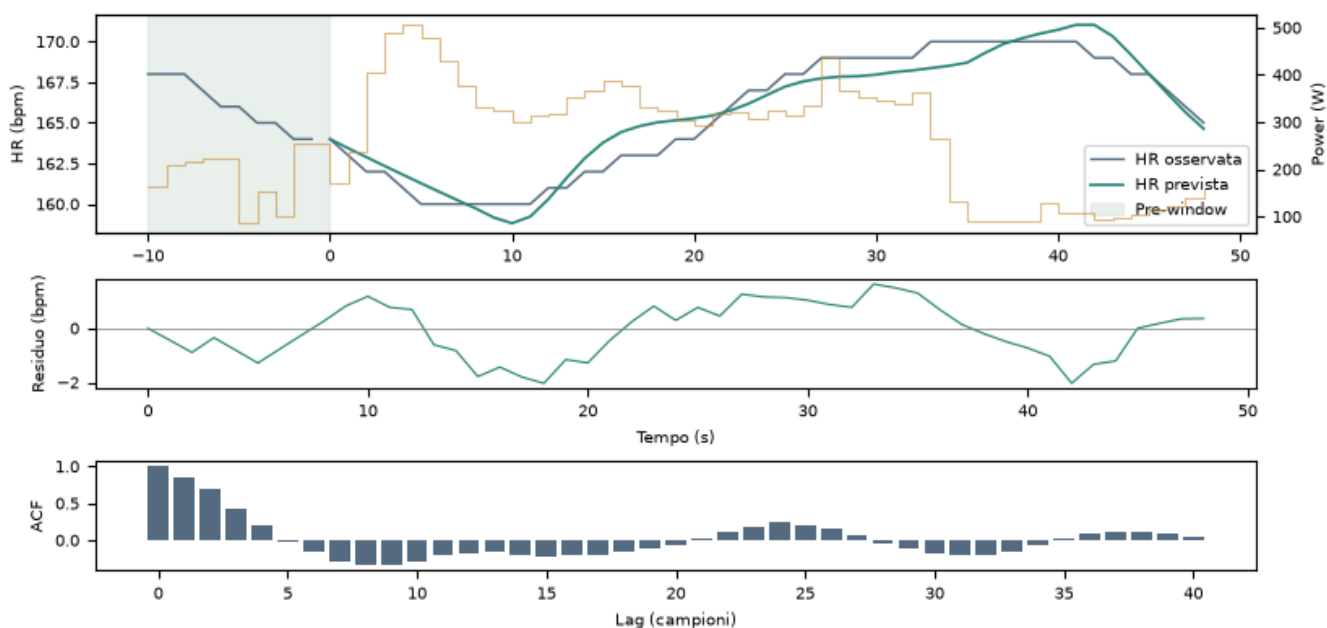
ID: 0f496556da5c7faed448db66 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=188.7 W; HR0=164 bpm.

HR(0) osservata fissata; B=122.64 bpm stimato; x(0)=41.361 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.99836	0.85344	-0.090362	0.92668	1.2213
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.090362	1.0046	0.85422	-0.012864	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.43126	0.26077	-0.079851 / 0.94237	60.467	no	no
L	8.3656	0.37826	7.6242 / 9.107	4.5216	no	no
tau	73.543	46.419	-17.438 / 164.52	63.118	no	no
B	122.64	25.552	72.557 / 172.72	20.835	no	no

rho=0.53893; Jacobian rank=4/4; cond=12747; accordo multistart=no.

Correlazioni: {"K_L": -0.8242726474403971, "K_tau": 0.9955607033797929, "K_B": -0.9995634619659175, "L_tau": -0.8177250500122137, "L_B": 0.8293048222692121, "tau_B": -0.9931848853659329}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 48.84 / 383.78

Griglia: 42441 coppie L/tau; SSE multistart originale: 48.84; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_068-069.csv

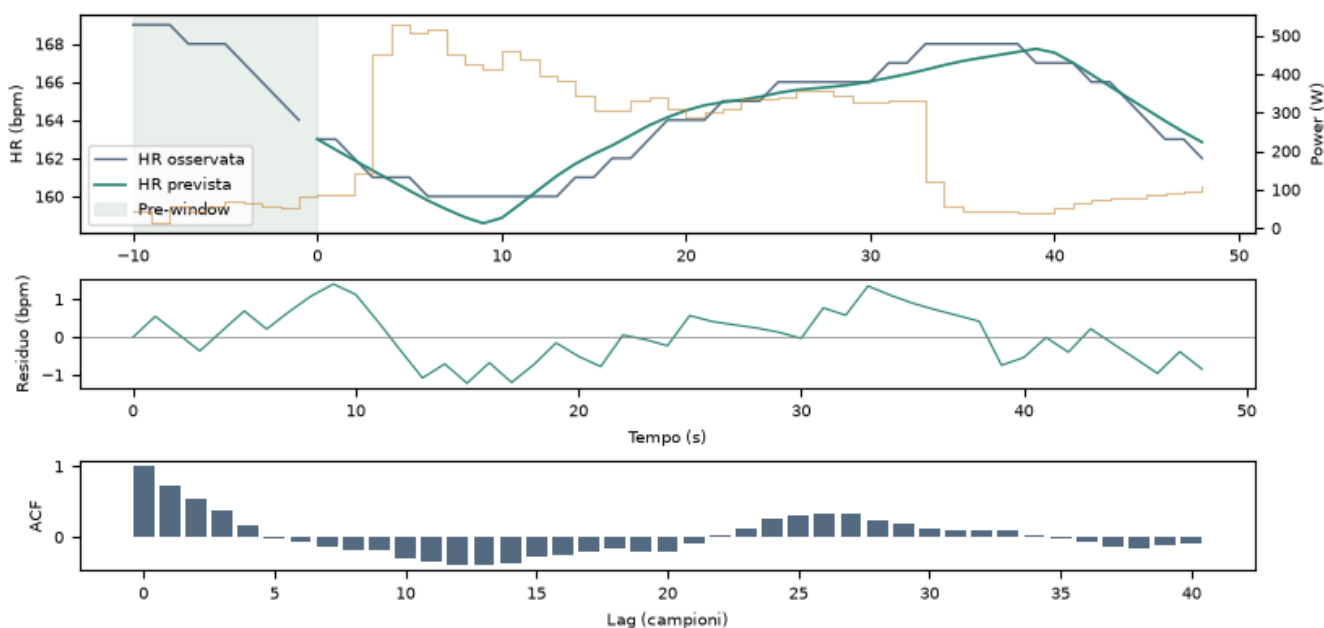
ID: 1651df43a12f3077d8734e4e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=53.5 W; HR0=163 bpm.

HR(0) osservata fissata; B=116.41 bpm stimato; x(0)=46.585 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.67703	0.5622	0.045326	0.93946	1.1437
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.045326	0.68251	0.72332	-0.0096045	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23175	0.17661	-0.11439 / 0.5779	76.204	no	no
L	6.3718	0.44145	5.5065 / 7.237	6.9283	no	no
tau	83.919	64.759	-43.01 / 210.85	77.169	no	no
B	116.41	36.148	45.564 / 187.27	31.051	no	no

rho=0.49606; Jacobian rank=4/4; cond=46137; accordo multistart=no.

Correlazioni: {"K_L": -0.8999468665172285, "K_tau": 0.9980733371412649, "K_B": -0.9998870287722018, "L_tau": -0.8950129854829916, "L_B": 0.9029213238620216, "tau_B": -0.9973954090478987}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 22.46 / 256.03

Griglia: 42441 coppie L/tau; SSE multistart originale: 22.46; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_074-075.csv

ID: 176da74b0342a10d297c514a | Stato: fitted

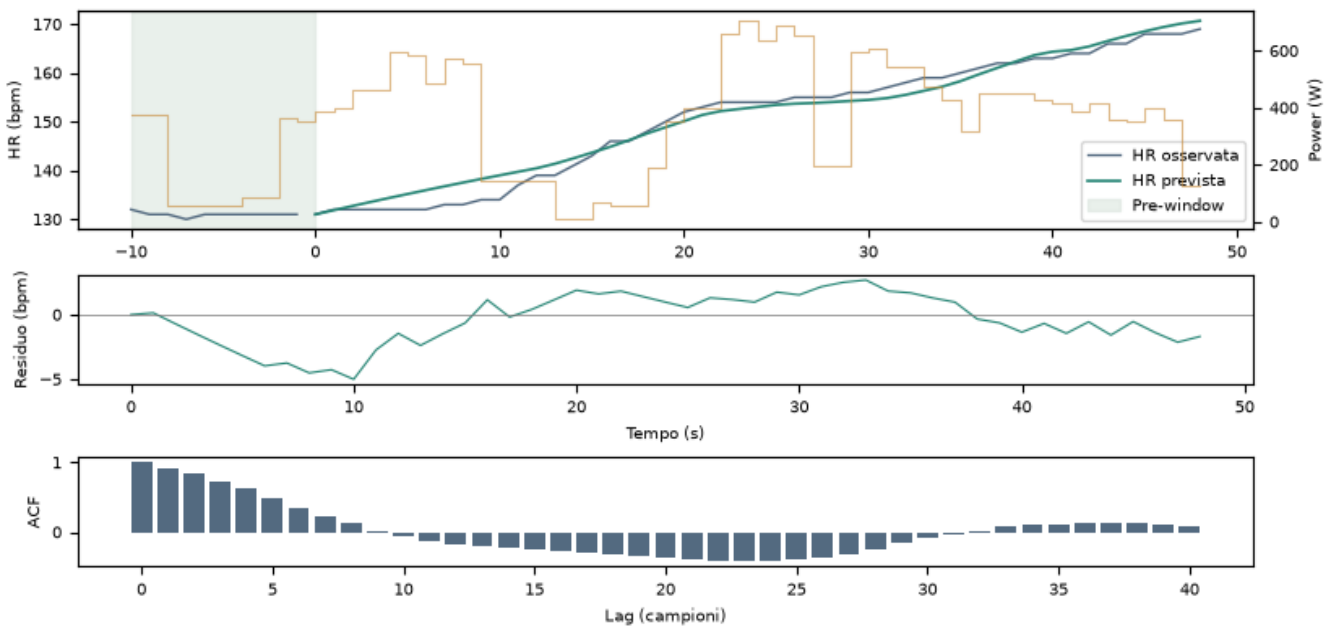
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=185.2 W; HR0=131 bpm.

HR(0) osservata fissata; B=182.33 bpm stimato; x(0)=-51.332 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0161	1.6629	-0.40092	0.97357	1.0445

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.40092	1.9963	0.91275	0.48603



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.1176	0.021864	0.074749 / 0.16046	18.591	no	no
L	12.428	0.81918	10.823 / 14.034	6.5912	no	no
tau	58.962	16.12	27.366 / 90.557	27.34	no	no
B	182.33	12.508	157.82 / 206.85	6.8601	no	no

rho=0.6033; Jacobian rank=4/4; cond=2965.9; accordo multistart=no.

Correlazioni: {"K_L": 0.024141480387395057, "K_tau": 0.40787417387266794, "K_B": 0.23248638130923865, "L_tau": -0.05327375107948356, "L_B": -0.014229200601931995, "tau_B": 0.9796392920636179}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 199.16 / 317.66

Griglia: 42441 coppie L/tau; SSE multistart originale: 199.16; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_072-073.csv

ID: 1ad5016c5e01c44ec0fd03f5 | Stato: fitted

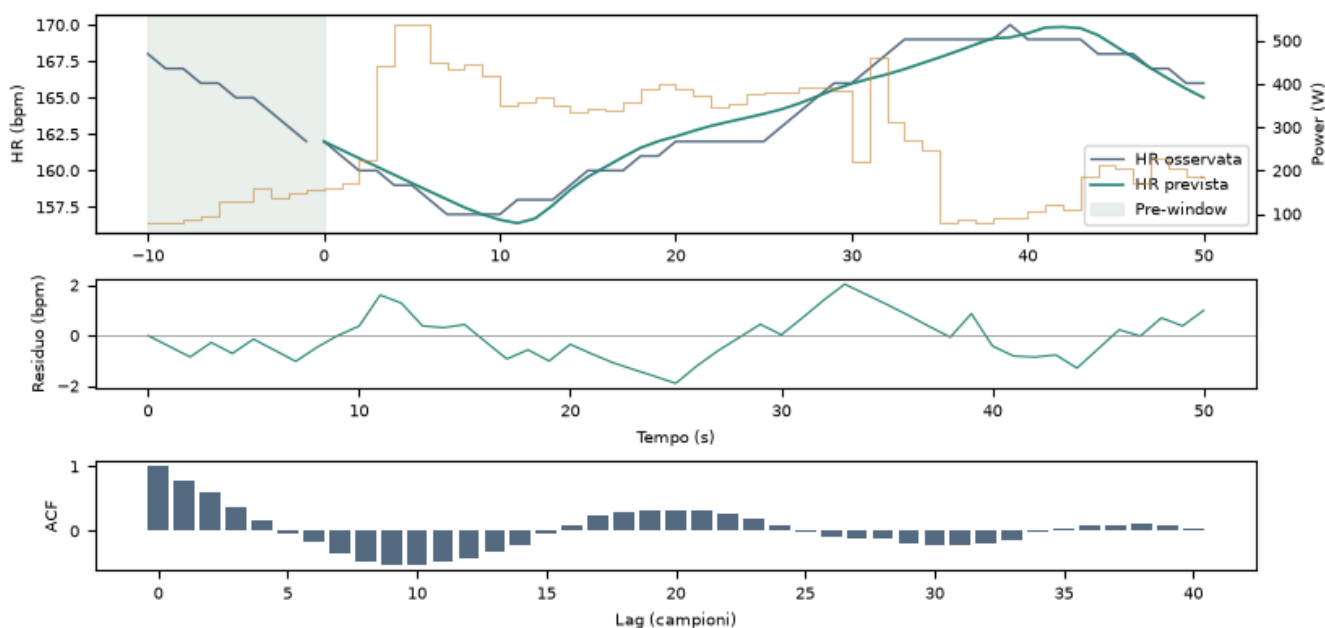
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=120.3 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.88671	0.72722	-0.090411	0.95718	1.0353

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.090411	0.89087	0.78526	-0.038796



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.6873	0.84102	-0.96111 / 2.3357	122.37	no	no
L	8.4592	0.44896	7.5793 / 9.3392	5.3073	no	no
tau	176.01	218.34	-251.95 / 603.96	124.06	no	no
B	60	126.01	-186.99 / 306.99	210.02	si	si

rho=0.23602; Jacobian rank=4/4; cond=52817; accordo multistart=no.

Correlazioni: {"K_L": -0.8693720647254185, "K_tau": 0.9993249315348479, "K_B": -0.9999172413180234, "L_tau": -0.8630021315118891, "L_B": 0.8732383367561093, "tau_B": -0.9988926284722041}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 40.099 / 527.66

Griglia: 42441 coppie L/tau; SSE multistart originale: 40.099; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_024-025.csv

ID: 1f616c303868d0569795f81a | Stato: fitted

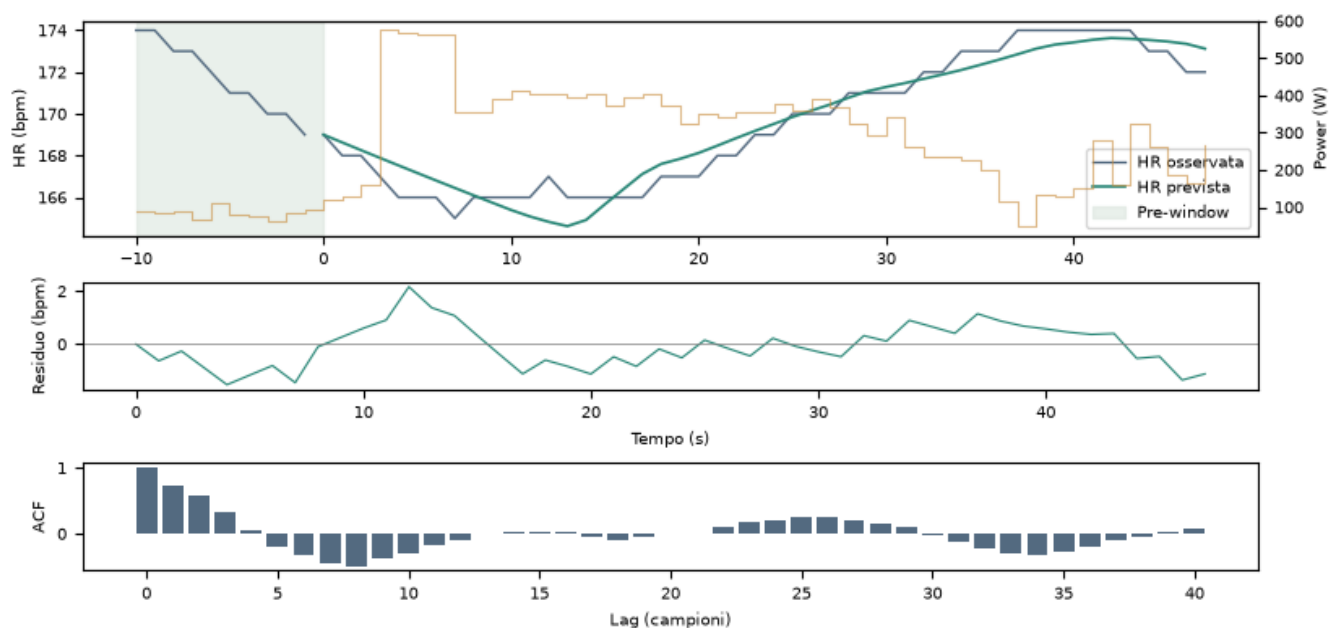
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=83.8 W; HR0=169 bpm.

HR(0) osservata fissata; B=112.35 bpm stimato; x(0)=56.65 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.80358	0.66421	-0.079392	0.9305	1.0009

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.079392	0.80811	0.7432	-0.20324



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.34211	0.50202	-0.64185 / 1.3261	146.74	no	no
L	10.494	0.64215	9.2352 / 11.752	6.1193	no	no
tau	152.14	228.25	-295.24 / 599.51	150.03	no	no
B	112.35	83.183	-50.689 / 275.39	74.039	no	no

rho=0.23996; Jacobian rank=4/4; cond=64937; accordo multistart=no.

Correlazioni: {"K_L": -0.7661587115293945, "K_tau": 0.9992151809618927, "K_B": -0.9998046759606922, "L_tau": -0.7535534102440555, "L_B": 0.7762844130691615, "tau_B": -0.9984691337571949}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 30.996 / 33.906

Griglia: 42441 coppie L/tau; SSE multistart originale: 30.996; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_066-067.csv

ID: 20cb88dcc3fc015110084812 | Stato: fitted

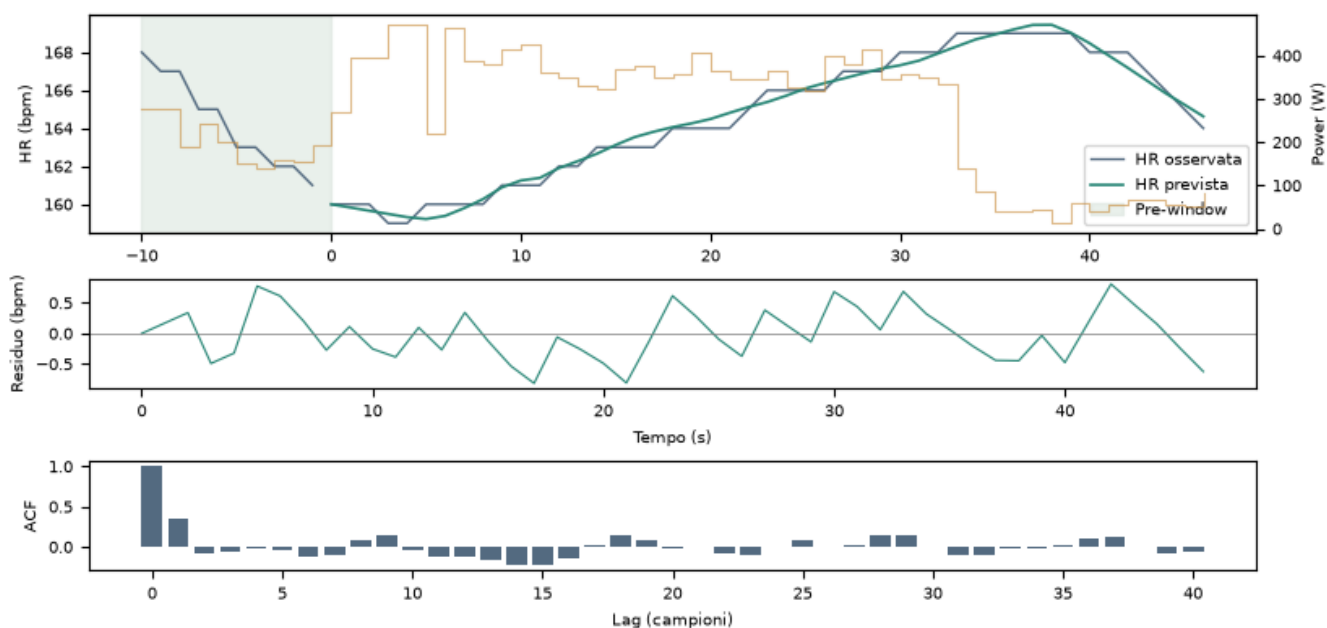
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=199.2 W; HR0=160 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=100 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.41314	0.3443	-0.0076468	0.98437	1.0223

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.0076468	0.41753	0.35094	-0.039549



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.7072	4.2654	-6.653 / 10.067	249.85	no	no
L	4.6509	0.31856	4.0265 / 5.2752	6.8496	no	no
tau	588.1	1471.4	-2295.9 / 3472.1	250.2	no	no
B	60	259.22	-448.07 / 568.07	432.03	si	si

rho=0.07031; Jacobian rank=4/4; cond=1.3847e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.878992797310612, "K_tau": 0.9998390357391258, "K_B": -0.9997803751362793, "L_tau": -0.8819677888547811, "L_B": 0.8759916069985689, "tau_B": -0.9992577671539886}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 8.0222 / 104.69

Griglia: 42441 coppie L/tau; SSE multistart originale: 8.0222; selezione: profile_grid.

Segnalazioni: at_bounds, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_078-079.csv

ID: 22c9cd899da4fae50a900da7 | Stato: fitted

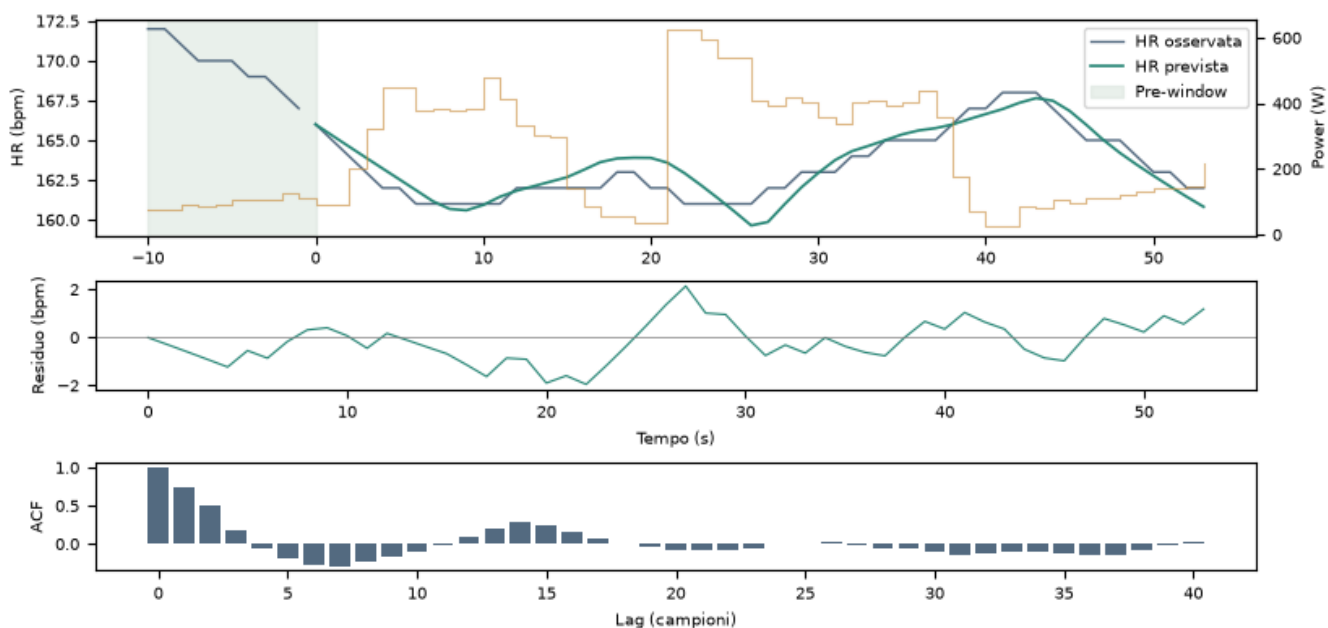
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=98.1 W; HR0=166 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=106 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.86316	0.69931	-0.1721	0.82995	1.1436

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.1721	0.85377	0.75196	-0.19079



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.51068	1.037	-1.5218 / 2.5432	203.06	no	no
L	5.4544	0.28531	4.8952 / 6.0136	5.2307	no	no
tau	150.89	307.38	-451.58 / 753.35	203.72	no	no
B	60	210.06	-351.72 / 471.72	350.1	si	si

rho=0.31511; Jacobian rank=4/4; cond=1.0396e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.7484516062102401, "K_tau": 0.9997326353803339, "K_B": -0.9999922333559254, "L_tau": -0.7444254960683444, "L_B": 0.7495070920066014, "tau_B": -0.9997506890171517}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 40.232 / 223.66

Griglia: 42441 coppie L/tau; SSE multistart originale: 40.232; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_060-061.csv

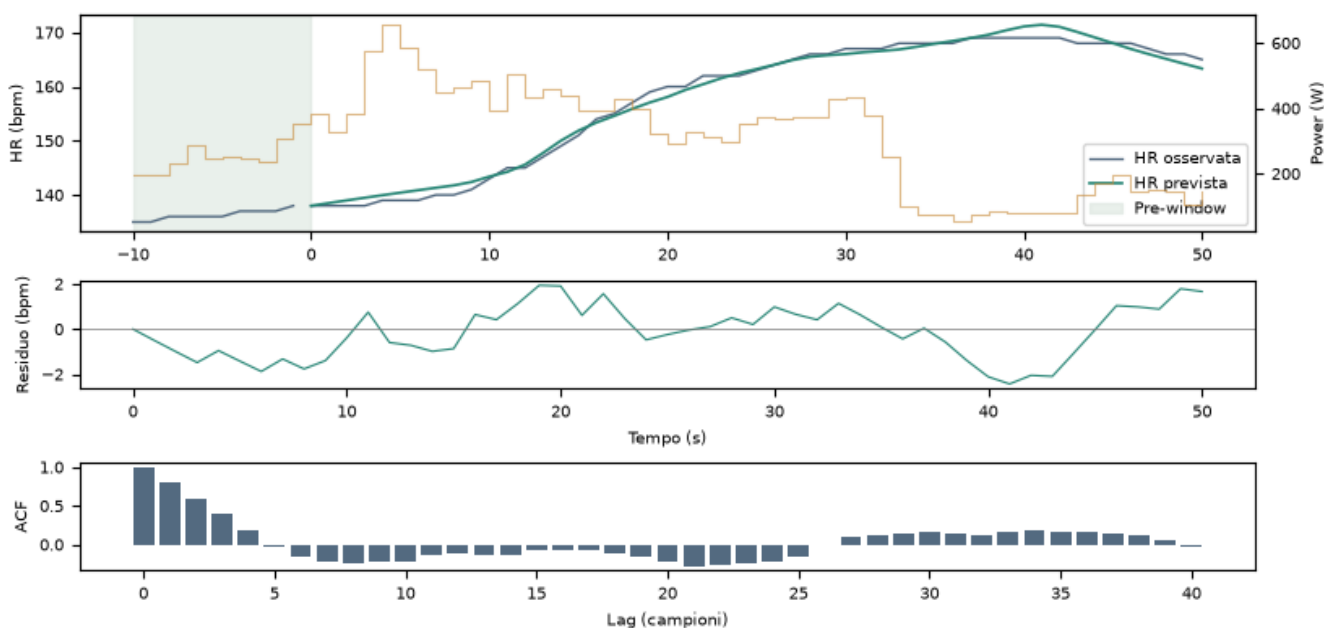
ID: 24f0cde09312a660fba76b6e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=255.3 W; HR0=138 bpm.

HR(0) osservata fissata; B=164.87 bpm stimato; x(0)=-26.87 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1485	0.95457	-0.14837	0.99008	1.0782
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.14837	1.1502	0.80125	-0.03346	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.27453	0.043244	0.18977 / 0.35929	15.752	no	no
L	8.6953	0.40919	7.8933 / 9.4974	4.7059	no	no
tau	53.069	7.9456	37.496 / 68.643	14.972	no	no
B	164.87	2.1693	160.62 / 169.12	1.3158	no	no

rho=0.77832; Jacobian rank=4/4; cond=2203.8; accordo multistart=no.

Correlazioni: {"K_L": -0.7379192297450845, "K_tau": 0.9471359704242572, "K_B": 0.4085713592351316, "L_tau": -0.8373498999441915, "L_B": -0.6896177990562808, "tau_B": 0.6729695456316994}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 67.273 / 637.35

Griglia: 42441 coppie L/tau; SSE multistart originale: 67.273; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_023-024.csv

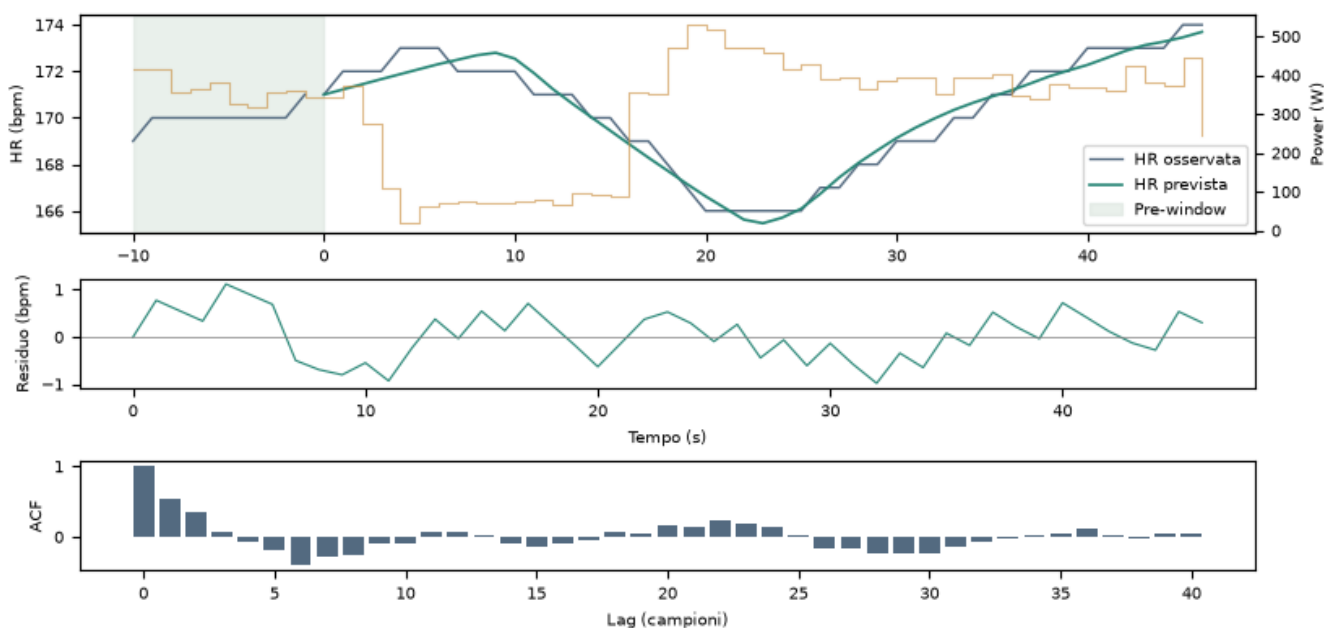
ID: 28f17f581c36976c89c202d0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=363.2 W; HR0=171 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-29 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.50909	0.42482	0.03612	0.95738	1.0284
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.03612	0.51329	0.53382		-0.17548



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.36566	0.52663	-0.66654 / 1.3979	144.02	no	no
L	6.5345	0.37153	5.8063 / 7.2626	5.6857	no	no
tau	129.92	188.46	-239.46 / 499.3	145.06	no	no
B	200	43.397	114.94 / 285.06	21.698	si	si

rho=0.30377; Jacobian rank=4/4; cond=42995; accordo multistart=no.

Correlazioni: {"K_L": -0.8732238513924379, "K_tau": 0.9997471344647754, "K_B": 0.9999496229677793, "L_tau": -0.8754964416507891, "L_B": -0.8733181823655551, "tau_B": 0.9996640389483379}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 12.181 / 271.31

Griglia: 42441 coppie L/tau; SSE multistart originale: 12.181; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_009-010.csv

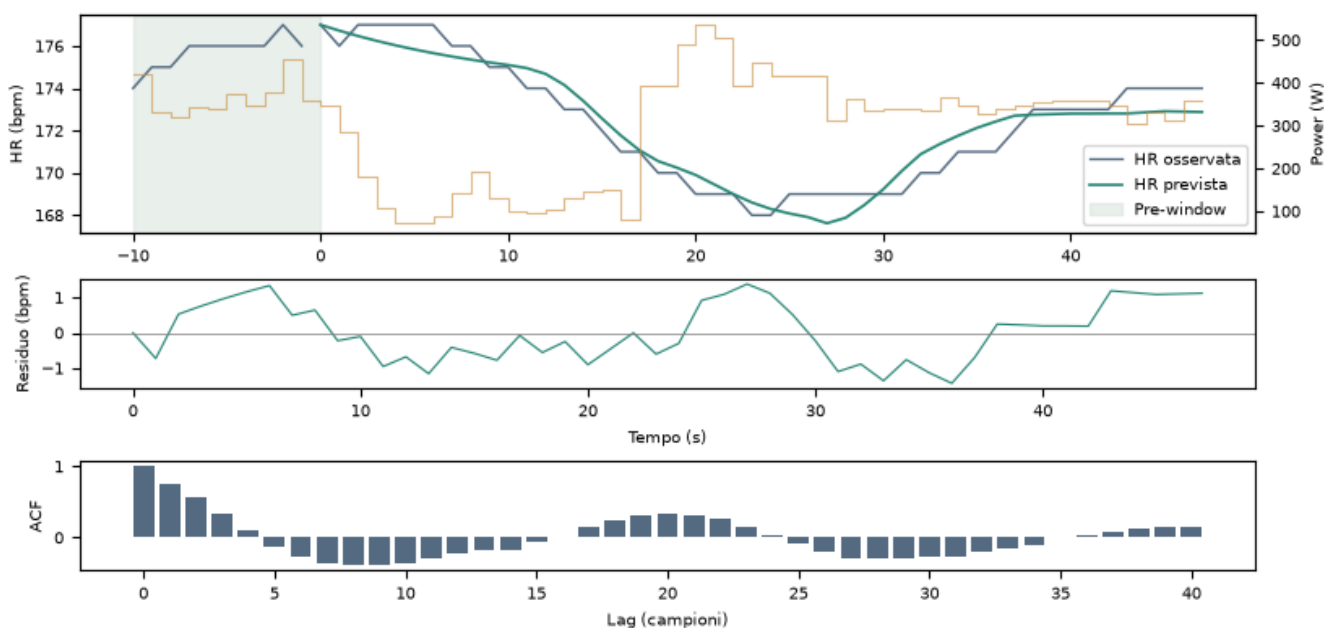
ID: 2a7d8e43dfc5ad32919d82a0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=366.4 W; HR0=177 bpm.

HR(0) osservata fissata; B=173.82 bpm stimato; x(0)=3.1786 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.81826	0.70678	0.02762	0.91701	1.0423
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.02762	0.82645	0.76138	-0.1241	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.035541	0.0043269	0.02706 / 0.044021	12.175	no	no
L	10.39	0.48277	9.4441 / 11.337	4.6464	no	no
tau	10.987	1.4444	8.1566 / 13.818	13.145	no	no
B	173.82	0.29227	173.25 / 174.39	0.16814	no	no

rho=3.3319; Jacobian rank=4/4; cond=1056.3; accordo multistart=no.

Correlazioni: {"K_L": -0.5983190743140532, "K_tau": 0.8419502431324986, "K_B": 0.7974274698418164, "L_tau": -0.7299675323808624, "L_B": -0.349110266648448, "tau_B": 0.5086391924962175}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 32.138 / 361.66

Griglia: 42441 coppie L/tau; SSE multistart originale: 34.852; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_025-026.csv

ID: 2fd0d9e138e25277b3e86300 | Stato: fitted

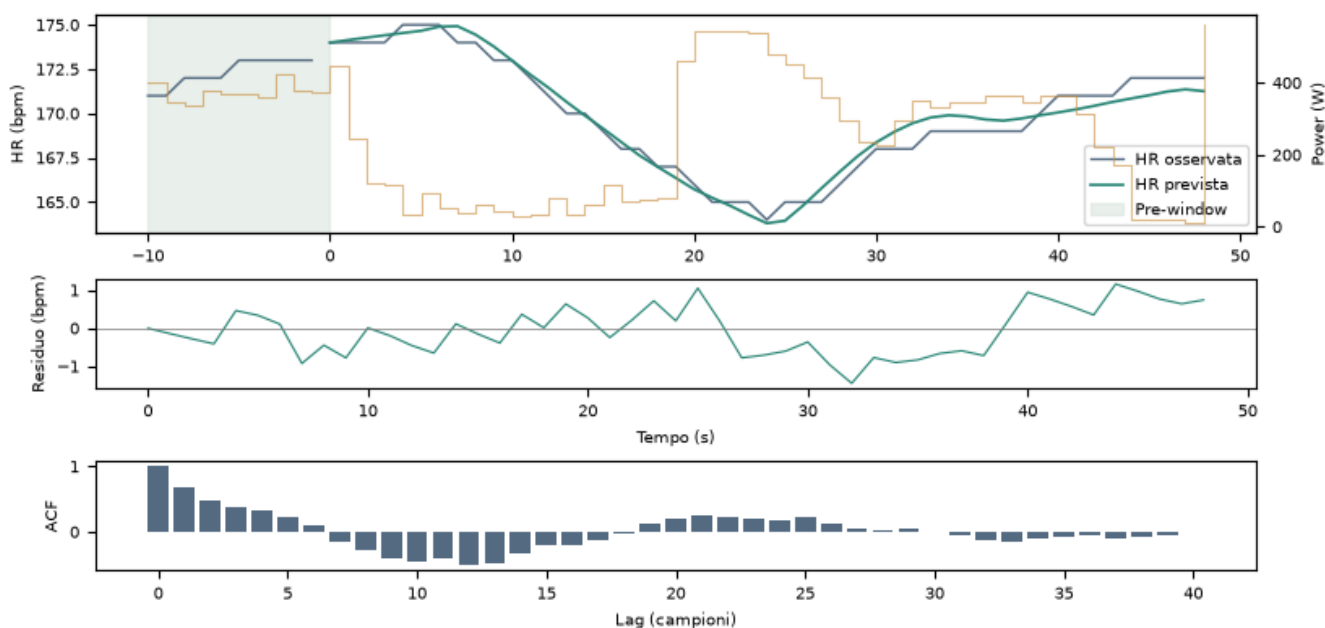
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=374.2 W; HR0=174 bpm.

HR(0) osservata fissata; B=178.41 bpm stimato; $x(0)=-4.4055$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.62745	0.53201	-0.054222	0.96052	1.0107

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.054222	0.63159	0.67407	0.22154



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.097575	0.023025	0.052447 / 0.1427	23.597	no	no
L	5.507	0.30678	4.9057 / 6.1083	5.5708	no	no
tau	30.777	7.0009	17.056 / 44.499	22.747	no	no
B	178.41	2.1313	174.23 / 182.58	1.1946	no	no

rho=1.3807; Jacobian rank=4/4; cond=5863.1; accordo multistart=no.

Correlazioni: {"K_L": -0.8086839835631382, "K_tau": 0.9895113489952155, "K_B": 0.9951215693928626, "L_tau": -0.8270608496233409, "L_B": -0.7971180366789371, "tau_B": 0.976568303860545}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 19.291 / 497.08

Griglia: 42441 coppie L/tau; SSE multistart originale: 19.291; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_031-032.csv

ID: 3191661f1d8e9009b460f004 | Stato: fitted

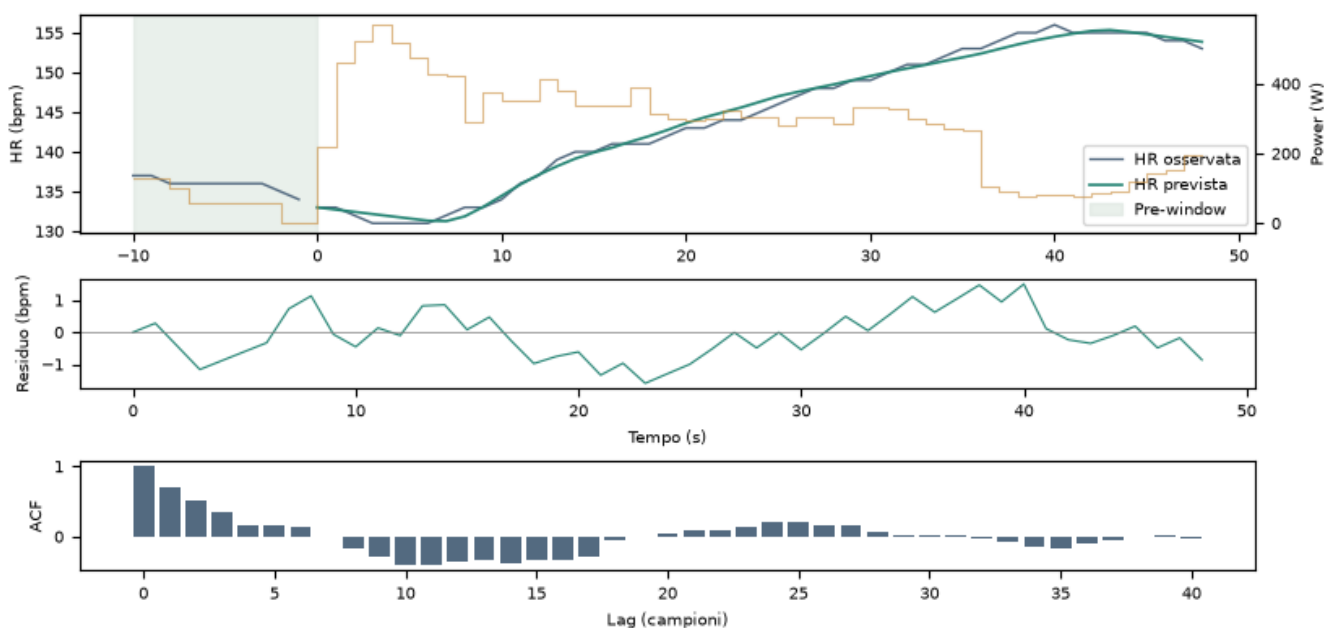
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=65.5 W; HR0=133 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=73 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.74279	0.59668	-0.08125	0.99241	0.96312

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.08125	0.74599	0.69866	0.17187



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.91296	0.66821	-0.39674 / 2.2227	73.192	no	no
L	6.6006	0.36972	5.8759 / 7.3252	5.6013	no	no
tau	258.9	187.01	-107.63 / 625.43	72.229	no	no
B	60	63.421	-64.305 / 184.3	105.7	si	si

rho=0.1599; Jacobian rank=4/4; cond=42155; accordo multistart=no.

Correlazioni: {"K_L": -0.8792280222026134, "K_tau": 0.9971422359017034, "K_B": -0.9928399981959226, "L_tau": -0.8691668290281908, "L_B": 0.891450184500017, "tau_B": -0.9811429362625793}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 27.035 / 286.46

Griglia: 42441 coppie L/tau; SSE multistart originale: 27.035; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_055-056.csv

ID: 3721270baf948942b57737f7 | Stato: fitted

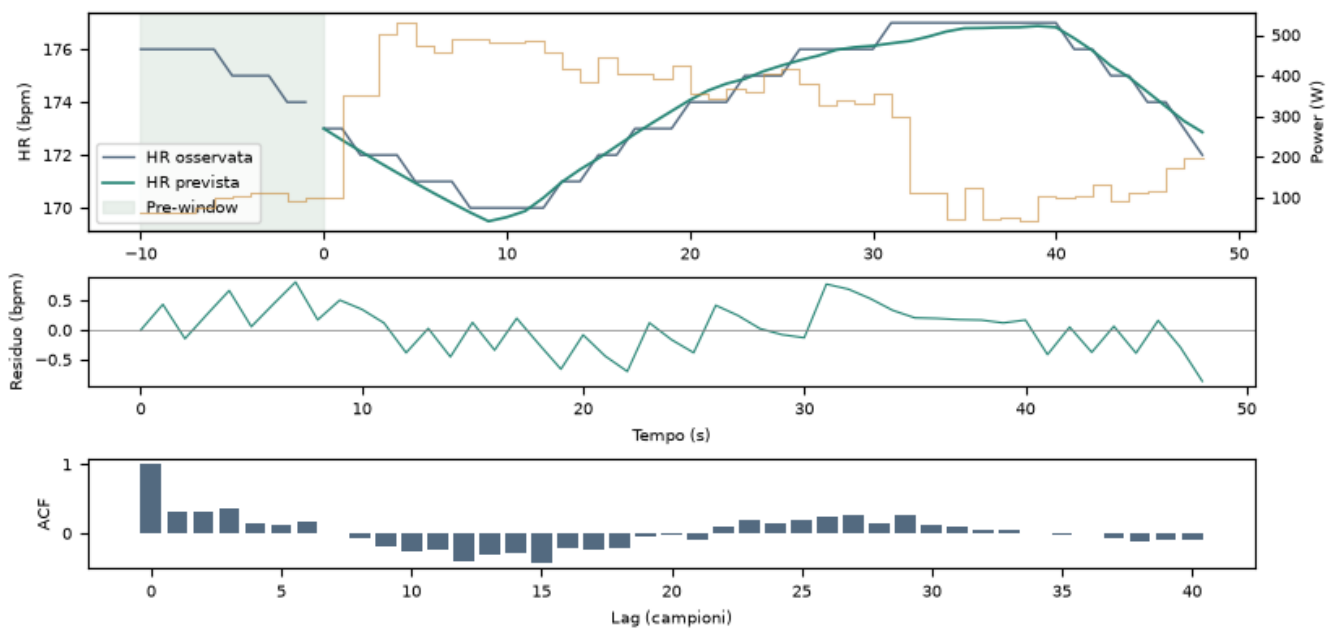
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=88.1 W; HR0=173 bpm.

HR(0) osservata fissata; B=155.75 bpm stimato; x(0)=17.253 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.38122	0.30892	0.042785	0.97435	1.0548

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.042785	0.38273	0.30786	0.12379



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.087183	0.024516	0.039132 / 0.13523	28.12	no	no
L	8.144	0.38293	7.3935 / 8.8946	4.7019	no	no
tau	39.374	11.668	16.506 / 62.243	29.633	no	no
B	155.75	5.0821	145.79 / 165.71	3.2631	no	no

rho=1.0122; Jacobian rank=4/4; cond=25307; accordo multistart=no.

Correlazioni: {"K_L": -0.9300866271485844, "K_tau": 0.9944172567906717, "K_B": -0.9995084014388647, "L_tau": -0.9265124368878196, "L_B": 0.9331686536908885, "tau_B": -0.9920860566603418}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 7.121 / 188.28

Griglia: 42441 coppie L/tau; SSE multistart originale: 7.121; selezione: multistart.

Segnalazioni: equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_086-087.csv

ID: 3f6df0991bbb2bc2387b781c | Stato: fitted

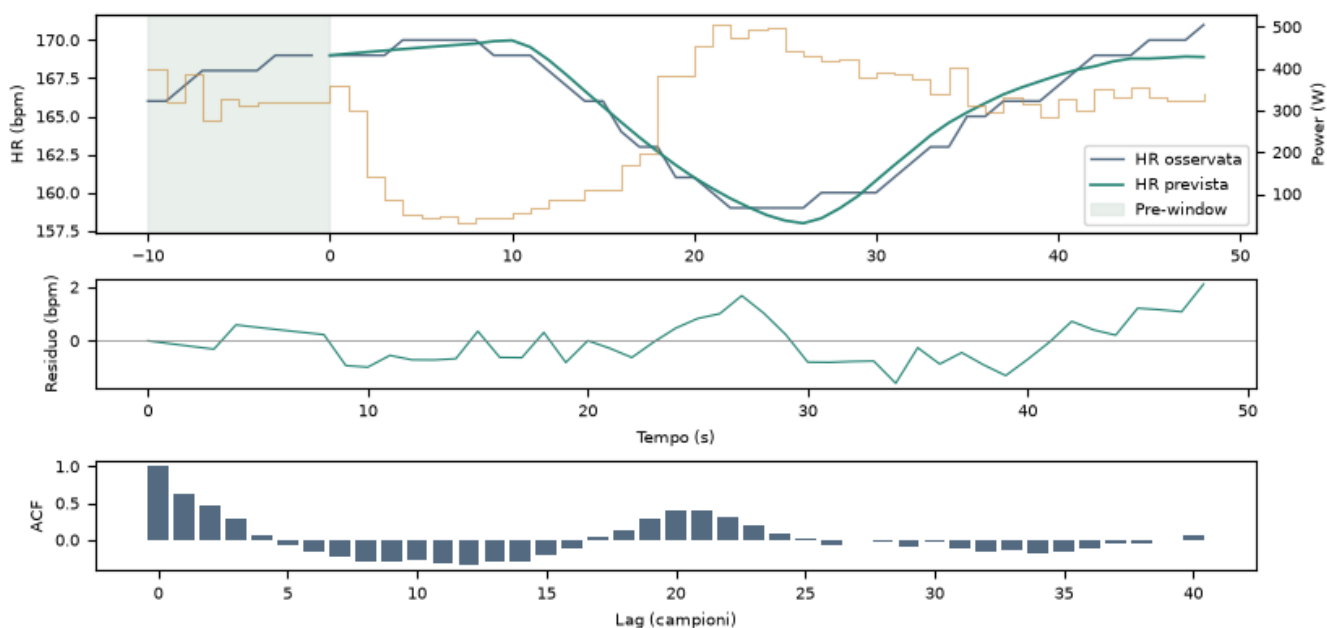
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=330.7 W; HR0=169 bpm.

HR(0) osservata fissata; B=172.6 bpm stimato; x(0)=-3.5963 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.7909	0.65704	-0.051053	0.9615	0.99883

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.051053	0.79743	0.63408	-0.062418



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.14151	0.043809	0.055649 / 0.22738	30.957	no	no
L	8.4277	0.42119	7.6022 / 9.2533	4.9976	no	no
tau	33.211	10.435	12.758 / 53.665	31.421	no	no
B	172.6	2.3296	168.03 / 177.16	1.3498	no	no

rho=1.1915; Jacobian rank=4/4; cond=4318.9; accordo multistart=no.

Correlazioni: {"K_L": -0.8788158878576593, "K_tau": 0.9949549462681306, "K_B": 0.9942484347091793, "L_tau": -0.8903731217473623, "L_B": -0.8635488761601533, "tau_B": 0.9848409400548357}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 30.651 / 643.81

Griglia: 42441 coppie L/tau; SSE multistart originale: 30.651; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_037-038.csv

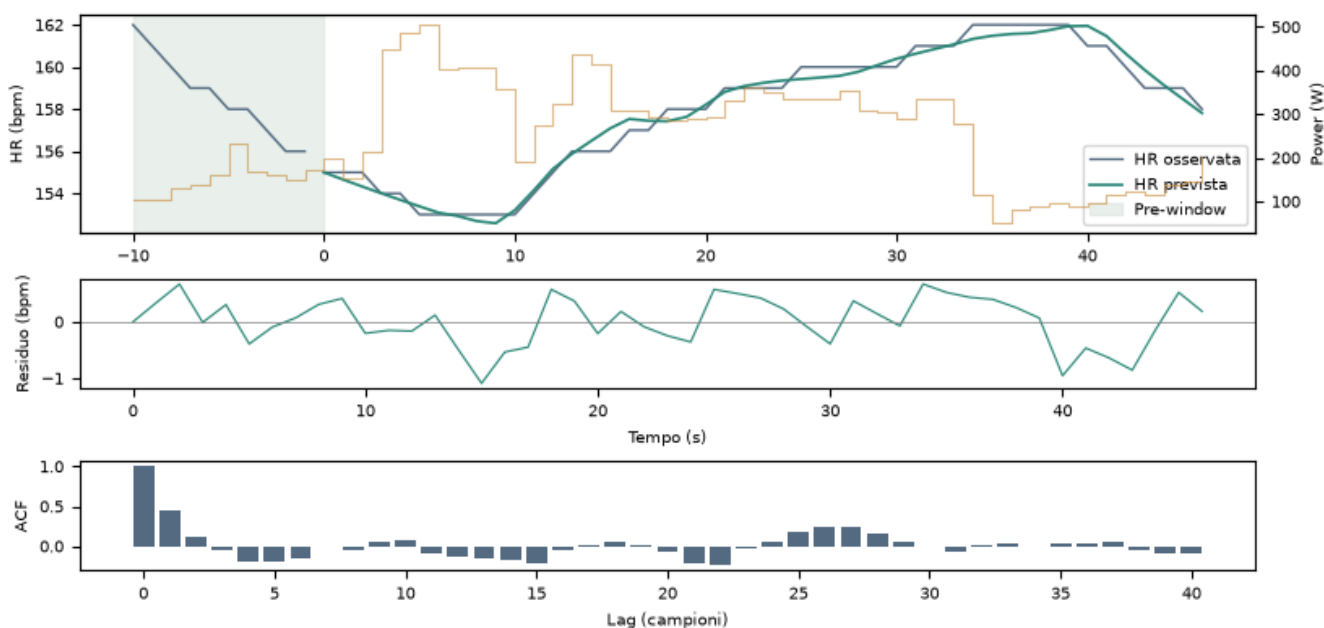
ID: 4879078e57cf3f0053d098d2 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=152.6 W; HR0=155 bpm.

HR(0) osservata fissata; B=141.55 bpm stimato; x(0)=13.447 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.43006	0.35547	0.01236	0.97949	1.0408
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.01236	0.43453	0.45352	-0.18957	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.15195	0.025089	0.10278 / 0.20113	16.511	no	no
L	6.2658	0.23257	5.81 / 6.7216	3.7117	no	no
tau	39.211	7.0291	25.434 / 52.989	17.926	no	no
B	141.55	2.6329	136.39 / 146.71	1.86	no	no

rho=1.0133; Jacobian rank=4/4; cond=7574.3; accordo multistart=no.

Correlazioni: {"K_L": -0.8145442057608618, "K_tau": 0.972915184899984, "K_B": -0.9941564434395379, "L_tau": -0.81882823240932, "L_B": 0.8100041632625299, "tau_B": -0.9461304417254593}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 8.6928 / 139.62

Griglia: 42441 coppie L/tau; SSE multistart originale: 8.6928; selezione: multistart.

Segnalazioni: equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_043-044.csv

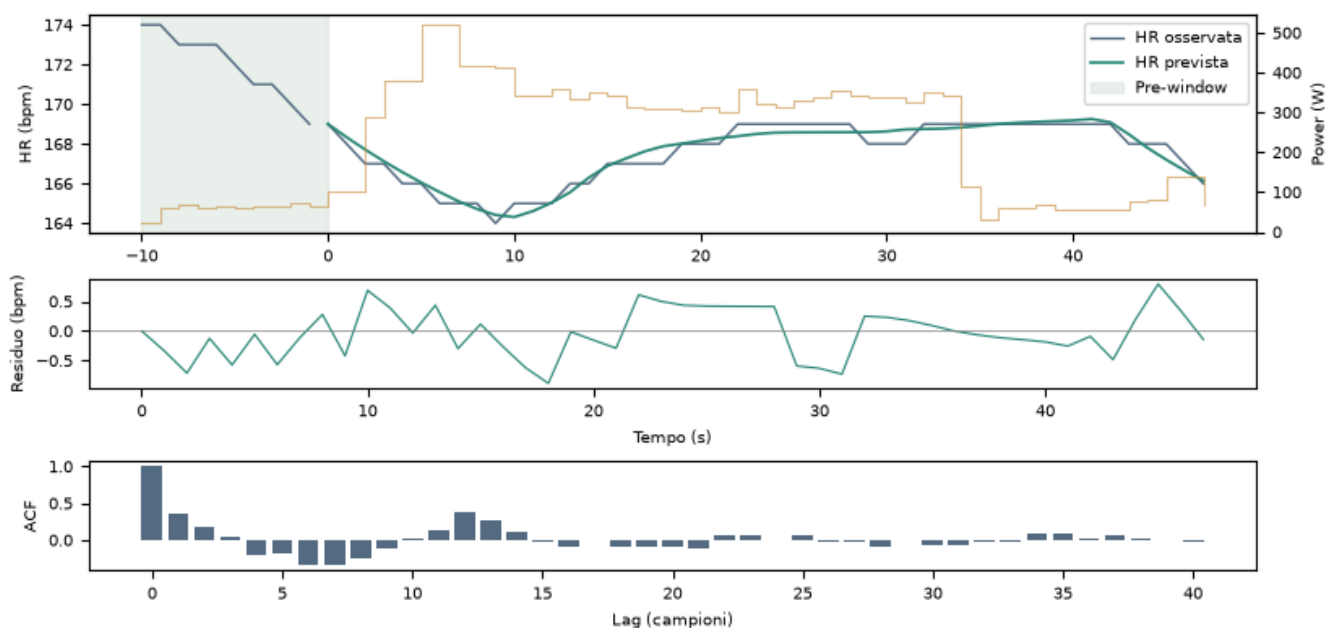
ID: 4fde6ee27c018254db01002a | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=60.2 W; HR0=169 bpm.

HR(0) osservata fissata; B=158.04 bpm stimato; x(0)=10.958 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.40639	0.33491	-0.034737	0.92637	0.98796
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.034737	0.40919	0.36416	-0.18998	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.042606	0.0082945	0.026348 / 0.058863	19.468	no	no
L	7.641	0.30758	7.0382 / 8.2439	4.0253	no	no
tau	15.971	3.3359	9.4329 / 22.51	20.887	no	no
B	158.04	1.8873	154.34 / 161.74	1.1942	no	no

rho=2.4643; Jacobian rank=4/4; cond=10461; accordo multistart=no.

Correlazioni: {"K_L": -0.8501636010392792, "K_tau": 0.9810752083557753, "K_B": -0.9989035861303771, "L_tau": -0.8256028541989894, "L_B": 0.8562594707707768, "tau_B": -0.9794756566792191}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 7.9274 / 105.51

Griglia: 42441 coppie L/tau; SSE multistart originale: 7.9274; selezione: multistart.

Segnalazioni: multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_045-046.csv

ID: 514ee79882a2a7f3ff8def37 | Stato: fitted

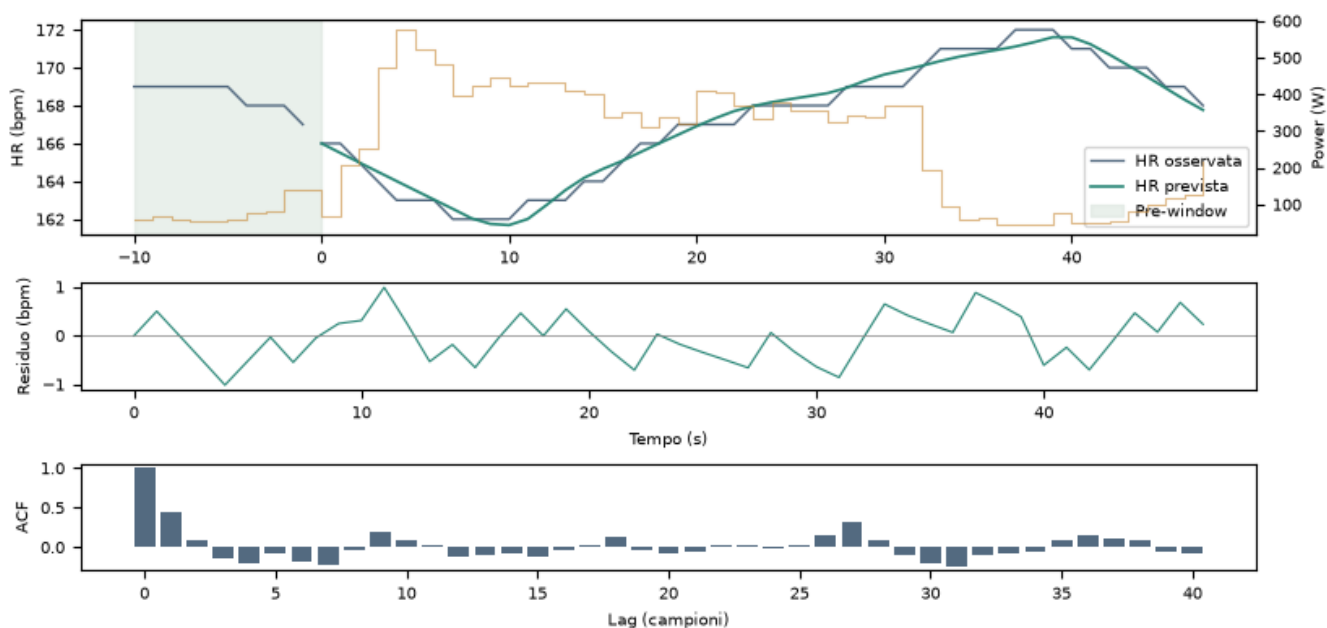
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=78.5 W; HR0=166 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=106 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.47715	0.38784	-0.04457	0.97678	0.99228

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.04457	0.4801	0.44447	-0.083997



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.57186	0.71575	-0.83102 / 1.9747	125.16	no	no
L	7.438	0.33531	6.7808 / 8.0952	4.5081	no	no
tau	211.22	266.3	-310.72 / 733.16	126.08	no	no
B	60	133.74	-202.13 / 322.13	222.9	si	si

rho=0.1873; Jacobian rank=4/4; cond=1.0962e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.8961918252255432, "K_tau": 0.9996497050688844, "K_B": -0.9999655172786897, "L_tau": -0.8922432199915964, "L_B": 0.8984657542842177, "tau_B": -0.9994559371562254}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 10.928 / 272.62

Griglia: 42441 coppie L/tau; SSE multistart originale: 10.928; selezione: profile_grid.

Segnalazioni: at_bounds, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_082-083.csv

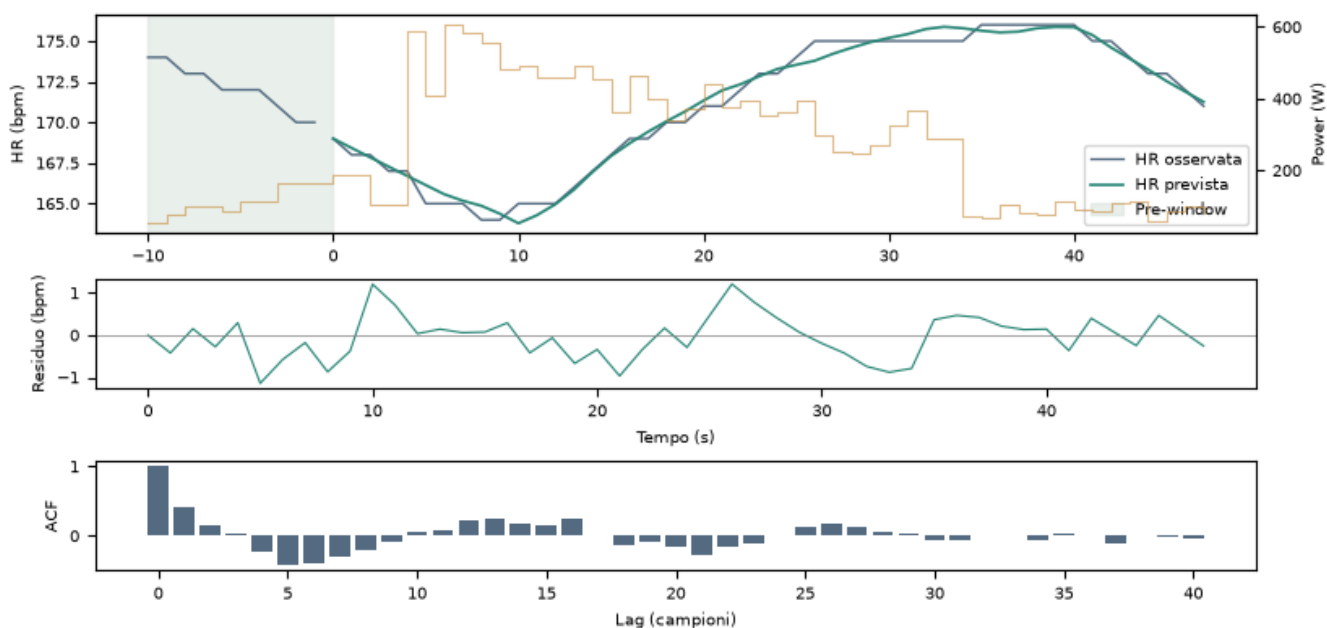
ID: 602f528f28e78481167bc724 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=113.6 W; HR0=169 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=109 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.51618	0.40985	-0.040963	0.9838	1.0071
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.040963	0.51999	0.41074		-0.43082



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.64701	0.54925	-0.42952 / 1.7235	84.891	no	no
L	6.3543	0.26576	5.8334 / 6.8752	4.1824	no	no
tau	188.04	160.66	-126.84 / 502.93	85.435	no	no
B	60	94.02	-124.28 / 244.28	156.7	si	si

rho=0.21615; Jacobian rank=4/4; cond=67094; accordo multistart=no.

Correlazioni: {"K_L": -0.8934544384667742, "K_tau": 0.9994747868182836, "K_B": -0.9999386411096169, "L_tau": -0.8906006257049401, "L_B": 0.8958089295689025, "tau_B": -0.9991483922158234}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 12.789 / 417.22

Griglia: 42441 coppie L/tau; SSE multistart originale: 12.789; selezione: multistart.

Segnalazioni: at_bounds, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_033-034.csv

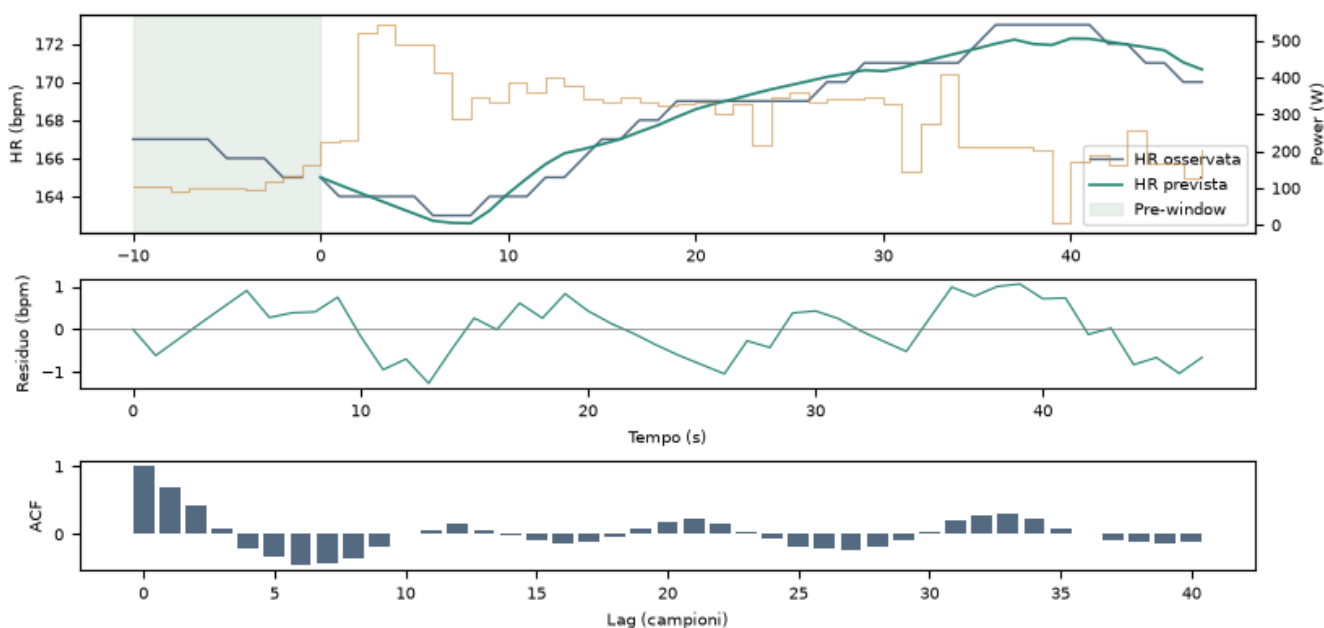
ID: 695b0904716b10d2f29f62fa | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=111.8 W; HR0=165 bpm.

HR(0) osservata fissata; B=122.27 bpm stimato; x(0)=42.728 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.60979	0.51448	0.0090915	0.96567	0.96914
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.0090915	0.61617	0.69215	-0.33831	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32924	0.19096	-0.045048 / 0.70353	58.001	no	no
L	6.2323	0.37919	5.4891 / 6.9755	6.0843	no	no
tau	109.6	64.01	-15.86 / 235.06	58.403	no	no
B	122.27	26.721	69.9 / 174.65	21.853	no	no

rho=0.37197; Jacobian rank=4/4; cond=31292; accordo multistart=no.

Correlazioni: {"K_L": -0.8295506648993671, "K_tau": 0.9934116497400352, "K_B": -0.9985720476798813, "L_tau": -0.8121735975379173, "L_B": 0.8404237997597923, "tau_B": -0.9863401397147028}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 17.848 / 118.73

Griglia: 42441 coppie L/tau; SSE multistart originale: 17.848; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_014-015.csv

ID: 6c124d7c4b0602592343e727 | Stato: fitted

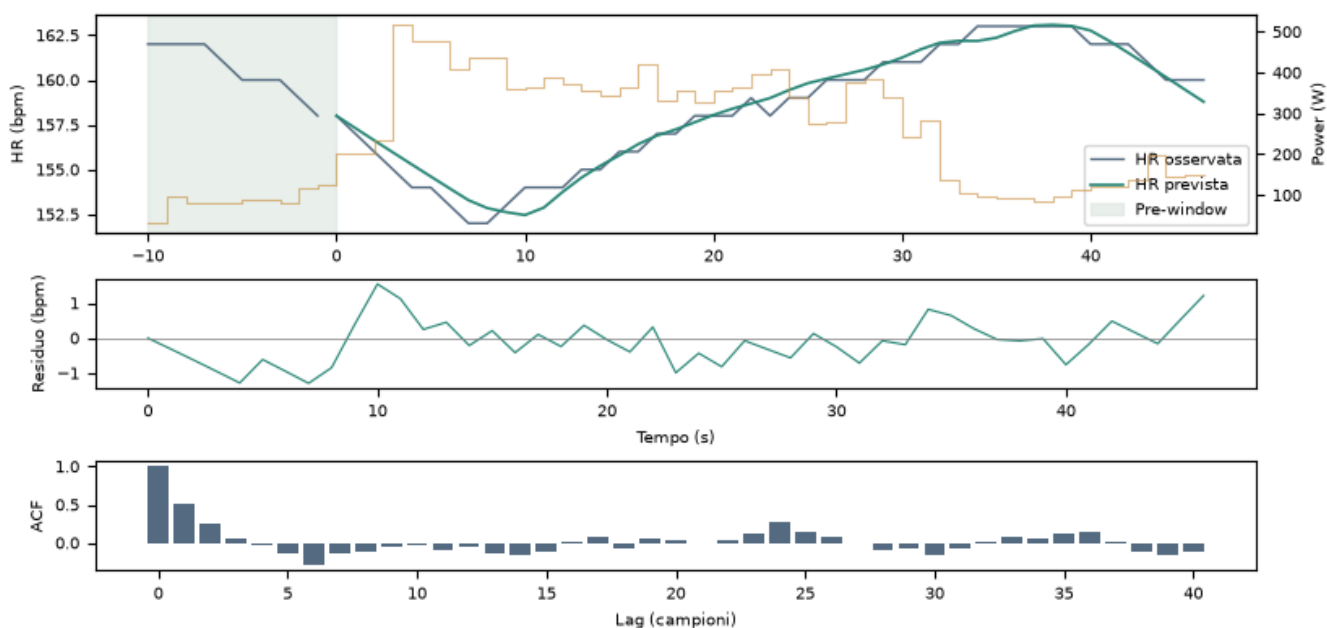
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=86.5 W; HR0=158 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=98 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.62221	0.48585	-0.10328	0.96572	0.96572

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.10328	0.62021	0.51093	-0.1365



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.53106	0.58849	-0.62237 / 1.6845	110.81	no	no
L	7.5093	0.38211	6.7603 / 8.2582	5.0885	no	no
tau	142.14	158.63	-168.77 / 453.05	111.6	no	no
B	60	108.9	-153.43 / 273.43	181.49	si	si

rho=0.27079; Jacobian rank=4/4; cond=69823; accordo multistart=no.

Correlazioni: {"K_L": -0.8979813973315836, "K_tau": 0.9994417132344483, "K_B": -0.9999625161041548, "L_tau": -0.8920498750681294, "L_B": 0.900573310515656, "tau_B": -0.9992227738322972}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 18.196 / 391.27

Griglia: 42441 coppie L/tau; SSE multistart originale: 18.196; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_062-063.csv

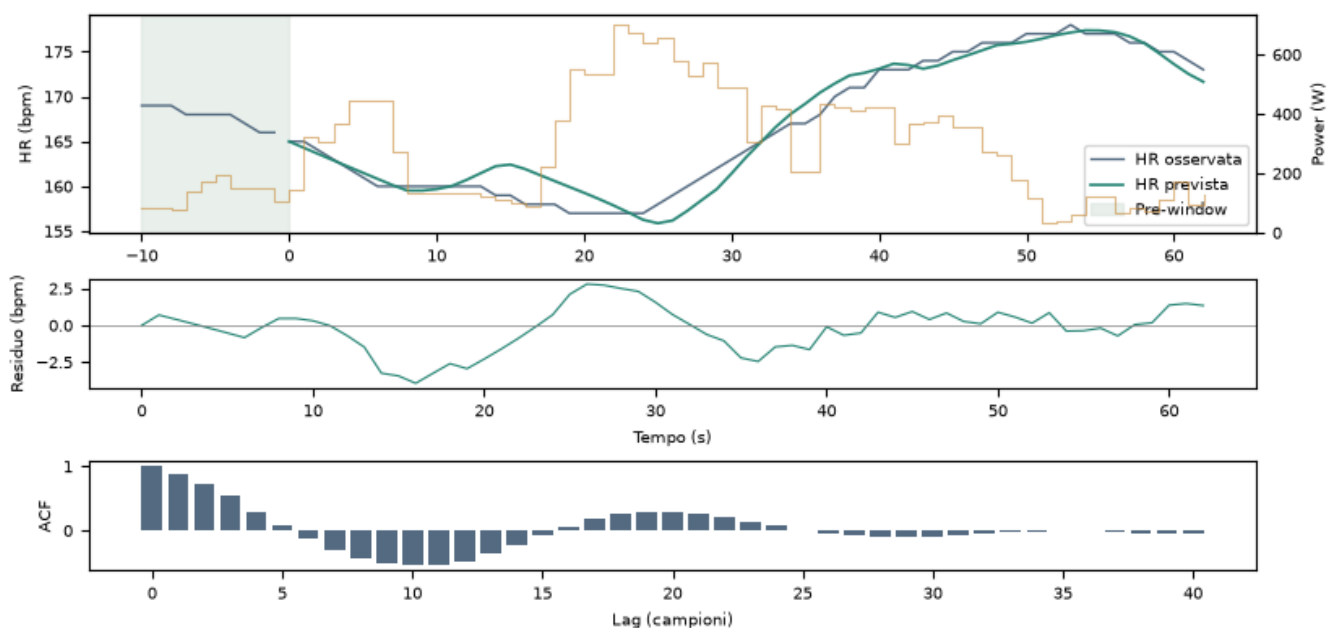
ID: 6d9a6d194e585a38fe2ced | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=129.7 W; HR0=165 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=105 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.5084	1.1234	-0.20421	0.95845	1.0237
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.20421	1.5065	0.8896	0.074272	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.68032	0.46206	-0.22532 / 1.586	67.918	no	no
L	7.2732	0.3598	6.568 / 7.9784	4.947	no	no
tau	148	104.16	-56.15 / 352.15	70.377	no	no
B	60	71.153	-79.461 / 199.46	118.59	si	si

rho=0.36977; Jacobian rank=4/4; cond=25473; accordo multistart=no.

Correlazioni: {"K_L": -0.7572935960764221, "K_tau": 0.9986965307995588, "K_B": -0.9997772336943226, "L_tau": -0.7479730228933583, "L_B": 0.7652683215944908, "tau_B": -0.9977853859568974}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 143.34 / 1469

Griglia: 42441 coppie L/tau; SSE multistart originale: 143.34; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_033-034.csv

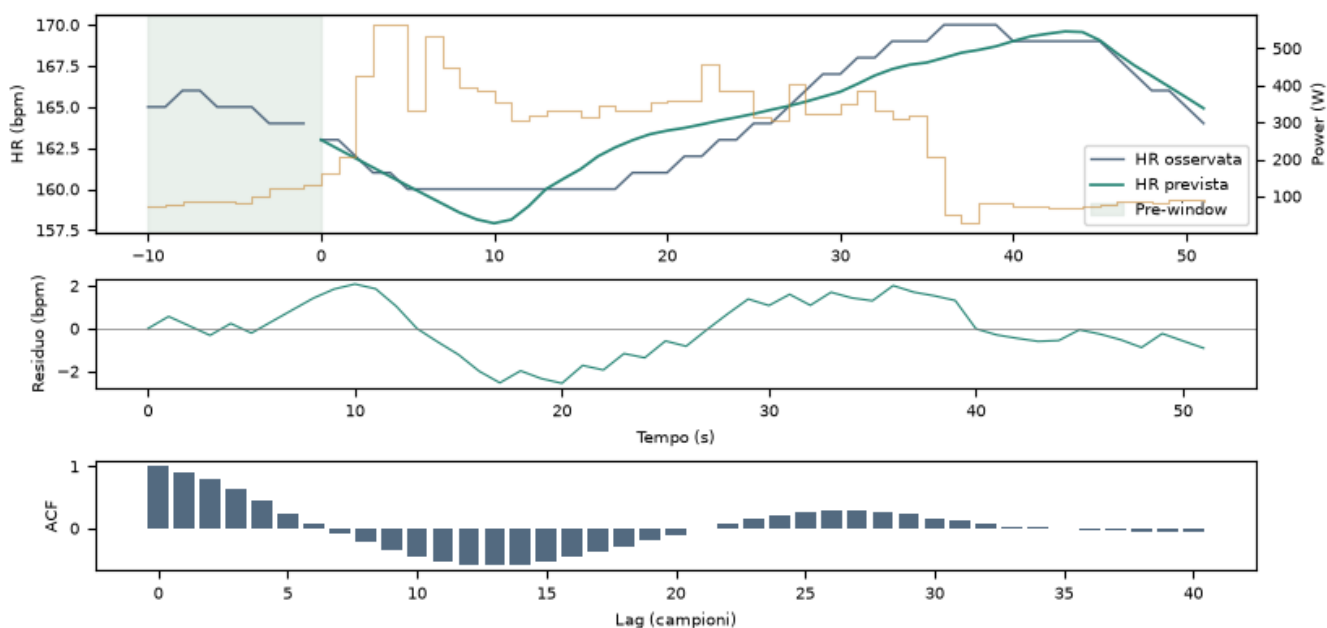
ID: 7315f0120a4ebfb61acadce1 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=97 W; HR0=163 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=103 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.264	1.0385	0.0067935	0.88437	1.1678
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.0067935	1.2764	0.90627	0.24532	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.61873	1.1986	-1.7305 / 2.9679	193.72	no	no
L	8.4823	0.6432	7.2216 / 9.743	7.5829	no	no
tau	182.21	357.1	-517.7 / 882.13	195.98	no	no
B	60	201.22	-334.38 / 454.38	335.36	si	si

rho=0.23334; Jacobian rank=4/4; cond=65236; accordo multistart=no.

Correlazioni: {"K_L": -0.8564312624559123, "K_tau": 0.9992620461057717, "K_B": -0.9999365670927973, "L_tau": -0.8497707110137105, "L_B": 0.8599077724124274, "tau_B": -0.9988823457613302}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 83.086 / 439.83

Griglia: 42441 coppie L/tau; SSE multistart originale: 83.086; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_068-069.csv

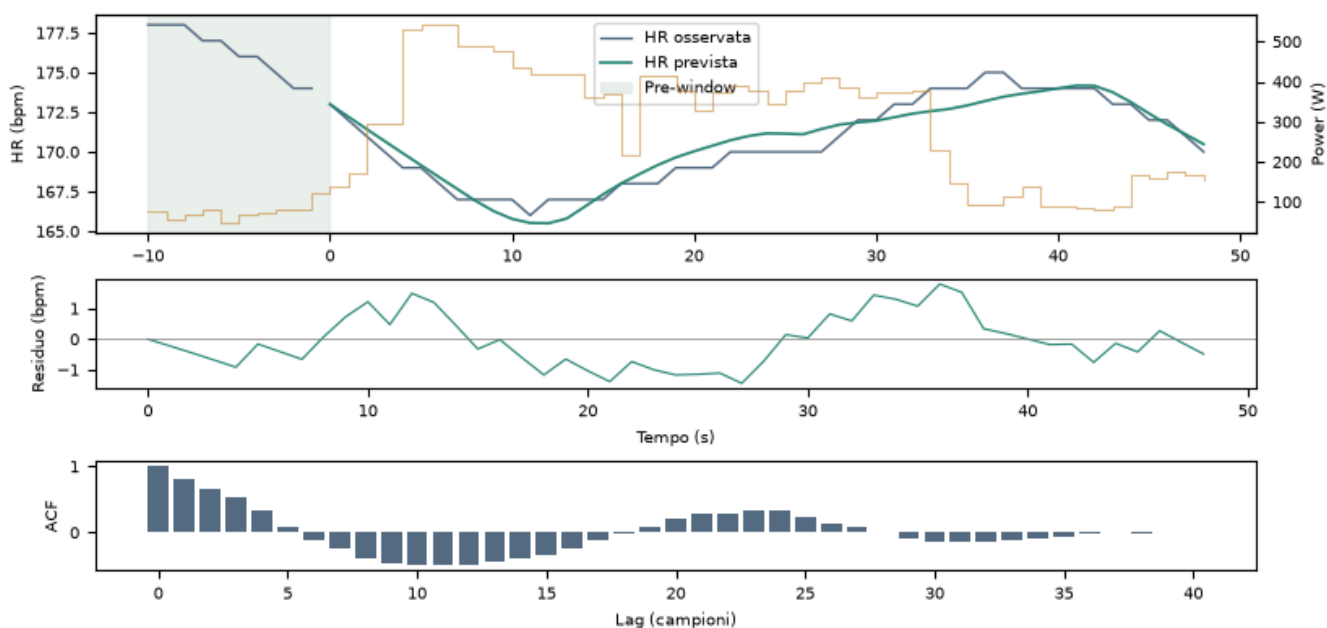
ID: 7615e04544d2f9da20308d2c | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=75.5 W; HR0=173 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=113 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.83796	0.68178	-0.060177	0.89816	0.96349
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.060177	0.84446	0.80302		0.080336



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.46785	0.90447	-1.3049 / 2.2406	193.33	no	no
L	8.5842	0.50787	7.5888 / 9.5796	5.9163	no	no
tau	144.52	279.55	-403.4 / 692.44	193.44	no	no
B	60	213.85	-359.14 / 479.14	356.41	si	si

rho=0.27274; Jacobian rank=4/4; cond=1.1026e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.8653491701738578, "K_tau": 0.9997262108712202, "K_B": -0.999988373764431, "L_tau": -0.8594338490650163, "L_B": 0.8669941401903014, "tau_B": -0.9996933547863138}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 34.407 / 330.24

Griglia: 42441 coppie L/tau; SSE multistart originale: 34.407; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_057-058.csv

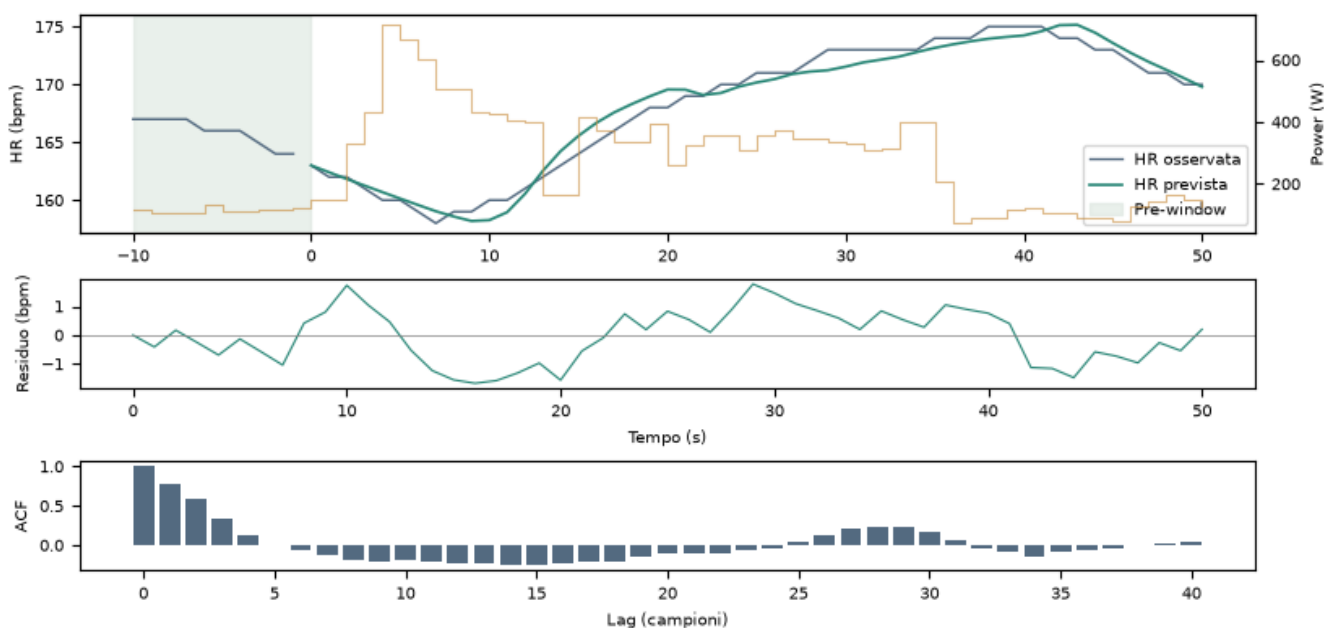
ID: 7ce6ac5eb655583acad1acf3 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=115.8 W; HR0=163 bpm.

HR(0) osservata fissata; B=119.74 bpm stimato; x(0)=43.256 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.91582	0.7804	-0.041648	0.97247	0.99763
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.041648	0.92398	0.78569	0.0074371	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32553	0.097421	0.13459 / 0.51648	29.927	no	no
L	7.4352	0.27678	6.8927 / 7.9777	3.7225	no	no
tau	72.87	22.202	29.354 / 116.39	30.468	no	no
B	119.74	14.374	91.571 / 147.92	12.004	no	no

rho=0.58412; Jacobian rank=4/4; cond=12476; accordo multistart=no.

Correlazioni: {"K_L": -0.8032284488538508, "K_tau": 0.9884803412001592, "K_B": -0.9977040466410342, "L_tau": -0.7922140206944062, "L_B": 0.8110010014052148, "tau_B": -0.9772898358289018}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 42.775 / 497.19

Griglia: 42441 coppie L/tau; SSE multistart originale: 42.775; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_033-034.csv

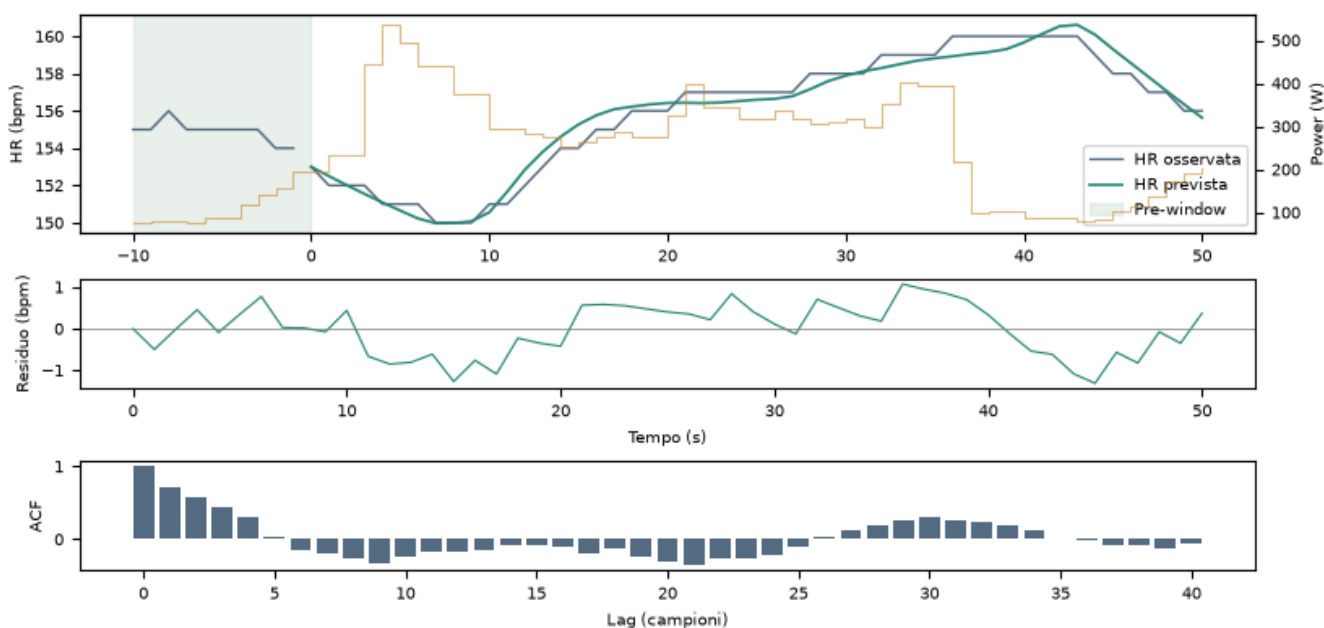
ID: 7eb77b4d33885e21dea0816d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=109.9 W; HR0=153 bpm.

HR(0) osservata fissata; B=137.69 bpm stimato; x(0)=15.307 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.6069	0.50693	-0.015447	0.96461	1.0643
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.015447	0.61274	0.70444	0.032798	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.12322	0.019295	0.0854 / 0.16104	15.659	no	no
L	6.5468	0.26976	6.018 / 7.0755	4.1206	no	no
tau	30.034	5.0635	20.109 / 39.958	16.859	no	no
B	137.69	2.8482	132.11 / 143.28	2.0685	no	no

rho=1.4468; Jacobian rank=4/4; cond=6631.1; accordo multistart=no.

Correlazioni: {"K_L": -0.8064278361038799, "K_tau": 0.9536137609612532, "K_B": -0.9953602612776308, "L_tau": -0.7917268682568223, "L_B": 0.8081580072037282, "tau_B": -0.9261992509762187}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 18.785 / 193.66

Griglia: 42441 coppie L/tau; SSE multistart originale: 18.785; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_041-042.csv

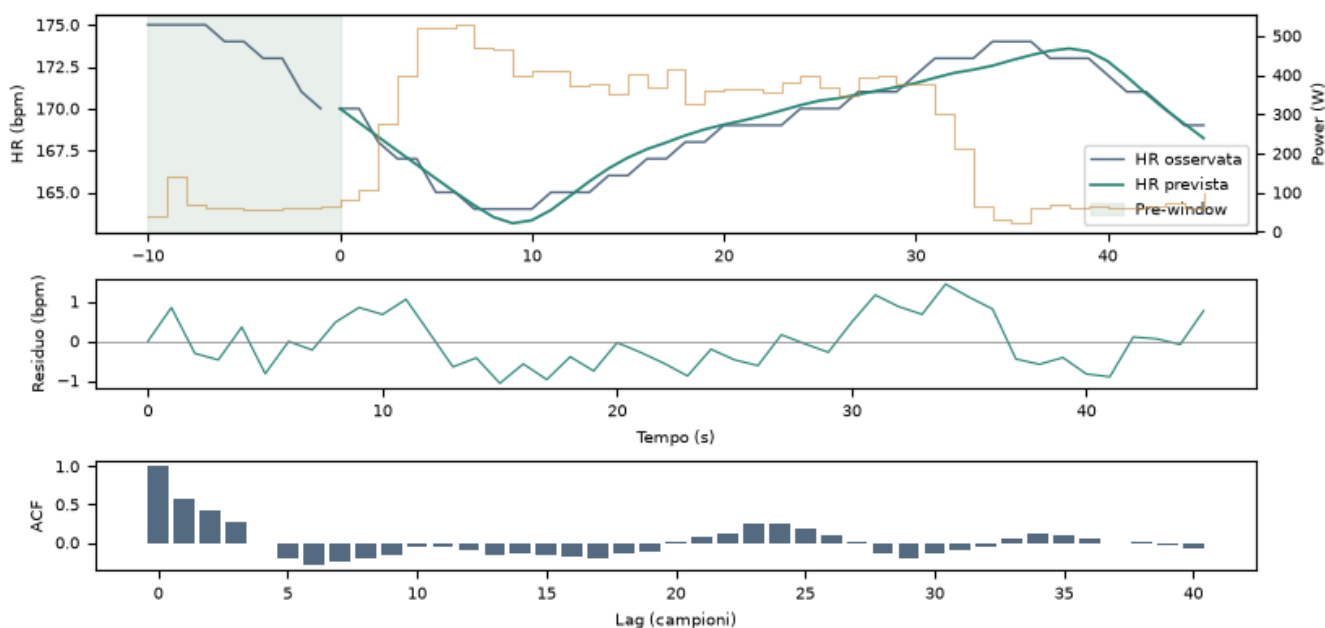
ID: 7f2d68830fc5621d92b88394 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=67.3 W; HR0=170 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=110 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.65548	0.55392	-0.025971	0.95421	1.0436
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.025971	0.6622	0.57791	-0.21188	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.47321	0.57759	-0.65887 / 1.6053	122.06	no	no
L	6.5458	0.34525	5.8691 / 7.2225	5.2743	no	no
tau	129.13	158.47	-181.47 / 439.72	122.72	no	no
B	60	133.05	-200.77 / 320.77	221.74	si	si

rho=0.2978; Jacobian rank=4/4; cond=86987; accordo multistart=no.

Correlazioni: {"K_L": -0.8897900552264215, "K_tau": 0.9995633431379688, "K_B": -0.9999823821828527, "L_tau": -0.8842409430253029, "L_B": 0.8916140709899647, "tau_B": -0.9994765086915706}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 19.764 / 464.28

Griglia: 42441 coppie L/tau; SSE multistart originale: 19.764; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_018-019.csv

ID: 808adc15cf9f44657bbcf7a1 | Stato: fitted

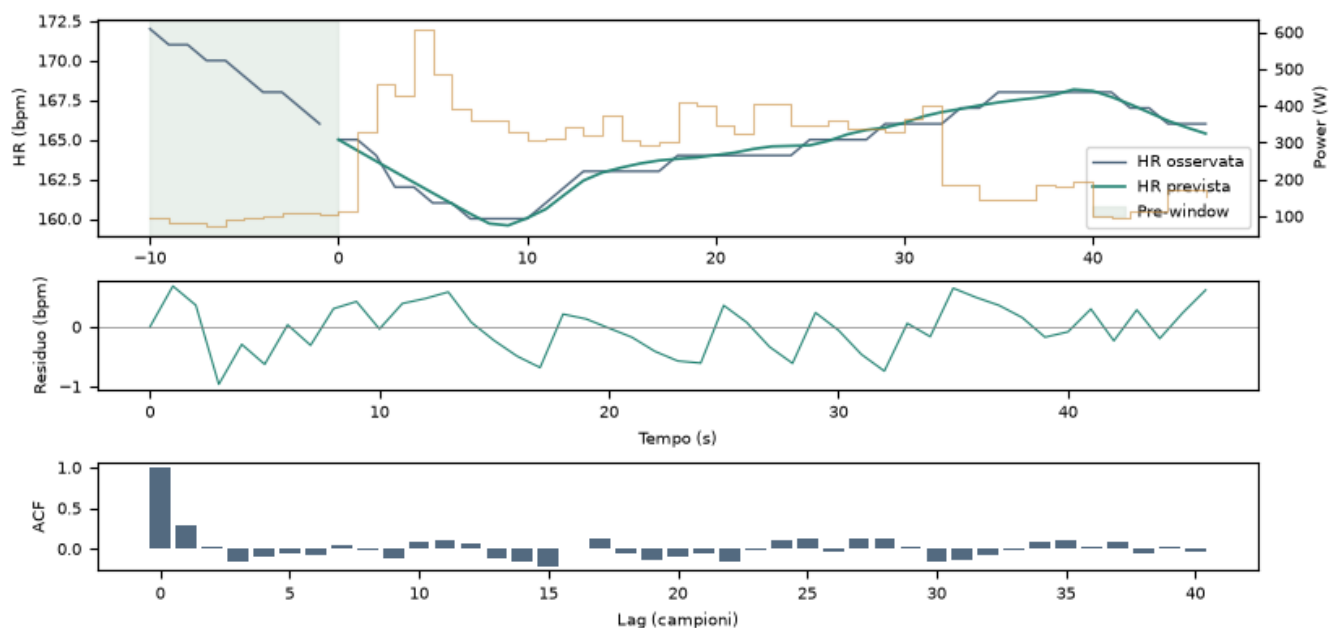
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=94.8 W; HR0=165 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=105 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.40852	0.33993	-0.020422	0.97176	1.0746

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.020422	0.41242	0.28659	-0.057027



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.54293	0.46945	-0.37718 / 1.4631	86.465	no	no
L	7.3886	0.22424	6.949 / 7.8281	3.035	no	no
tau	153.15	133.51	-108.52 / 414.82	87.173	no	no
B	60	90.242	-116.87 / 236.87	150.4	si	si

rho=0.25211; Jacobian rank=4/4; cond=81474; accordo multistart=si.

Correlazioni: {"K_L": -0.7916079465590072, "K_tau": 0.99937039990498, "K_B": -0.9999718939091496, "L_tau": -0.7805521172920095, "L_B": 0.7951792108579875, "tau_B": -0.9991962191616387}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 7.8437 / 7.8437

Griglia: 42441 coppie L/tau; SSE multistart originale: 7.8437; selezione: profile_grid.

Segnalazioni: at_bounds

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_005-006.csv

ID: 846dc1cdc252d93dd92df54a | Stato: fitted

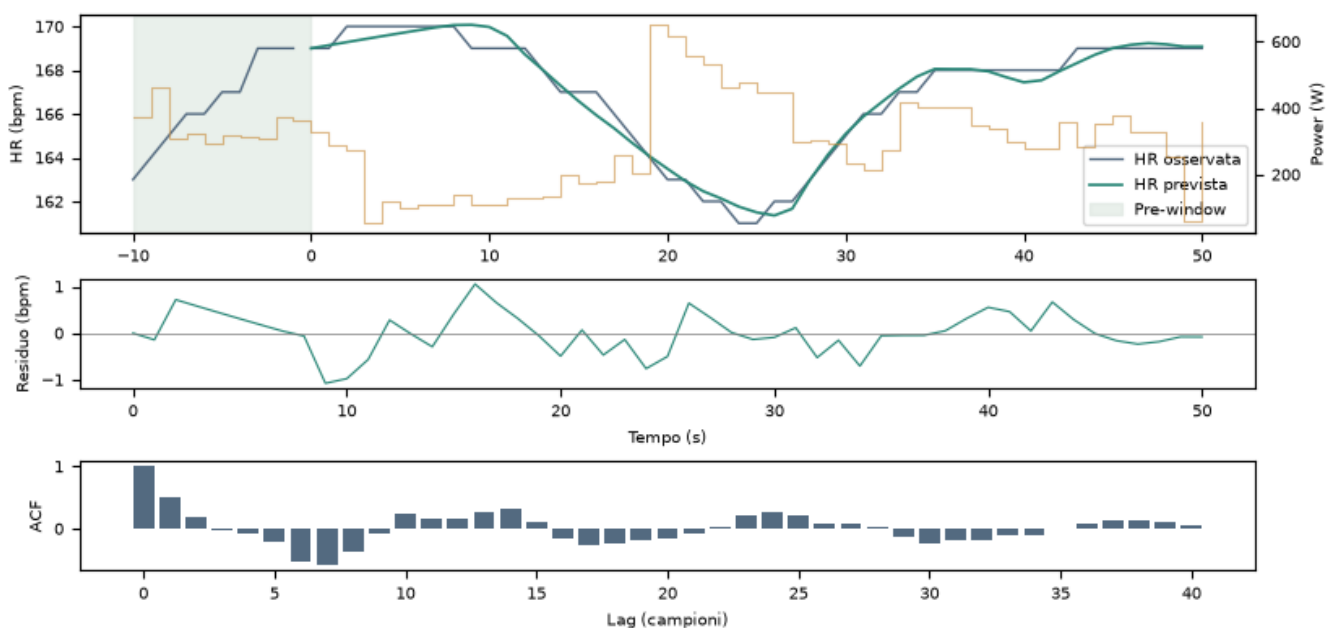
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=343.4 W; HR0=169 bpm.

HR(0) osservata fissata; B=179.55 bpm stimato; x(0)=-10.552 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.4329	0.3252	0.010105	0.97465	0.96875

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.010105	0.43709	0.50515	-0.19902



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.27184	0.11066	0.054941 / 0.48874	40.709	no	no
L	7.6509	0.19112	7.2763 / 8.0255	2.498	no	no
tau	73.787	29.828	15.325 / 132.25	40.424	no	no
B	179.55	5.2521	169.26 / 189.85	2.9251	no	no

rho=0.57393; Jacobian rank=4/4; cond=10339; accordo multistart=no.

Correlazioni: {"K_L": -0.768342653324626, "K_tau": 0.9983587681570891, "K_B": 0.9990925400018544, "L_tau": -0.773022026649969, "L_B": -0.7665270153568072, "tau_B": 0.9964631530017719}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 9.5575 / 323.31

Griglia: 42441 coppie L/tau; SSE multistart originale: 9.5575; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_060-061.csv

ID: 84bd4cf291c32ffb4052ce | Stato: fitted

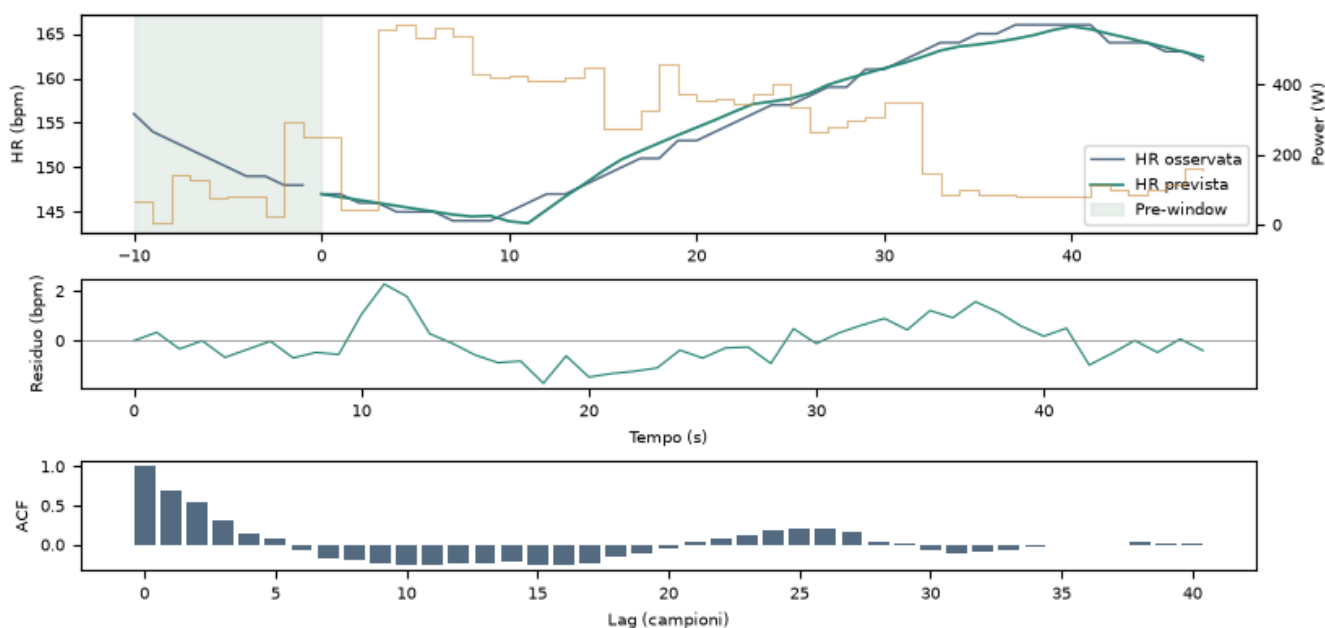
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=114.3 W; HR0=147 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=87 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.85345	0.6819	-0.070942	0.98844	1.0047

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.070942	0.8595	0.69398	0.072613



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.0639	0.89726	-0.69473 / 2.8225	84.337	no	no
L	7.8173	0.31982	7.1905 / 8.4442	4.0911	no	no
tau	260.98	222.38	-174.89 / 696.86	85.21	no	no
B	60	78.828	-94.502 / 214.5	131.38	si	si

rho=0.15013; Jacobian rank=4/4; cond=34261; accordo multistart=no.

Correlazioni: {"K_L": -0.8342601097938936, "K_tau": 0.9987654387134691, "K_B": -0.9979426050743283, "L_tau": -0.8260903113627893, "L_B": 0.8470859417921841, "tau_B": -0.9936834066699783}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 34.962 / 594.3

Griglia: 42441 coppie L/tau; SSE multistart originale: 34.962; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_053-054.csv

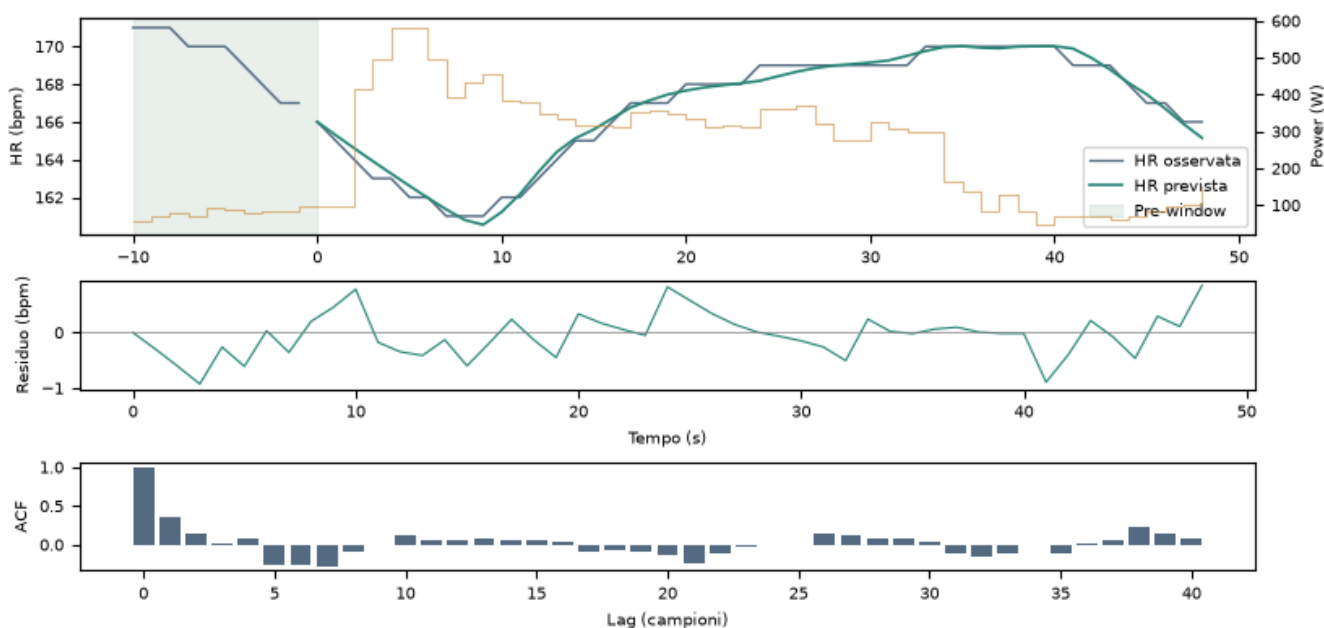
ID: 85a45d0dc7865978a59f34a0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=80.3 W; HR0=166 bpm.

HR(0) osservata fissata; B=129.13 bpm stimato; x(0)=36.872 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.39113	0.29506	-0.046582	0.98179	1.0528
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.046582	0.39238	0.34928	-0.26337	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.18795	0.045137	0.099485 / 0.27642	24.015	no	no
L	6.742	0.19297	6.3638 / 7.1202	2.8622	no	no
tau	51.805	12.59	27.128 / 76.482	24.303	no	no
B	129.13	9.0506	111.39 / 146.87	7.009	no	no

rho=0.7964; Jacobian rank=4/4; cond=22916; accordo multistart=no.

Correlazioni: {"K_L": -0.8778066672378553, "K_tau": 0.9945855321030349, "K_B": -0.9997574334431039, "L_tau": -0.866966725350921, "L_B": 0.8820956787493377, "tau_B": -0.9930797029977613}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 7.4963 / 294.26

Griglia: 42441 coppie L/tau; SSE multistart originale: 7.4963; selezione: multistart.

Segnalazioni: multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_039-040.csv

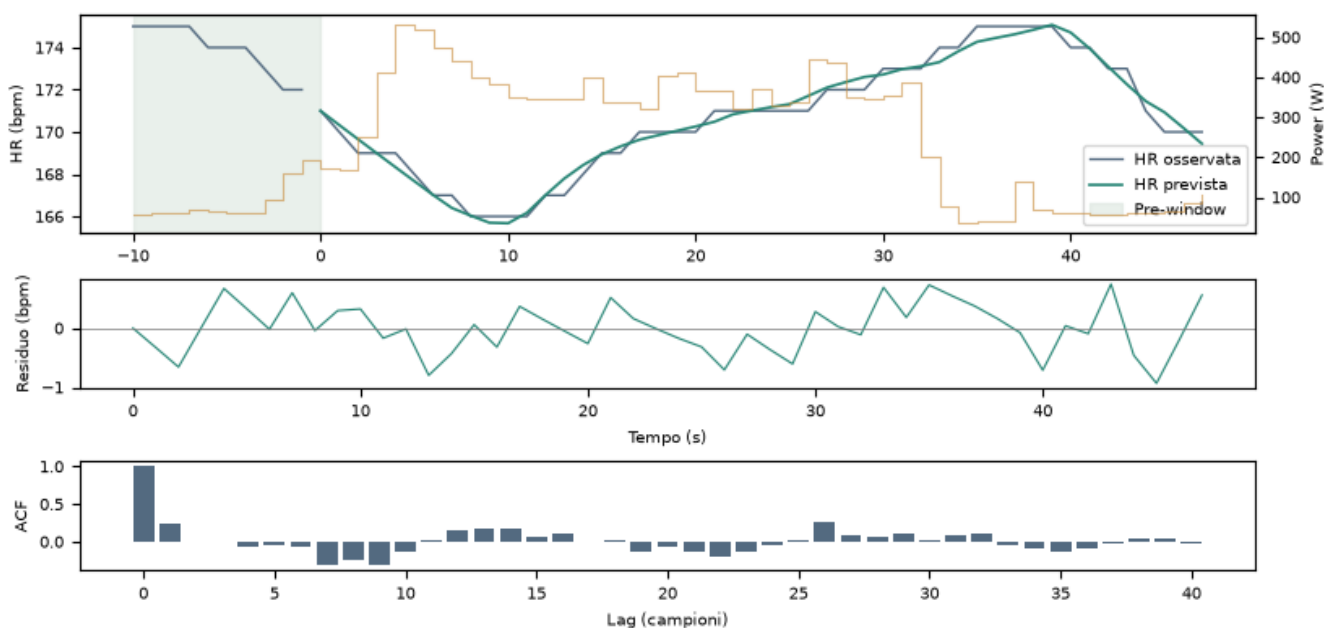
ID: 8b47513cc3c457398fd78b1d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=87.4 W; HR0=171 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=111 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.41754	0.32835	-0.0018307	0.9757	1.0445
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.0018307	0.42196	0.24108	-0.056174	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.55489	0.54486	-0.51304 / 1.6228	98.193	no	no
L	6.8828	0.23773	6.4168 / 7.3487	3.454	no	no
tau	164.22	161.81	-152.92 / 481.36	98.532	no	no
B	60	108.55	-152.75 / 272.75	180.91	si	si

rho=0.24429; Jacobian rank=4/4; cond=97407; accordo multistart=no.

Correlazioni: {"K_L": -0.8851790649736838, "K_tau": 0.9996330571913118, "K_B": -0.9999858762871412, "L_tau": -0.8804465613077647, "L_B": 0.8867462957516146, "tau_B": -0.9995540226664149}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 8.3684 / 325

Griglia: 42441 coppie L/tau; SSE multistart originale: 8.3684; selezione: profile_grid.

Segnalazioni: at_bounds, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_078-079.csv

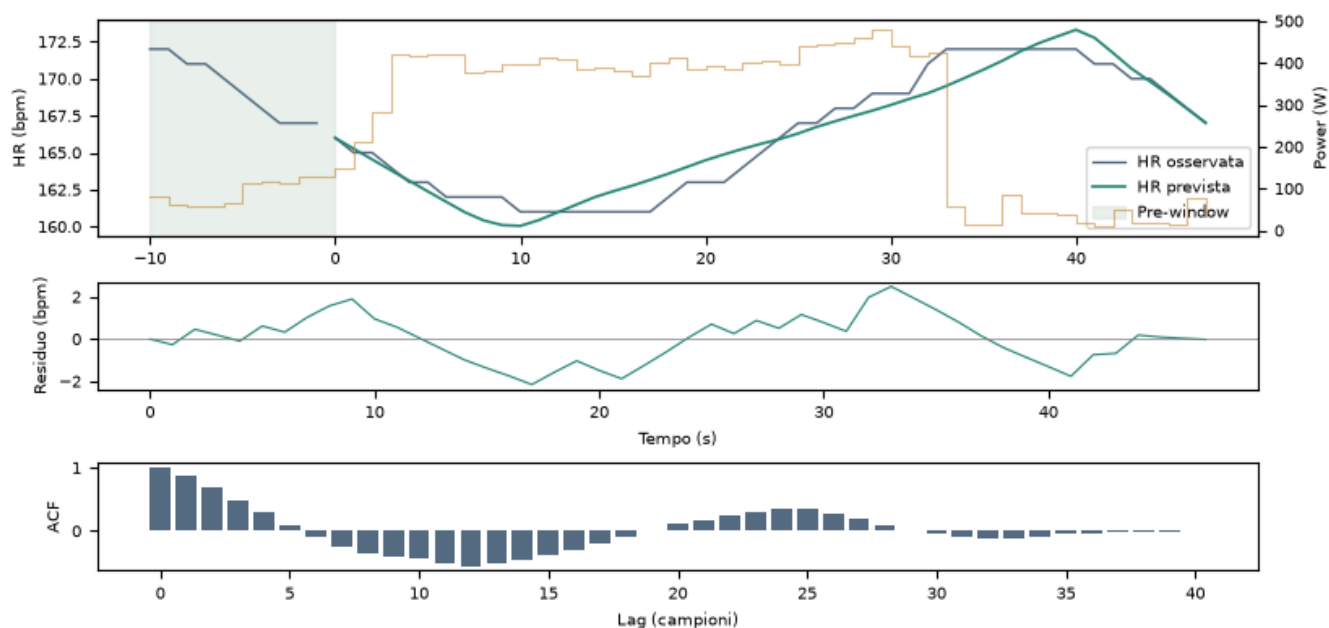
ID: 901dca47525dc1d3e9ba3ca5 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=92.1 W; HR0=166 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=106 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1013	0.8787	0.014494	0.92536	1.2055
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.014494	1.1128	0.86545	0.0967	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.53657	0.93032	-1.2869 / 2.36	173.38	no	no
L	7.2728	0.57167	6.1523 / 8.3933	7.8604	no	no
tau	144.06	253.77	-353.34 / 641.45	176.16	no	no
B	60	183.12	-298.92 / 418.92	305.21	si	si

rho=0.27577; Jacobian rank=4/4; cond=69703; accordo multistart=no.

Correlazioni: {"K_L": -0.8861627705554952, "K_tau": 0.9995455525690783, "K_B": -0.9999631101018069, "L_tau": -0.8808015103444046, "L_B": 0.8888698225650067, "tau_B": -0.9993577148313677}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 58.214 / 69.472

Griglia: 42441 coppie L/tau; SSE multistart originale: 58.214; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_049-050.csv

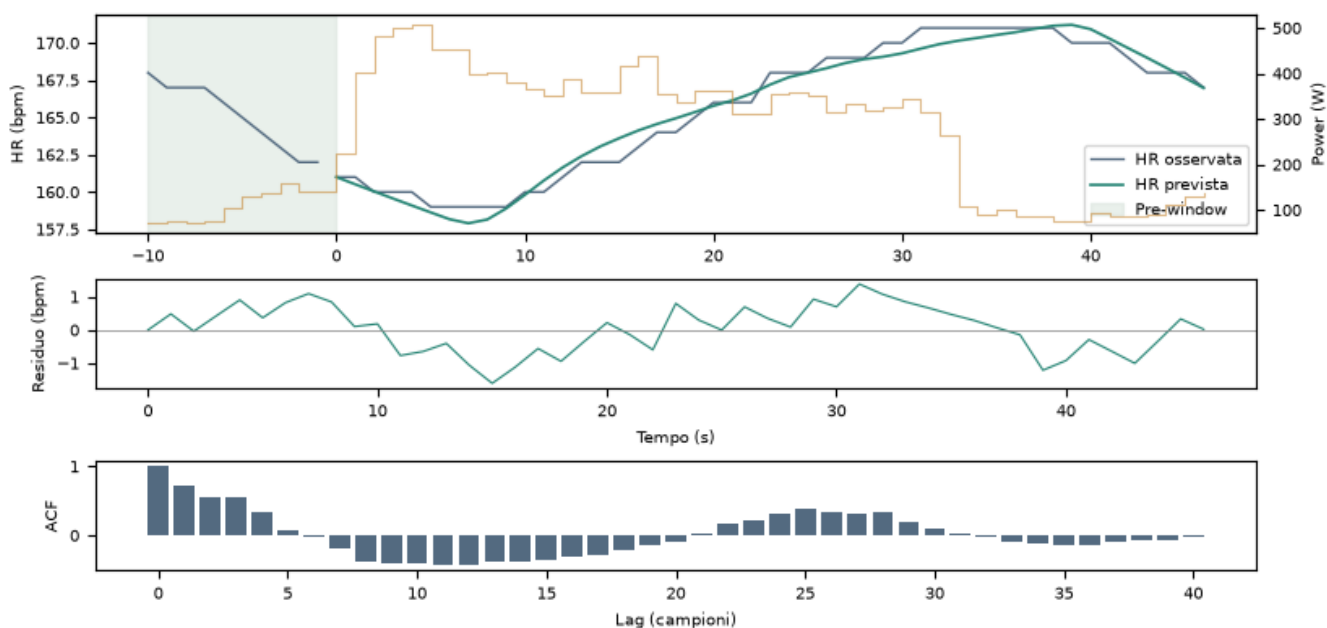
ID: 91f24db9c6e295398ce74422 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=110.8 W; HR0=161 bpm.

HR(0) osservata fissata; B=117.4 bpm stimato; x(0)=43.602 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.70333	0.58089	0.036811	0.97423	1.109
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.036811	0.70996	0.72014	0.070051	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32773	0.18194	-0.028871 / 0.68434	55.515	no	no
L	6.5579	0.42549	5.7239 / 7.3918	6.4882	no	no
tau	89.515	49.929	-8.3446 / 187.38	55.776	no	no
B	117.4	26.663	65.138 / 169.66	22.712	no	no

rho=0.44062; Jacobian rank=4/4; cond=27226; accordo multistart=no.

Correlazioni: {"K_L": -0.9112725393972212, "K_tau": 0.9958314491574318, "K_B": -0.9990242013376842, "L_tau": -0.906481135627008, "L_B": 0.914742623948982, "tau_B": -0.9912156749772668}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.25 / 241.63

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.25; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_031-032.csv

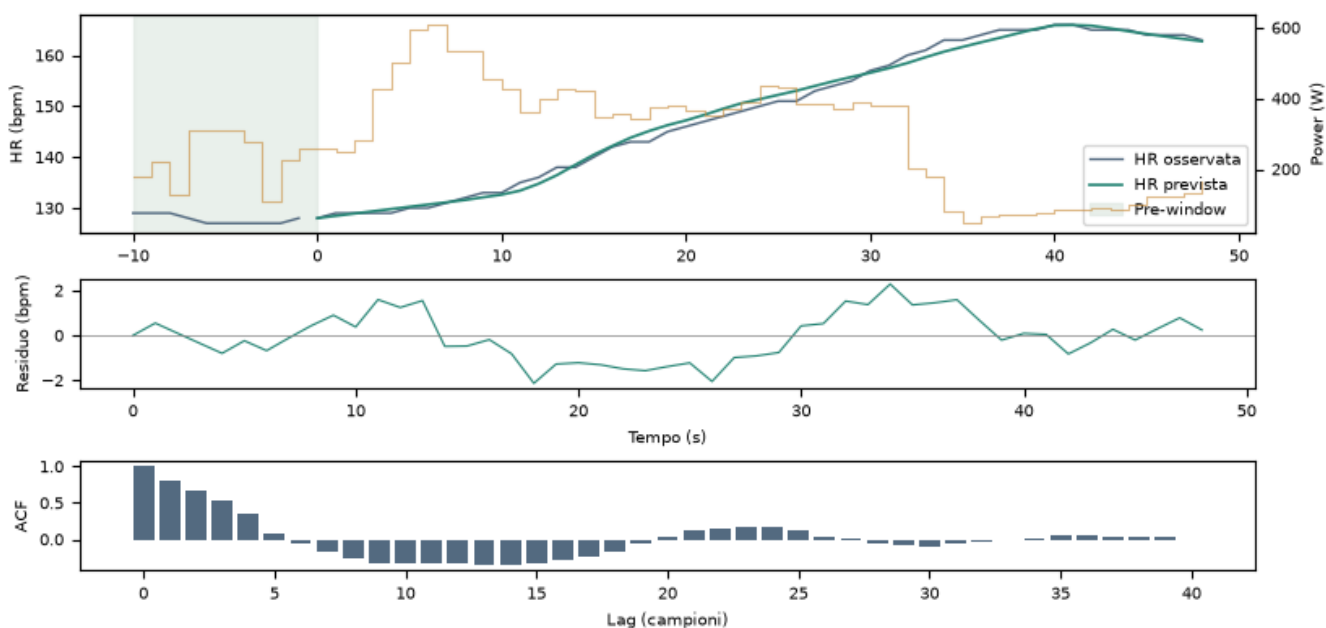
ID: 950eb08f423b8524064a04eb | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=233.8 W; HR0=128 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-72 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0377	0.8491	-0.048094	0.99402	0.99883
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.048094	1.0473	0.79582		0.086853



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.70906	0.2608	0.1979 / 1.2202	36.781	no	no
L	7.7839	0.44481	6.9121 / 8.6557	5.7144	no	no
tau	158.38	59.263	42.226 / 274.54	37.418	no	no
B	200	21.959	156.96 / 243.04	10.98	si	si

rho=0.25392; Jacobian rank=4/4; cond=7852.7; accordo multistart=no.

Correlazioni: {"K_L": -0.8473840683152953, "K_tau": 0.9875337610726013, "K_B": 0.9261700399000061, "L_tau": -0.8638329794207773, "L_B": -0.8438894085349287, "tau_B": 0.9735412312510417}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 52.761 / 463.27

Griglia: 42441 coppie L/tau; SSE multistart originale: 52.761; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_004-005.csv

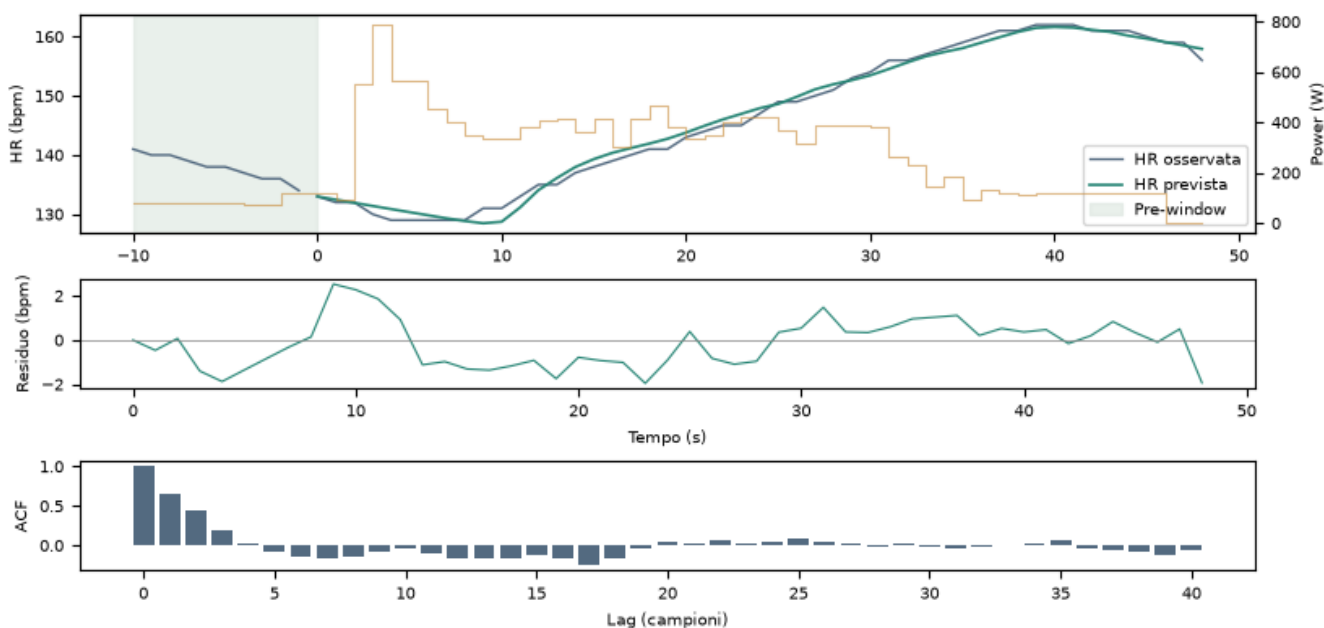
ID: 957de74a530862ea5ea4bebb | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=88.8 W; HR0=133 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=73 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0801	0.89443	-0.14382	0.99176	1.005
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.14382	1.0816	0.66325	-0.080074	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.74055	0.25741	0.23602 / 1.2451	34.76	no	no
L	7.7099	0.27328	7.1743 / 8.2455	3.5445	no	no
tau	135.2	48.078	40.964 / 229.43	35.561	no	no
B	60	29.371	2.4332 / 117.57	48.951	si	si

rho=0.29801; Jacobian rank=4/4; cond=14494; accordo multistart=si.

Correlazioni: {"K_L": -0.8171335151650179, "K_tau": 0.993415171566461, "K_B": -0.9912953820740839, "L_tau": -0.7935323651877475, "L_B": 0.8423975380174915, "tau_B": -0.9703113773066742}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 57.165 / 57.165

Griglia: 42441 coppie L/tau; SSE multistart originale: 57.165; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_070-071.csv

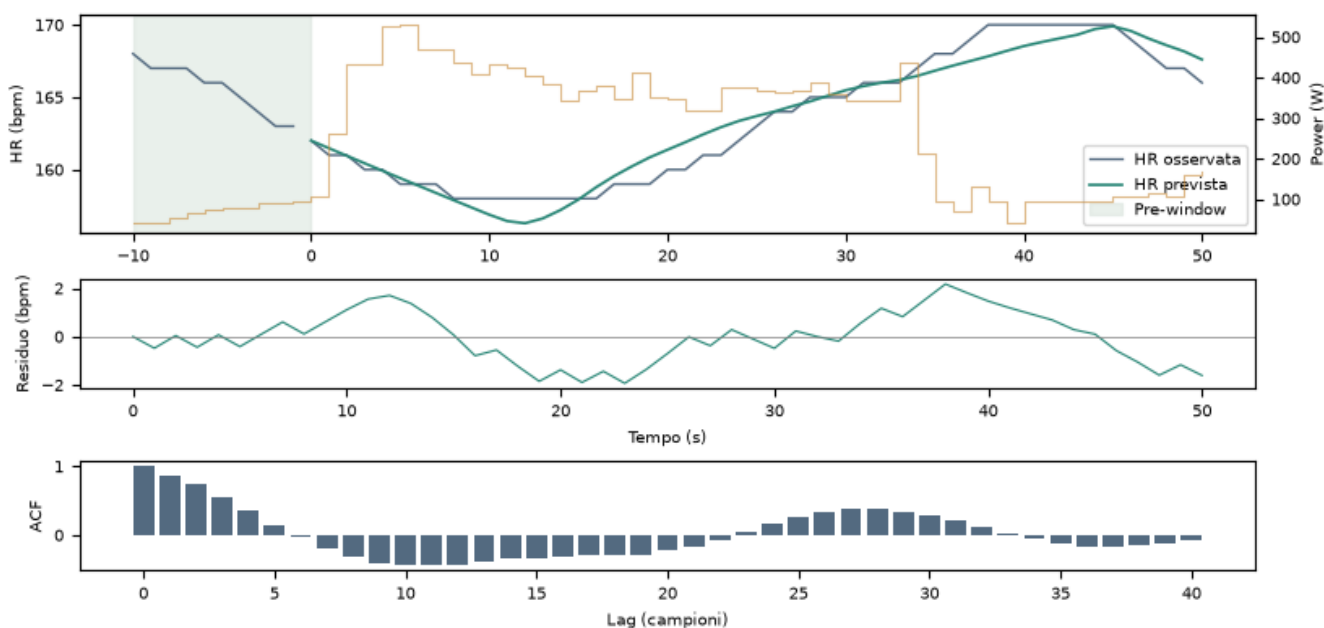
ID: 97b68b6089bf373dcd178a48 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=70.4 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0471	0.84186	-0.003214	0.94413	1.1335
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.003214	1.0575	0.86161		0.13817



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.57829	0.95607	-1.2956 / 2.4522	165.33	no	no
L	10.487	0.63884	9.2349 / 11.739	6.0917	no	no
tau	194.43	326.68	-445.85 / 834.72	168.01	no	no
B	60	169.81	-272.82 / 392.82	283.01	si	si

rho=0.20322; Jacobian rank=4/4; cond=72206; accordo multistart=no.

Correlazioni: {"K_L": -0.8361390141189883, "K_tau": 0.9994505147391836, "K_B": -0.9998874266542276, "L_tau": -0.8269946878341212, "L_B": 0.8424188248161224, "tau_B": -0.9989561834312753}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 55.914 / 69.559

Griglia: 42441 coppie L/tau; SSE multistart originale: 55.914; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_049-050.csv

ID: 9f508134d10608bb4f303c9d | Stato: fitted

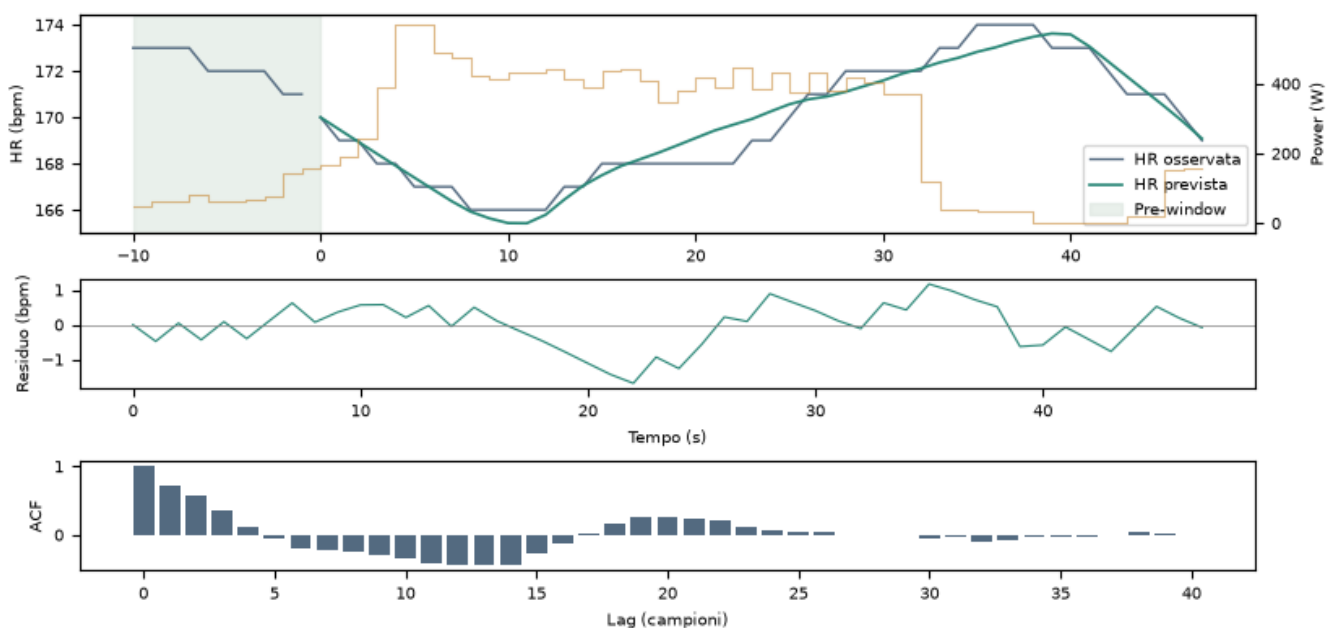
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=83.5 W; HR0=170 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=110 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.63059	0.49766	-0.022721	0.93742	1.0253

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.022721	0.63684	0.72206	-0.047059



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.50211	1.1807	-1.812 / 2.8162	235.14	no	no
L	7.6942	0.49483	6.7243 / 8.6641	6.4313	no	no
tau	208.75	493.32	-758.16 / 1175.7	236.32	no	no
B	60	257.61	-444.91 / 564.91	429.35	si	si

rho=0.18829; Jacobian rank=4/4; cond=1.6478e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.9108689328887188, "K_tau": 0.9998266625207206, "K_B": -0.9999908641541242, "L_tau": -0.9081047250838247, "L_B": 0.9119988678774489, "tau_B": -0.9997806088930032}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 19.087 / 304.94

Griglia: 42441 coppie L/tau; SSE multistart originale: 19.087; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_074-075.csv

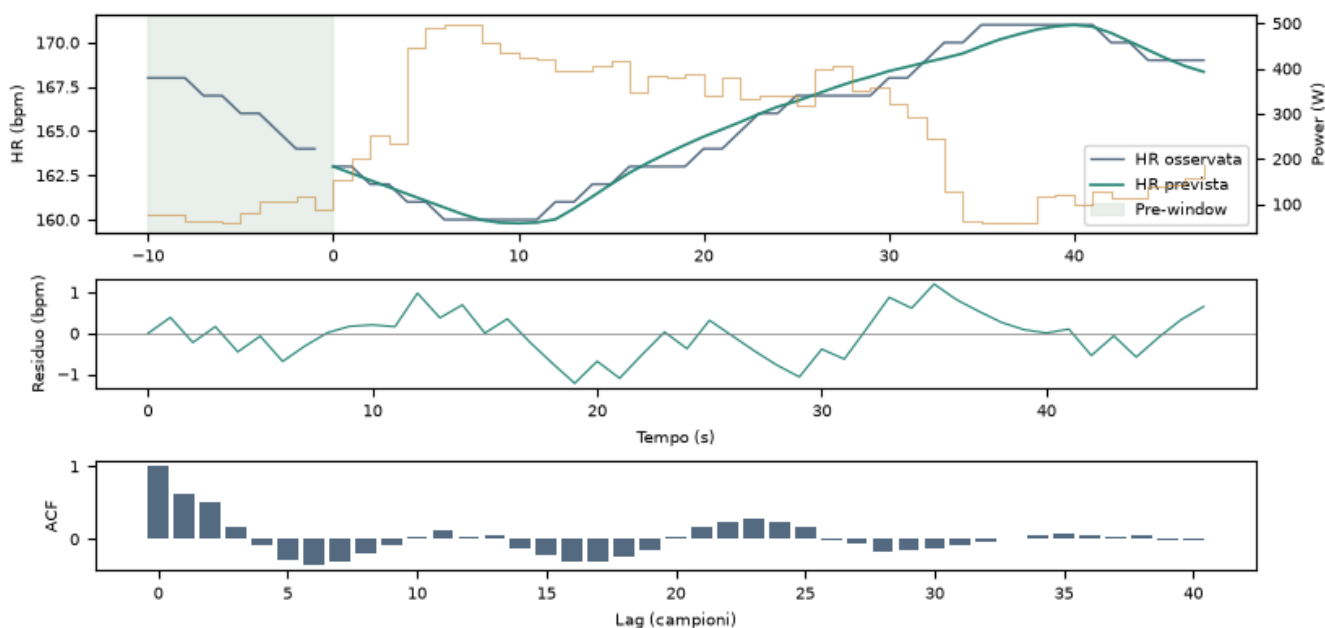
ID: a2c2d7c10c607c1ad7629c01 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=84 W; HR0=163 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=103 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.54693	0.43097	-0.033885	0.98038	1.0179
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.033885	0.55166	0.61678	-0.27999	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.68673	1.1106	-1.4901 / 2.8636	161.73	no	no
L	7.6905	0.48533	6.7392 / 8.6418	6.3108	no	no
tau	263.86	429.55	-578.06 / 1105.8	162.8	no	no
B	60	169.92	-273.04 / 393.04	283.2	si	si

rho=0.14898; Jacobian rank=4/4; cond=1.1567e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.9110702673827993, "K_tau": 0.9996993985876939, "K_B": -0.9999123889447844, "L_tau": -0.906681878023738, "L_B": 0.9145629644241416, "tau_B": -0.99933143598422}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 14.358 / 278.43

Griglia: 42441 coppie L/tau; SSE multistart originale: 14.358; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_047-048.csv

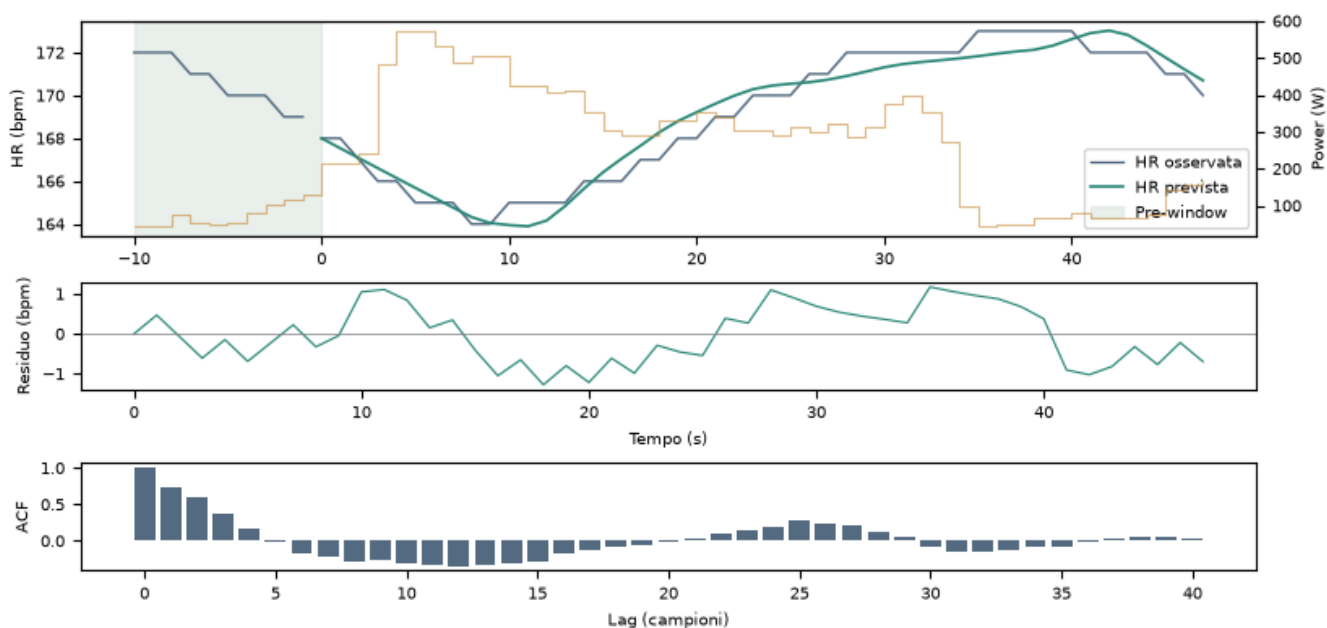
ID: a35c4370963792f42710cf72 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=74.5 W; HR0=168 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=108 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.70282	0.61173	-0.018087	0.94578	1.0142
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.018087	0.71002	0.73434		-0.012546



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.58411	1.1004	-1.5726 / 2.7408	188.38	no	no
L	8.5192	0.47733	7.5836 / 9.4548	5.6029	no	no
tau	231.02	438.16	-627.77 / 1089.8	189.66	no	no
B	60	204.94	-341.68 / 461.68	341.57	no	si

rho=0.16657; Jacobian rank=4/4; cond=1.1138e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.8514337996754346, "K_tau": 0.9995915164229168, "K_B": -0.9999541419055682, "L_tau": -0.8451291532492723, "L_B": 0.8548987799664309, "tau_B": -0.9993392530500376}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.71 / 251.51

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.71; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_045-046.csv

ID: a389fbf2c6767e2eb5a01768 | Stato: fitted

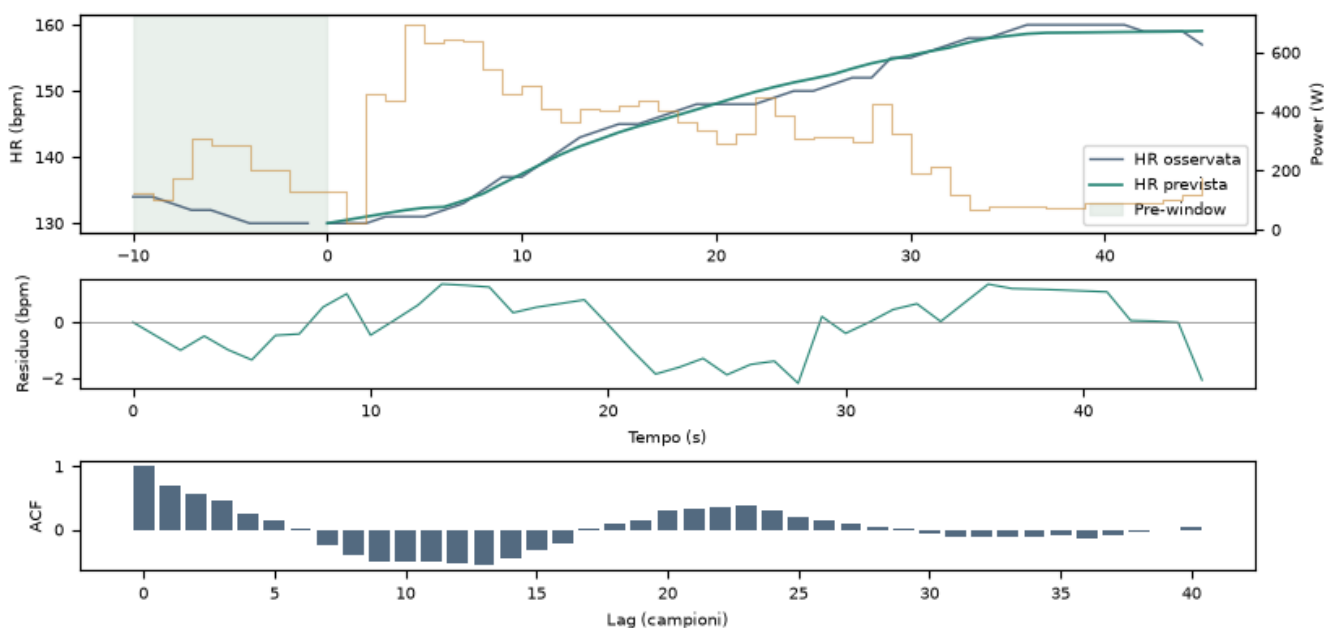
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=195.1 W; HR0=130 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-70 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0263	0.84146	-0.06824	0.99	0.96877

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.06824	1.0354	0.71066	0.15152



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.3184	0.17149	-0.017718 / 0.65452	53.86	no	no
L	4.139	0.66418	2.8372 / 5.4408	16.047	no	no
tau	139.15	67.909	6.0511 / 272.26	48.802	no	no
B	200	25.079	150.84 / 249.16	12.54	si	si

rho=0.29364; Jacobian rank=4/4; cond=12737; accordo multistart=no.

Correlazioni: {"K_L": -0.7706318886935591, "K_tau": 0.9854895900467765, "K_B": 0.9356235499127505, "L_tau": -0.8190646951793658, "L_B": -0.8402415700896708, "tau_B": 0.9816040757631402}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 48.455 / 157.24

Griglia: 42441 coppie L/tau; SSE multistart originale: 48.455; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_037-038.csv

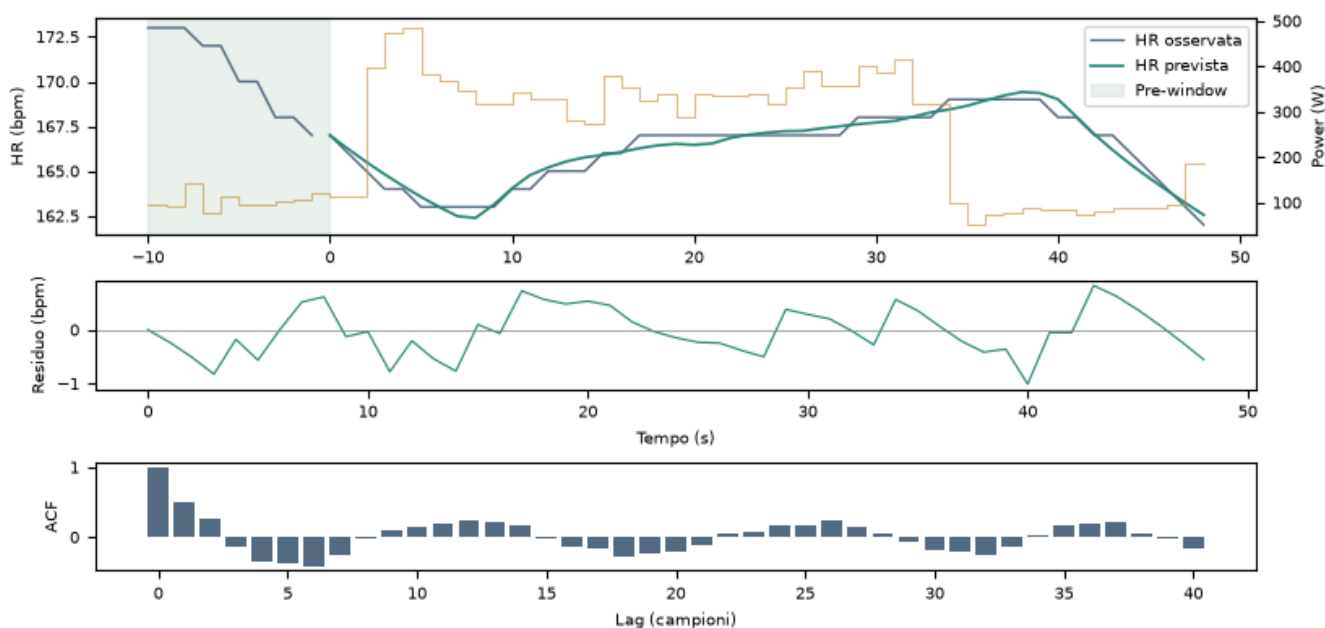
ID: a5390da6fb99a4bba9ec5877 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=103.7 W; HR0=167 bpm.

HR(0) osservata fissata; B=152.85 bpm stimato; x(0)=14.154 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.43852	0.35629	-0.032918	0.9504	1.0039
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.032918	0.44181	0.50526	-0.38626	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.072312	0.011782	0.049219 / 0.095405	16.294	no	no
L	5.6576	0.20798	5.25 / 6.0653	3.6761	no	no
tau	17.972	3.1996	11.7 / 24.243	17.804	no	no
B	152.85	2.2156	148.5 / 157.19	1.4496	no	no

rho=2.3561; Jacobian rank=4/4; cond=7998.2; accordo multistart=no.

Correlazioni: {"K_L": -0.8200186944119724, "K_tau": 0.9801535316879699, "K_B": -0.9990834560353731, "L_tau": -0.8191492044376207, "L_B": 0.8229355367053385, "tau_B": -0.9788757647206191}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 9.4227 / 189.31

Griglia: 42441 coppie L/tau; SSE multistart originale: 9.4227; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_047-048.csv

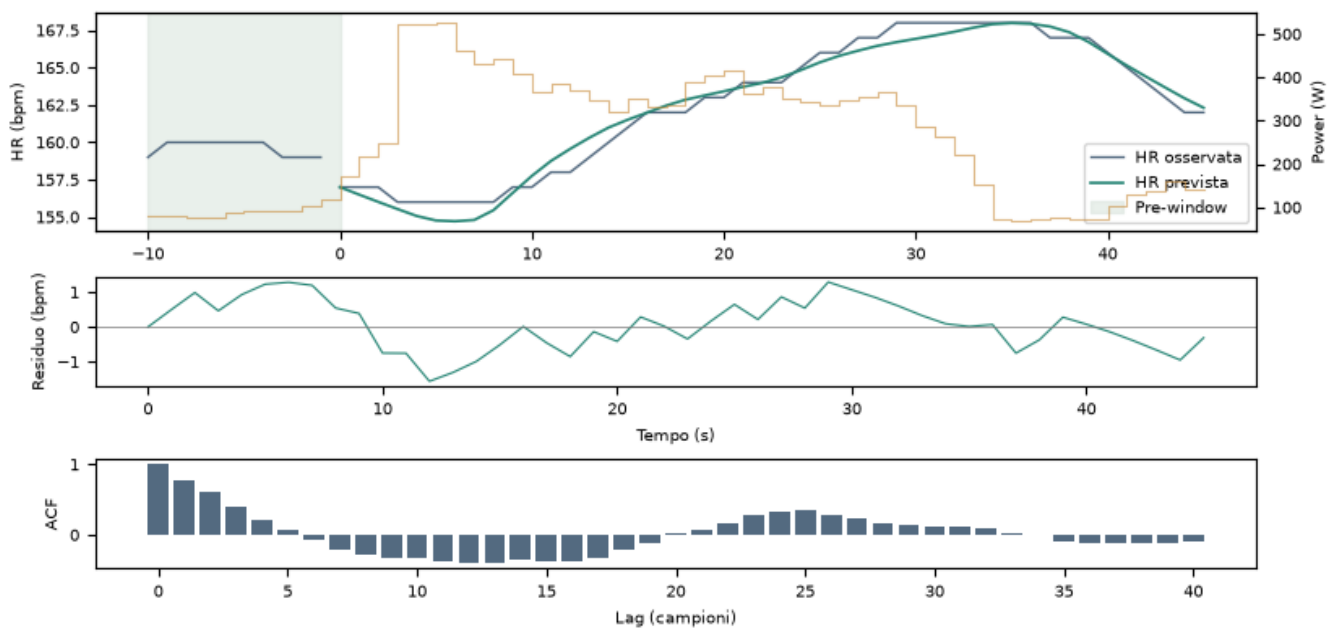
ID: a65cf16c7fa7a99b473eaf29 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=89.7 W; HR0=157 bpm.

HR(0) osservata fissata; B=135.3 bpm stimato; x(0)=21.701 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.71003	0.57937	0.071872	0.97332	1.1059
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.071872	0.71419	0.78219	0.06427	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.16319	0.04857	0.067998 / 0.25839	29.762	no	no
L	4.508	0.41239	3.6997 / 5.3162	9.1482	no	no
tau	42.923	12.759	17.915 / 67.931	29.726	no	no
B	135.3	8.1556	119.31 / 151.28	6.0278	no	no

rho=0.94336; Jacobian rank=4/4; cond=13516; accordo multistart=no.

Correlazioni: {"K_L": -0.907902709401299, "K_tau": 0.9865068907176342, "K_B": -0.9975018463925247, "L_tau": -0.9160984984282403, "L_B": 0.9028295580188486, "tau_B": -0.9739027874129651}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.19 / 256.58

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.19; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_049-050.csv

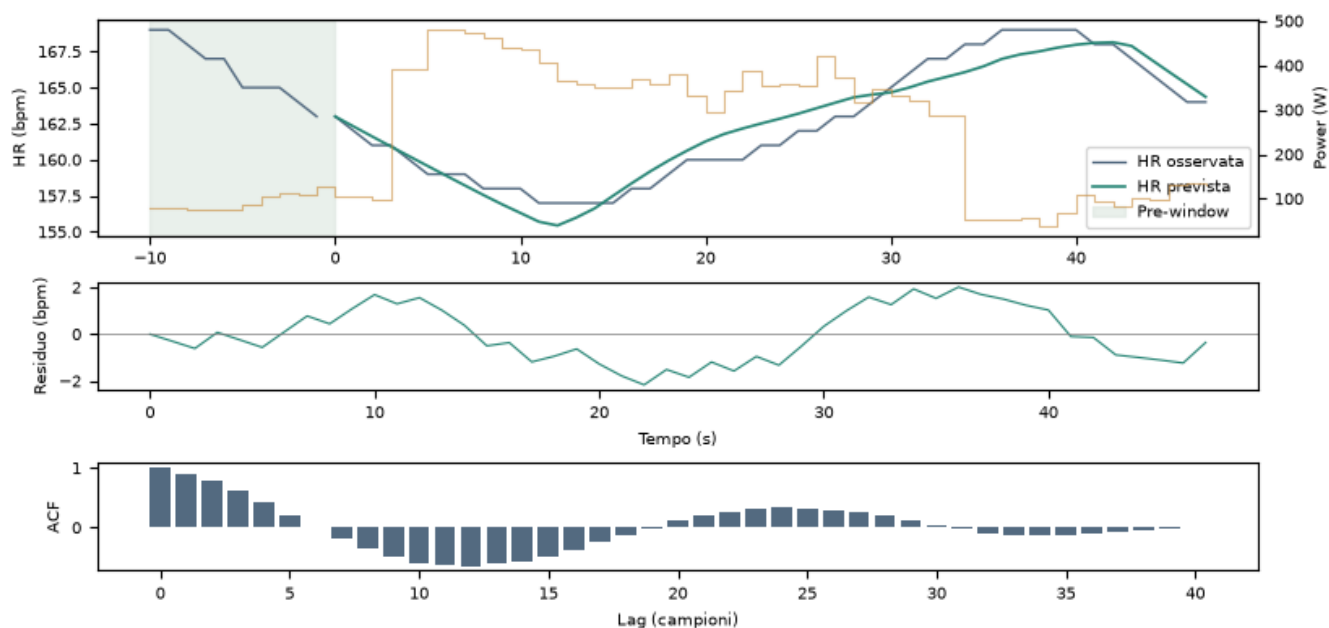
ID: a6db87c9f323a2c0abcd5a81 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=93.7 W; HR0=163 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=103 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1518	0.99595	-0.016854	0.91835	1.058
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.016854	1.1638	0.89198	0.20344	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.58613	1.0486	-1.4691 / 2.6414	178.9	no	no
L	8.6879	0.59913	7.5136 / 9.8622	6.8962	no	no
tau	147.45	266.79	-375.46 / 670.35	180.94	no	no
B	60	182.6	-297.89 / 417.89	304.33	si	si

rho=0.25984; Jacobian rank=4/4; cond=58609; accordo multistart=no.

Correlazioni: {"K_L": -0.8632638655314788, "K_tau": 0.9995606532942639, "K_B": -0.9999589710756182, "L_tau": -0.8564414449367175, "L_B": 0.8665537267159535, "tau_B": -0.9993742335629054}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 63.677 / 721.73

Griglia: 42441 coppie L/tau; SSE multistart originale: 63.677; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_051-052.csv

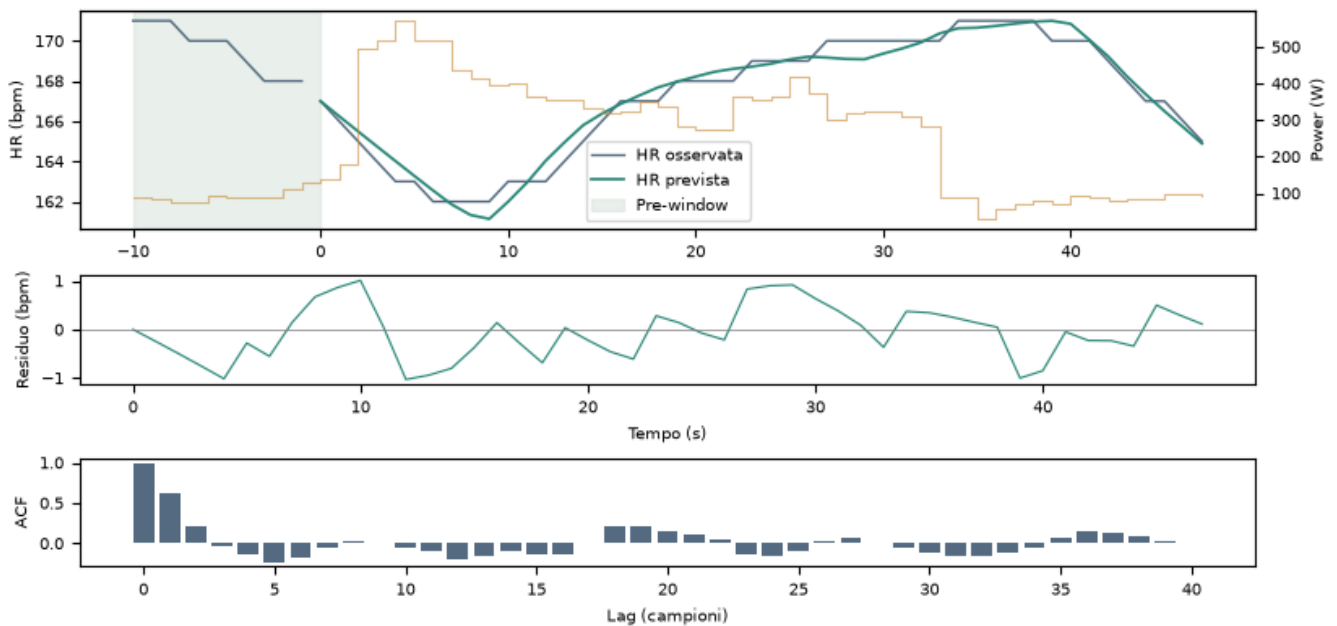
ID: a89892f2004122df4cc06649 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=92.4 W; HR0=167 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=107 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.54481	0.44277	-0.060678	0.9649	1.0958
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.060678	0.54715	0.6149		-0.22794



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.54687	0.52441	-0.48097 / 1.5747	95.893	no	no
L	6.8535	0.25095	6.3617 / 7.3454	3.6616	no	no
tau	141.28	135.72	-124.74 / 407.3	96.065	no	no
B	60	102.83	-141.56 / 261.56	171.39	si	si

rho=0.28416; Jacobian rank=4/4; cond=74425; accordo multistart=no.

Correlazioni: {"K_L": -0.8826068086350594, "K_tau": 0.9994186747464471, "K_B": -0.999980781686481, "L_tau": -0.8787345593566059, "L_B": 0.8841857517925166, "tau_B": -0.9993002066691785}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 14.247 / 340.1

Griglia: 42441 coppie L/tau; SSE multistart originale: 14.247; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_045-046.csv

ID: a8cb189f1f1334b618162a6c | Stato: fitted

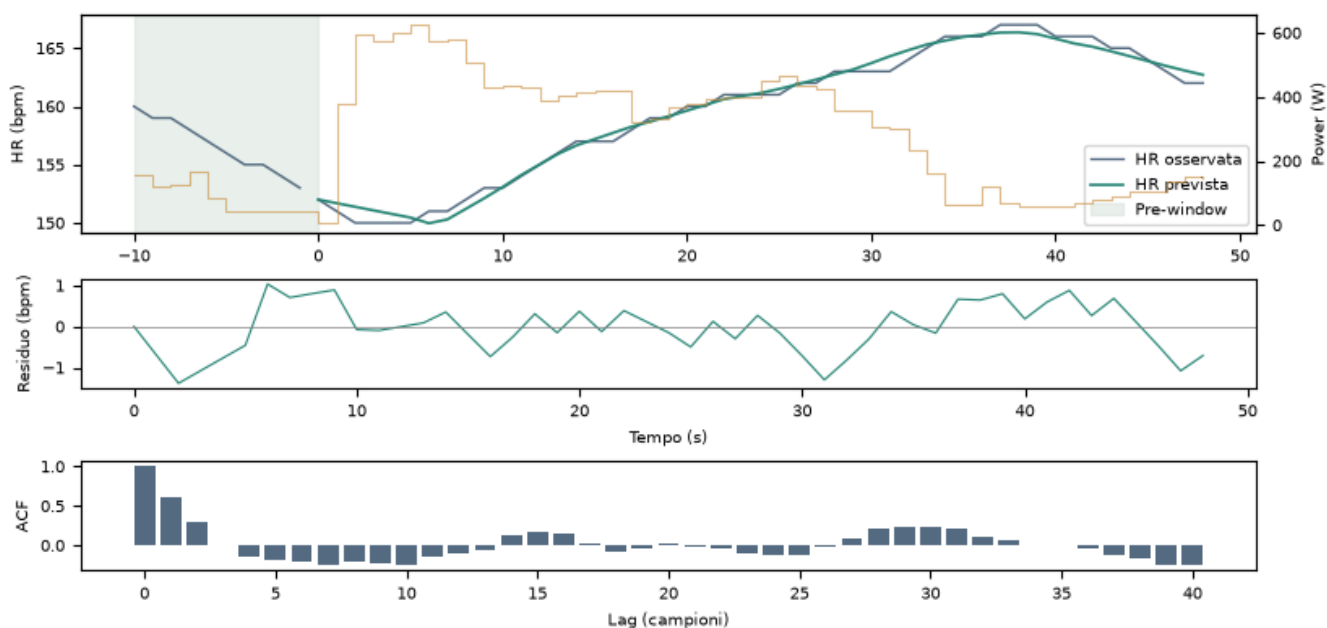
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=87.5 W; HR0=152 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=92 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.59206	0.47537	-0.039713	0.98861	0.96403

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.039713	0.59685	0.61272	-0.18458



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.73897	0.80655	-0.84187 / 2.3198	109.14	no	no
L	5.0994	0.35063	4.4122 / 5.7866	6.8759	no	no
tau	295.51	320.05	-331.8 / 922.82	108.31	no	no
B	60	109.06	-153.77 / 273.77	181.77	si	si

rho=0.14517; Jacobian rank=4/4; cond=90550; accordo multistart=no.

Correlazioni: {"K_L": -0.8980522798067915, "K_tau": 0.999244055514066, "K_B": -0.9992725974324167, "L_tau": -0.9006082829225391, "L_B": 0.8967550673480237, "tau_B": -0.9970940393961227}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 17.176 / 201.75

Griglia: 42441 coppie L/tau; SSE multistart originale: 17.176; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_080-083.csv

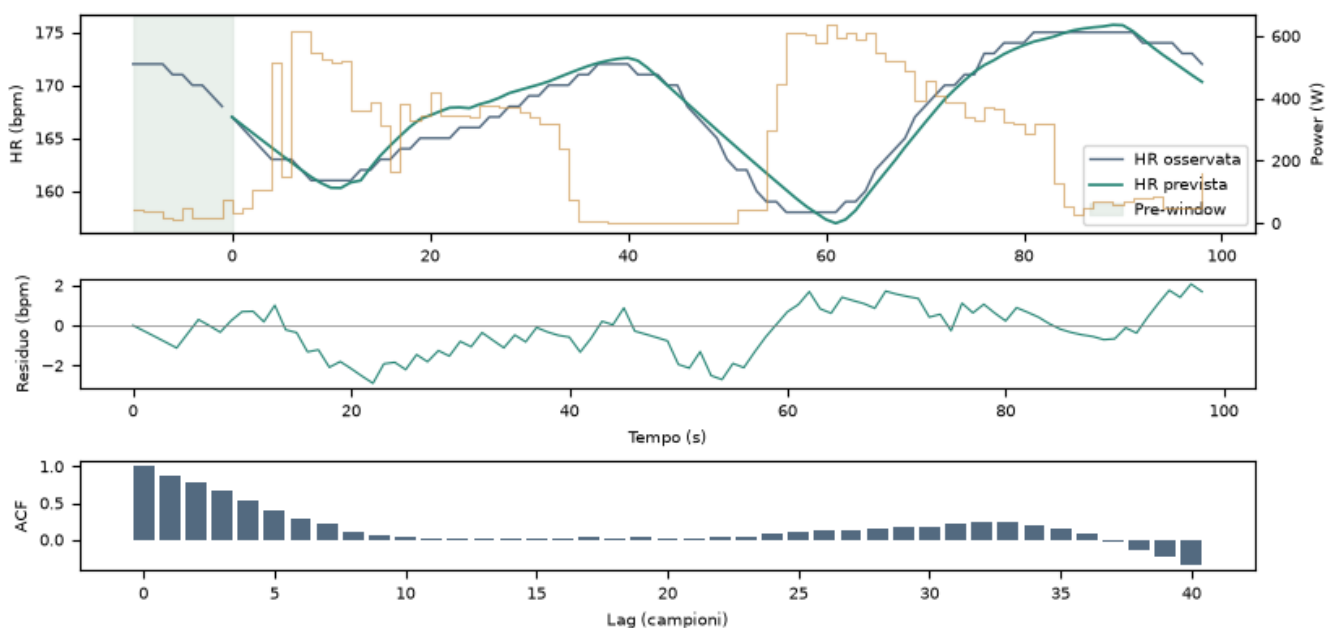
ID: a925df34213980cec0dc3595 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=32.8 W; HR0=167 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=107 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1857	0.96719	-0.26504	0.95119	1.1027
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.26504	1.1615	0.86566		0.40339



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.48125	0.40366	-0.30993 / 1.2724	83.878	no	no
L	6.6866	0.33701	6.026 / 7.3471	5.0402	no	no
tau	146.56	124.3	-97.069 / 390.19	84.812	no	no
B	60	89.733	-115.88 / 235.88	149.56	si	si

rho=0.62304; Jacobian rank=4/4; cond=90448; accordo multistart=no.

Correlazioni: {"K_L": -0.8778480523854556, "K_tau": 0.999566943651037, "K_B": -0.9999846382164626, "L_tau": -0.8756104015027371, "L_B": 0.879253739295432, "tau_B": -0.999496488710933}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 139.18 / 1311

Griglia: 42441 coppie L/tau; SSE multistart originale: 139.18; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_060-061.csv

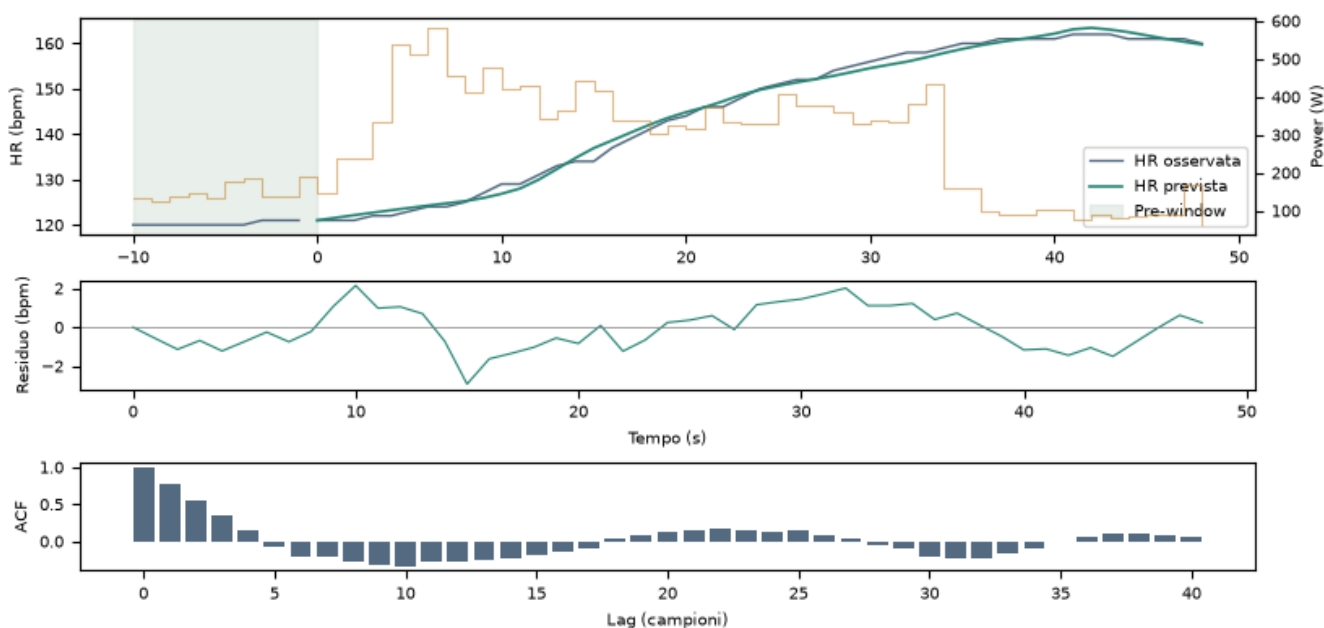
ID: a9f2276aa44bf2f4b2e803df | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=150 W; HR0=121 bpm.

HR(0) osservata fissata; B=144.86 bpm stimato; x(0)=-23.857 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0807	0.90979	-0.077041	0.99471	1.0352
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.077041	1.0892	0.76653	-0.074118	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21999	0.019475	0.18182 / 0.25816	8.8529	no	no
L	7.4946	0.37641	6.7569 / 8.2324	5.0224	no	no
tau	40.953	3.986	33.141 / 48.766	9.7332	no	no
B	144.86	2.2555	140.44 / 149.28	1.5571	no	no

rho=0.98906; Jacobian rank=4/4; cond=2081.5; accordo multistart=no.

Correlazioni: {"K_L": -0.7643896627941498, "K_tau": 0.8209338967168994, "K_B": -0.12060998359111397, "L_tau": -0.8224933237549421, "L_B": -0.20647455213313756, "tau_B": 0.45954091568854244}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 57.233 / 510.12

Griglia: 42441 coppie L/tau; SSE multistart originale: 57.233; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_051-052.csv

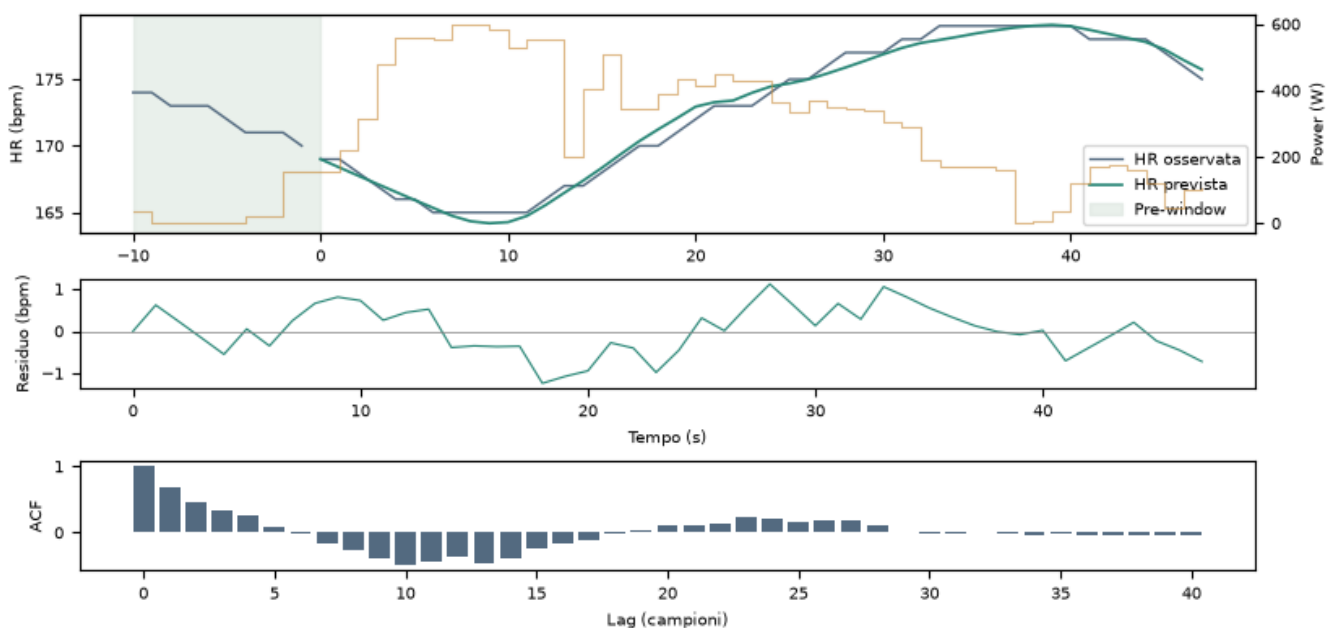
ID: ae1be1a785ce6a699fec1365 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=38.5 W; HR0=169 bpm.

HR(0) osservata fissata; B=80.212 bpm stimato; x(0)=88.788 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.55698	0.45761	0.017851	0.98846	1.0637
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.017851	0.56259	0.678	0.095785	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.41902	0.25343	-0.077695 / 0.91574	60.481	no	no
L	7.491	0.31277	6.878 / 8.1041	4.1752	no	no
tau	142.88	87.143	-27.925 / 313.68	60.992	no	no
B	80.212	55.702	-28.964 / 189.39	69.444	no	no

rho=0.27653; Jacobian rank=4/4; cond=61194; accordo multistart=no.

Correlazioni: {"K_L": -0.9070594375028217, "K_tau": 0.9988193969050078, "K_B": -0.999756690834609, "L_tau": -0.8995055568369622, "L_B": 0.9122305301015314, "tau_B": -0.9976607373281038}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 14.891 / 485.84

Griglia: 42441 coppie L/tau; SSE multistart originale: 14.891; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_011-012.csv

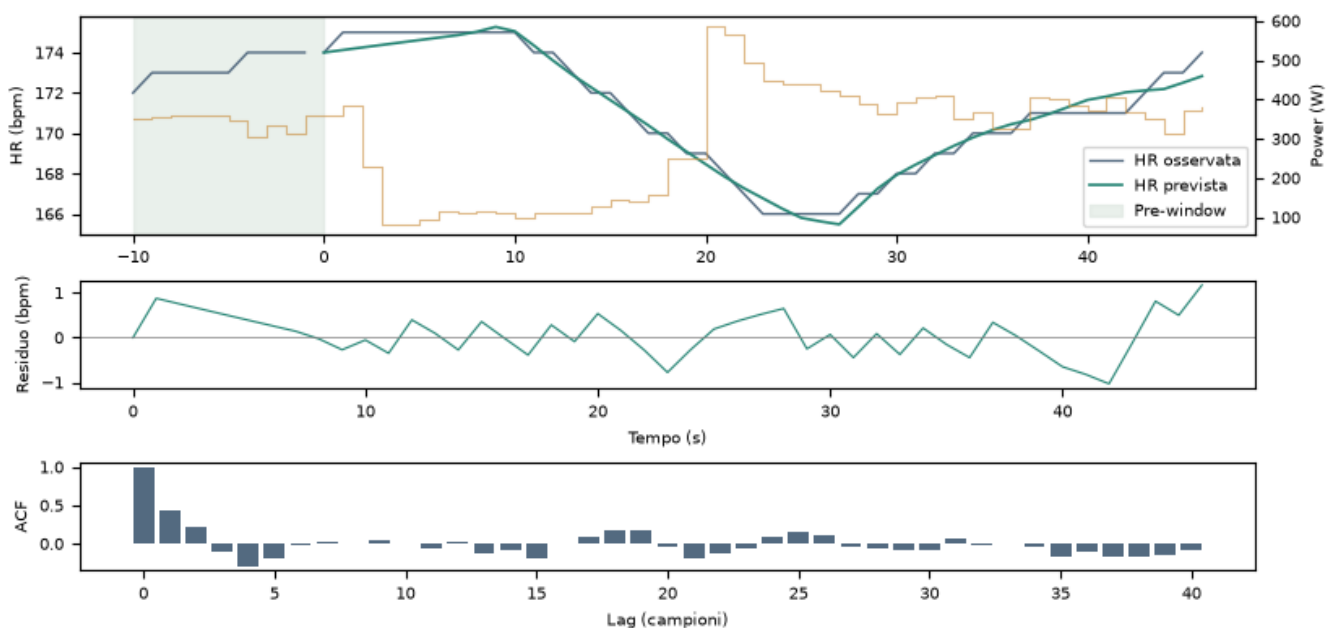
ID: b1cd47229a5980173116a923 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=343.7 W; HR0=174 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-26 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.46762	0.37665	0.063225	0.9762	1.0873
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.063225	0.46834	0.43715	-0.18498	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.69244	0.76796	-0.81277 / 2.1976	110.91	no	no
L	7.0867	0.27007	6.5574 / 7.6161	3.8109	no	no
tau	208.73	230.37	-242.78 / 660.25	110.36	no	no
B	200	32.174	136.94 / 263.06	16.087	si	si

rho=0.18643; Jacobian rank=4/4; cond=32375; accordo multistart=no.

Correlazioni: {"K_L": -0.8183885479669126, "K_tau": 0.999678556396382, "K_B": 0.9995928235733623, "L_tau": -0.822484887588098, "L_B": -0.8149781947028854, "tau_B": 0.9987281180683241}

Multistart: 16 tentativi, 13 convergenti; SSE min/max: 10.277 / 319.95

Griglia: 42441 coppie L/tau; SSE multistart originale: 10.277; selezione: multistart.

Segnalazioni: at_bounds, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_078-079.csv

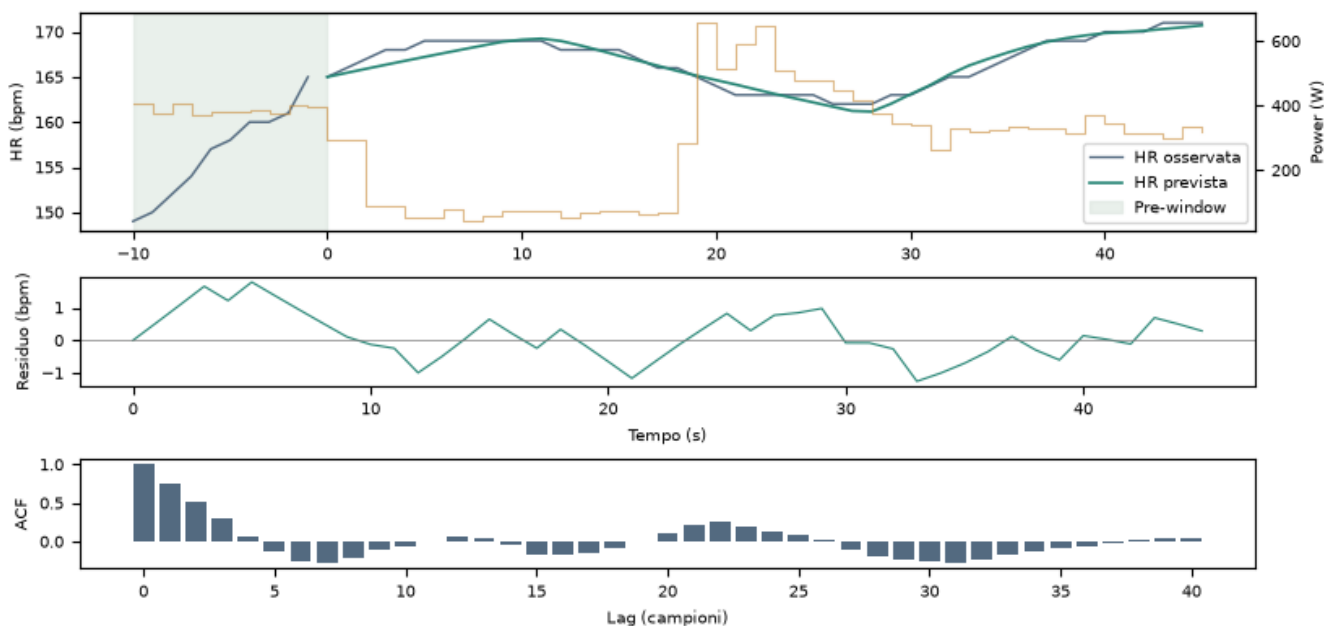
ID: b320d6f5c39fc00f0b645bcb | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=387.8 W; HR0=165 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-35 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.71793	0.56074	0.14545	0.93094	1.0617
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.14545	0.71081	0.76118		-0.1257



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23232	0.32732	-0.40922 / 0.87386	140.89	no	no
L	9.4004	0.43144	8.5548 / 10.246	4.5896	no	no
tau	76.347	109.48	-138.24 / 290.94	143.4	no	no
B	200	47.972	105.97 / 294.03	23.986	si	si

rho=0.46628; Jacobian rank=4/4; cond=39893; accordo multistart=no.

Correlazioni: {"K_L": -0.894132085295374, "K_tau": 0.999585888840376, "K_B": 0.999955531721829, "L_tau": -0.8950531730692856, "L_B": -0.8950870441610932, "tau_B": 0.9996567346014739}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.709 / 363.44

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.709; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_080-081.csv

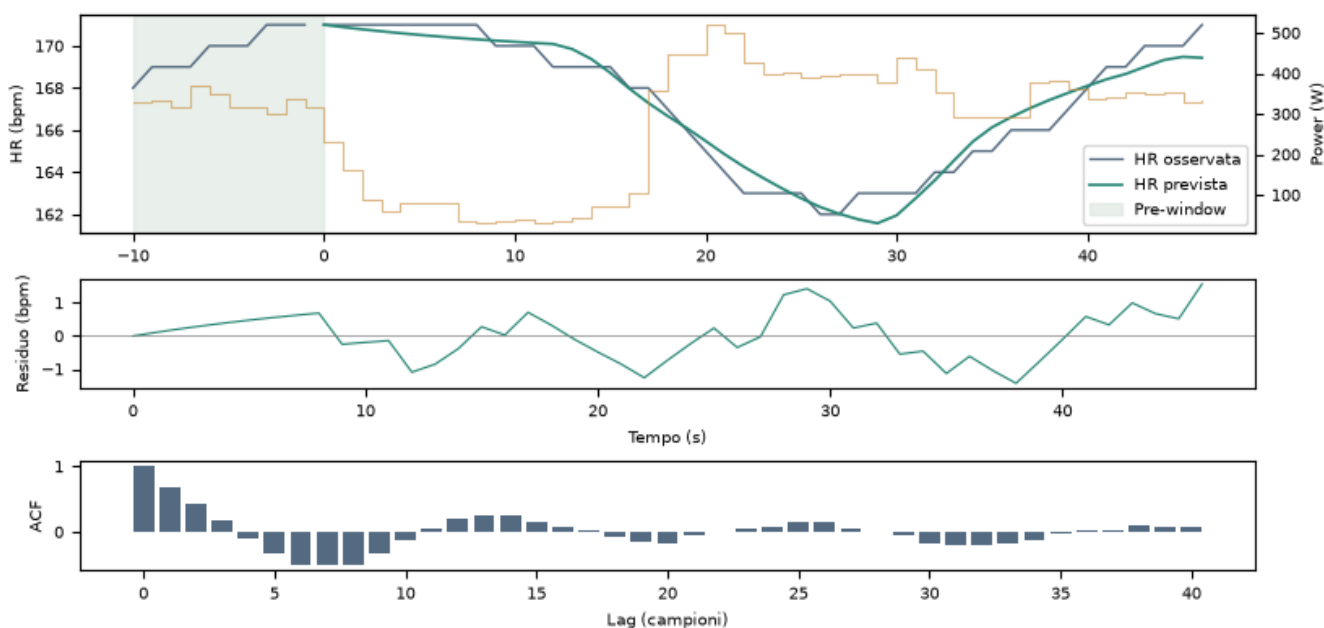
ID: b360d7d5e1b260062391045d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=328.5 W; HR0=171 bpm.

HR(0) osservata fissata; B=169.46 bpm stimato; x(0)=1.5425 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.70411	0.5745	0.020917	0.94923	1.0474
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.020917	0.71141	0.67915	-0.33419	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.043494	0.0051199	0.033459 / 0.05353	11.771	no	no
L	12.364	0.424	11.533 / 13.195	3.4293	no	no
tau	13.486	1.8449	9.8702 / 17.102	13.68	no	no
B	169.46	0.32193	168.83 / 170.09	0.18998	no	no

rho=2.4941; Jacobian rank=4/4; cond=1562.2; accordo multistart=no.

Correlazioni: {"K_L": -0.7312868986483789, "K_tau": 0.9248493342556896, "K_B": 0.8640187897081538, "L_tau": -0.8192841789448563, "L_B": -0.5212063889430495, "tau_B": 0.7062493526324779}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.301 / 297.19

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.443; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_055-056.csv

ID: b76122cf81b1029fd7b87e21 | Stato: fitted

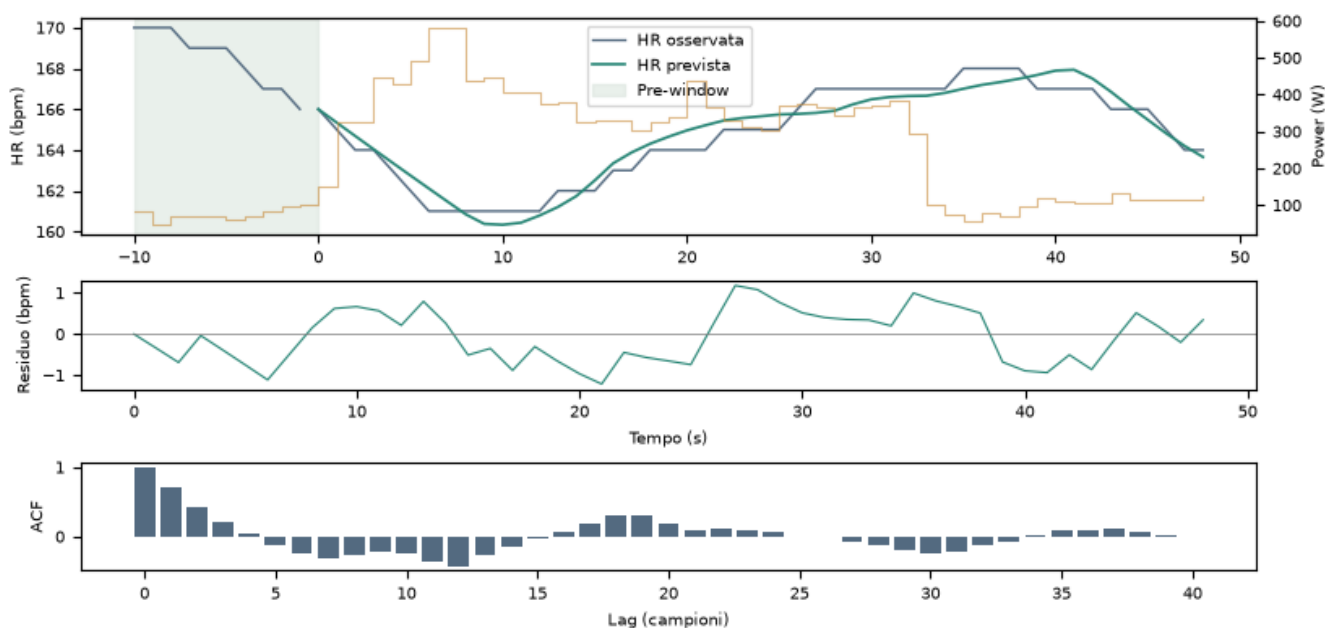
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=75.5 W; HR0=166 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=106 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.64038	0.56436	-0.059004	0.92075	1.0858

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.059004	0.64427	0.72038	-0.12254



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.4693	0.85023	-1.1971 / 2.1357	181.17	no	no
L	8.29	0.41886	7.469 / 9.1109	5.0526	no	no
tau	160.49	291.92	-411.68 / 732.66	181.89	no	no
B	60	189.84	-312.09 / 432.09	316.41	si	si

rho=0.24743; Jacobian rank=4/4; cond=1.2694e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.8842409233382605, "K_tau": 0.9997373800455353, "K_B": -0.999990720541637, "L_tau": -0.8802159774246364, "L_B": 0.8855610470998284, "tau_B": -0.9997067727603473}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 20.094 / 251.88

Griglia: 42441 coppie L/tau; SSE multistart originale: 20.094; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_039-040.csv

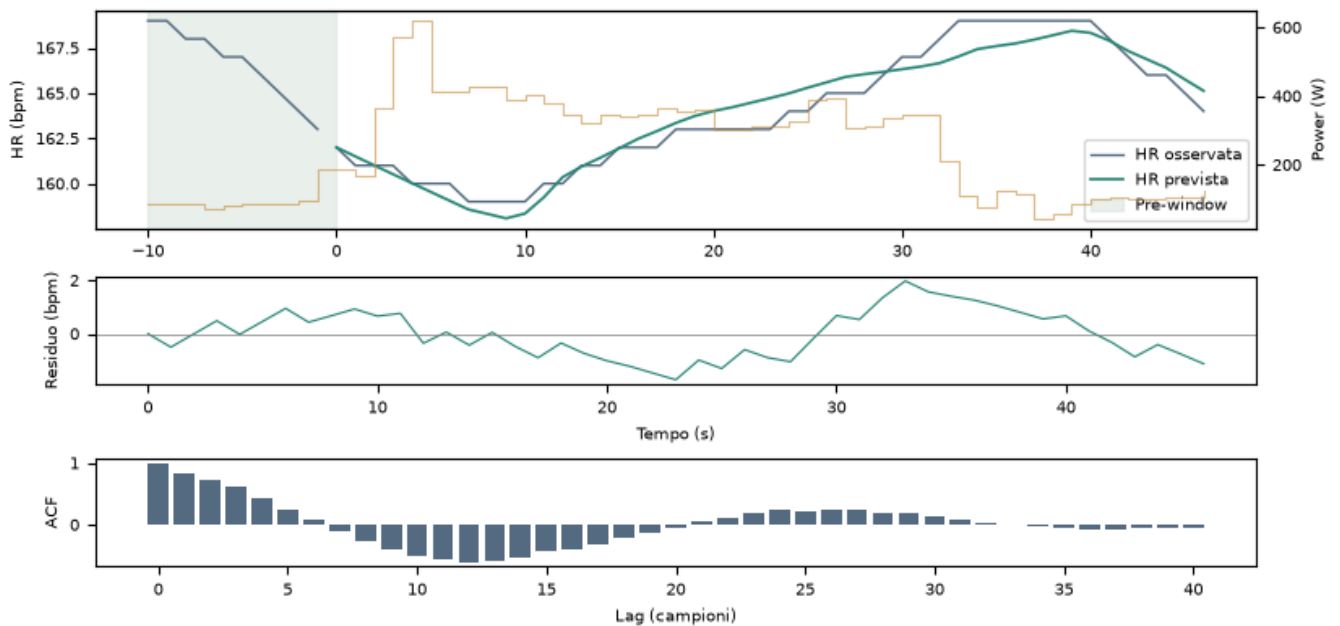
ID: b9cdcc7461e98ec0166fa300 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=95.6 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.8812	0.74595	-0.0064489	0.93007	1.037
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.0064489	0.8907	0.83967	0.24856	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.63578	1.2295	-1.7741 / 3.0456	193.39	no	no
L	7.1429	0.53895	6.0865 / 8.1992	7.5453	no	no
tau	204.57	399.87	-579.17 / 988.32	195.46	no	no
B	60	200.08	-332.17 / 452.17	333.47	si	si

rho=0.18994; Jacobian rank=4/4; cond=81170; accordo multistart=no.

Correlazioni: {"K_L": -0.8654020895484082, "K_tau": 0.9993816374893978, "K_B": -0.9999250329182554, "L_tau": -0.8601085726050614, "L_B": 0.8688653311616242, "tau_B": -0.9989584995990007}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 36.496 / 242.57

Griglia: 42441 coppie L/tau; SSE multistart originale: 36.496; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_072-073.csv

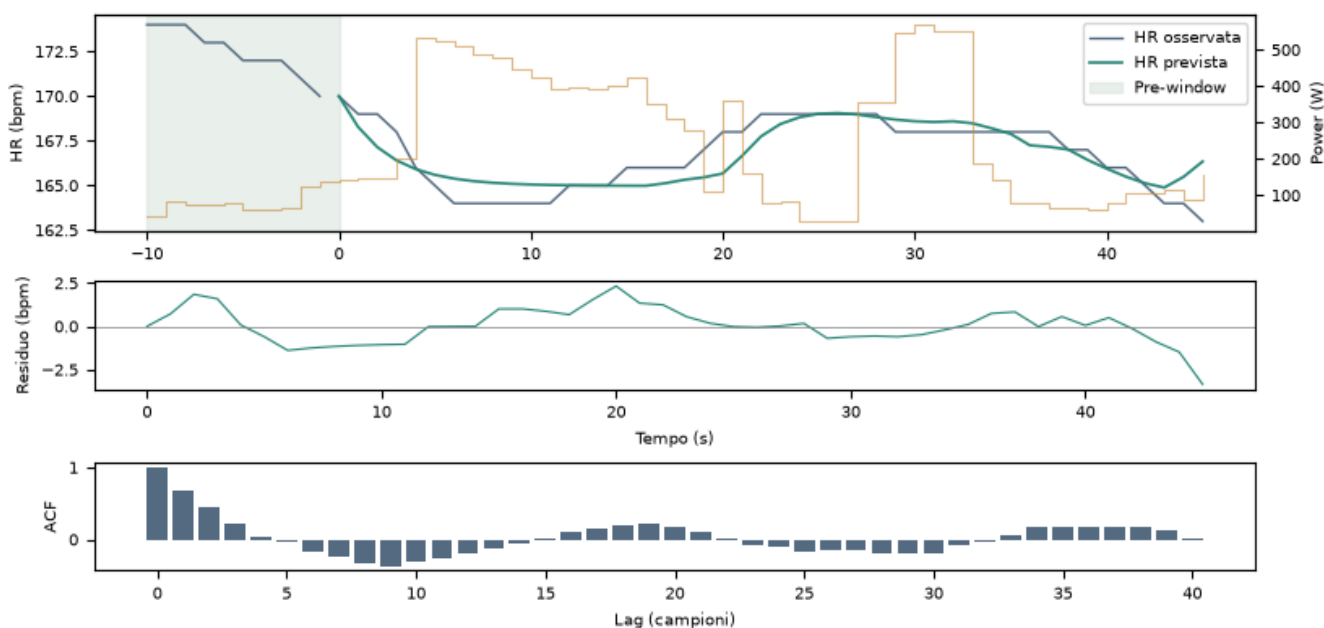
ID: ba9a731adfd3534485f3573b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=80.2 W; HR0=170 bpm.

HR(0) osservata fissata; B=164.98 bpm stimato; x(0)=5.0189 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0177	0.75364	0.030524	0.71787	0.73058
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.030524	1.0285	0.69049	-0.036972	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.010947	0.0014182	0.0081676 / 0.013727	12.955	no	no
L	16.436	0.52207	15.413 / 17.459	3.1764	no	no
tau	2.3825	0.68361	1.0427 / 3.7224	28.693	no	no
B	164.98	0.29221	164.41 / 165.55	0.17712	no	no

rho=11.989; Jacobian rank=4/4; cond=989.47; accordo multistart=no.

Correlazioni: {"K_L": -0.3831401600986195, "K_tau": 0.5953889451930287, "K_B": -0.8082469848814475, "L_tau": -0.6310422025403557, "L_B": 0.4329678499857409, "tau_B": -0.6135068605531573}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 47.643 / 149.62

Griglia: 42441 coppie L/tau; SSE multistart originale: 47.643; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_031-032.csv

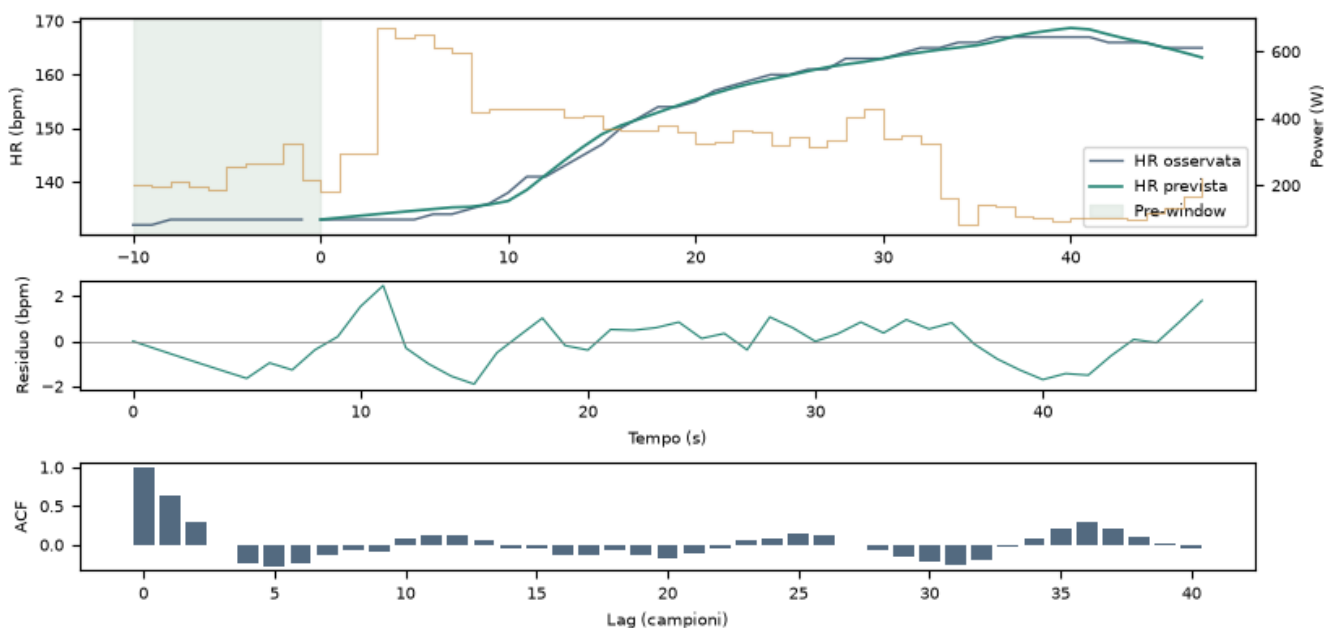
ID: c5859d387285d355acd61e56 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=232.8 W; HR0=133 bpm.

HR(0) osservata fissata; B=155.3 bpm stimato; x(0)=-22.302 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.97952	0.79374	-0.099479	0.99417	1.0501
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.099479	0.98477	0.64305	-0.26924	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.39378	0.055686	0.28464 / 0.50292	14.141	no	no
L	7.3756	0.26564	6.8549 / 7.8962	3.6017	no	no
tau	65.138	8.822	47.847 / 82.43	13.543	no	no
B	155.3	2.3478	150.7 / 159.9	1.5118	no	no

rho=0.60831; Jacobian rank=4/4; cond=2631.7; accordo multistart=no.

Correlazioni: {"K_L": -0.7383946823883937, "K_tau": 0.9528671653031067, "K_B": -0.05516569571281063, "L_tau": -0.8031779031837472, "L_B": -0.258301710867559, "tau_B": 0.2443689262866597}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 46.054 / 575.76

Griglia: 42441 coppie L/tau; SSE multistart originale: 46.054; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_013-014.csv

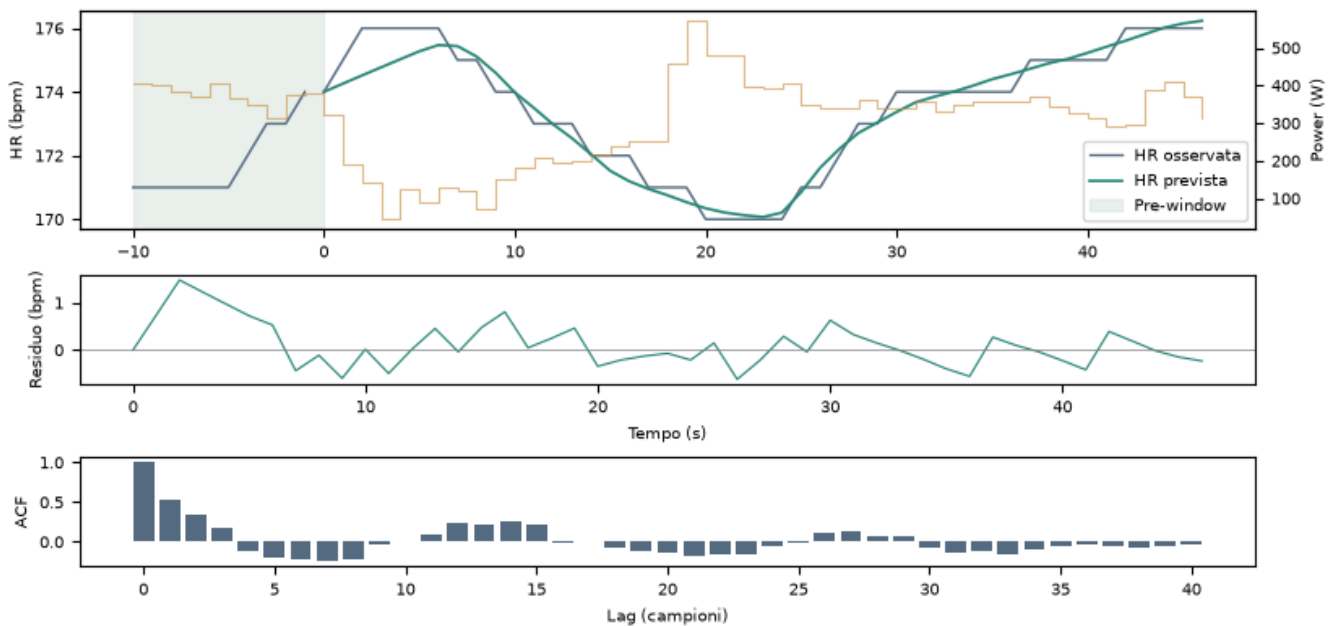
ID: c5914a330a5691f55a0e048c | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=375.6 W; HR0=174 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-26 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.47399	0.35015	0.10544	0.94307	1.027
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.10544	0.46711	0.53512		-0.19473



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.28588	0.34005	-0.38061 / 0.95237	118.95	no	no
L	5.6694	0.35321	4.9771 / 6.3617	6.2301	no	no
tau	99.449	120.03	-135.81 / 334.7	120.69	no	no
B	200	32.129	137.03 / 262.97	16.065	si	si

rho=0.40554; Jacobian rank=4/4; cond=39107; accordo multistart=no.

Correlazioni: {"K_L": -0.8245397690890981, "K_tau": 0.9994962875834746, "K_B": 0.9999491431549751, "L_tau": -0.8282880944595153, "L_B": -0.8255195317482921, "tau_B": 0.9994445921202587}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 10.559 / 177.61

Griglia: 42441 coppie L/tau; SSE multistart originale: 10.559; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_053-054.csv

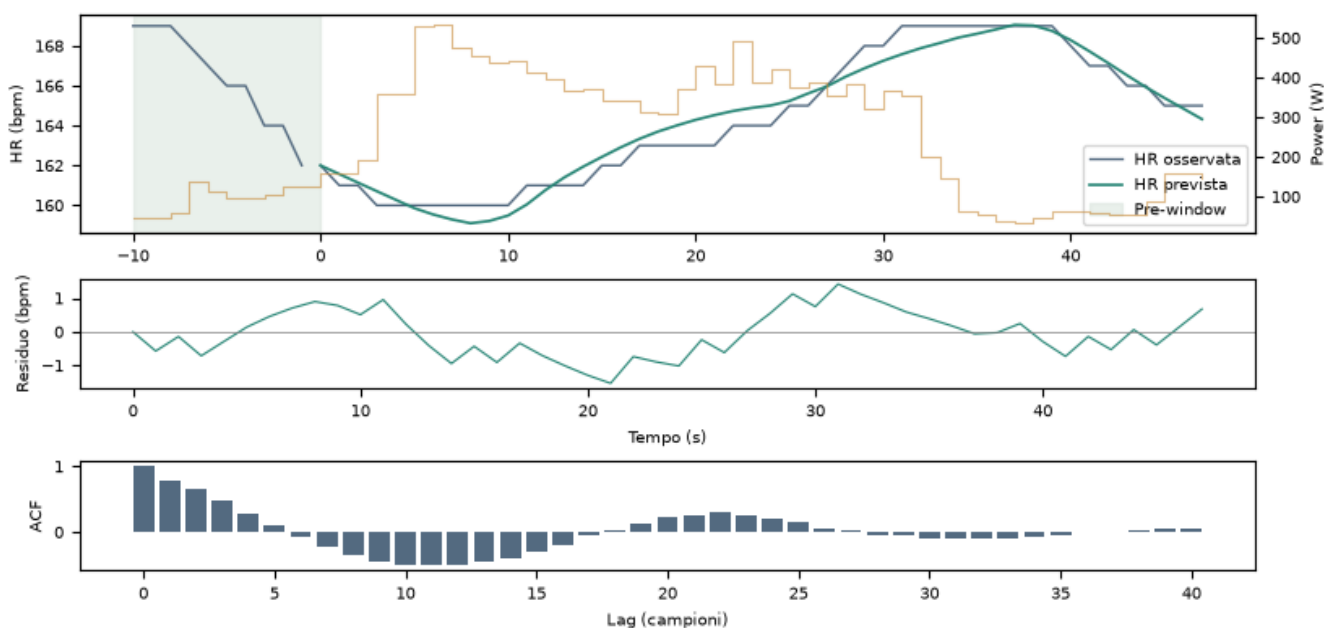
ID: c744b619c6f2cb6fb8fe5654 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=95.2 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.69239	0.57678	-0.041982	0.95494	1.1061
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.041982	0.69843	0.77862		0.09858



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.62254	1.3158	-1.9564 / 3.2014	211.36	no	no
L	5.3658	0.55584	4.2763 / 6.4552	10.359	no	no
tau	235.68	500.77	-745.82 / 1217.2	212.47	no	no
B	60	219.8	-370.8 / 490.8	366.33	si	si

rho=0.17665; Jacobian rank=4/4; cond=1.22e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.9267074343995223, "K_tau": 0.9996907464843603, "K_B": -0.9999576202133438, "L_tau": -0.925452975454549, "L_B": 0.9280323059662853, "tau_B": -0.9994605787167619}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.012 / 219.18

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.012; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_008-009.csv

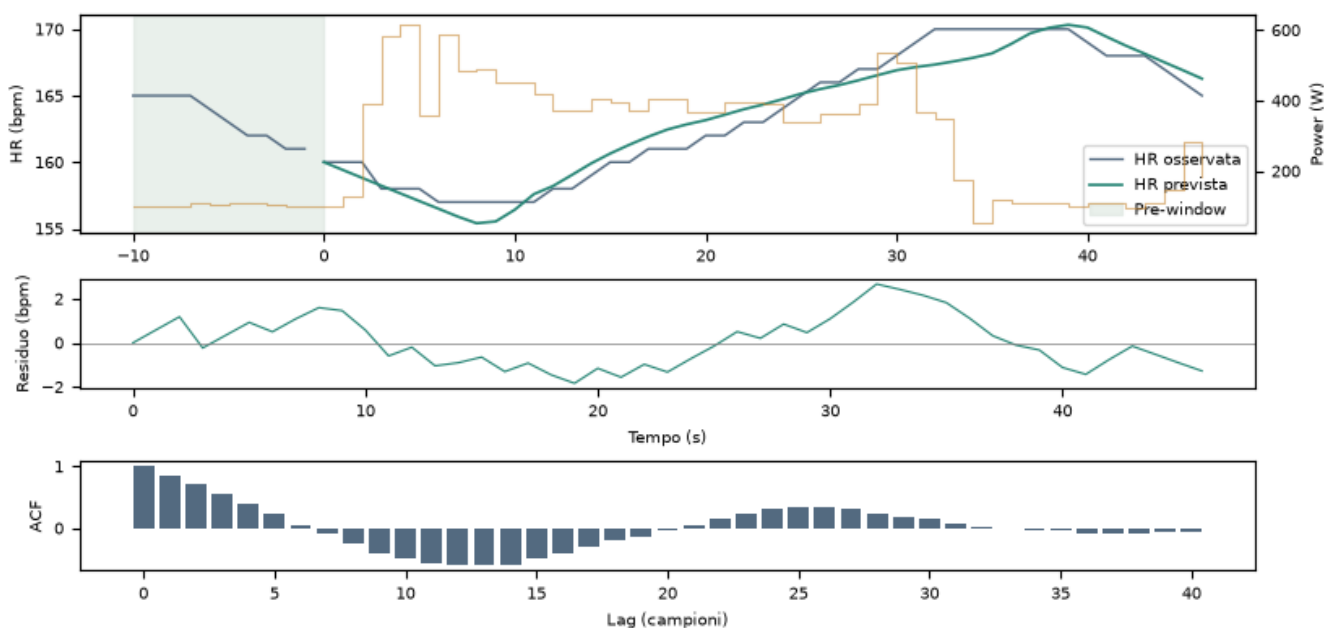
ID: c8c597cfe2f8f3a99b75bd63 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=103.2 W; HR0=160 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=100 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1478	0.96552	0.044565	0.94101	1.1478
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.044565	1.1593	0.85343		0.23414



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.58809	0.79984	-0.9796 / 2.1558	136.01	no	no
L	6.3258	0.51463	5.3172 / 7.3345	8.1353	no	no
tau	168.34	232.49	-287.34 / 624.02	138.11	no	no
B	60	139.57	-213.56 / 333.56	232.62	si	si

rho=0.23568; Jacobian rank=4/4; cond=51798; accordo multistart=no.

Correlazioni: {"K_L": -0.8417478852278091, "K_tau": 0.9986342335893968, "K_B": -0.9997388924131071, "L_tau": -0.8319170018805087, "L_B": 0.8484669605291633, "tau_B": -0.997334326767674}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 61.917 / 394.24

Griglia: 42441 coppie L/tau; SSE multistart originale: 61.917; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_041-042.csv

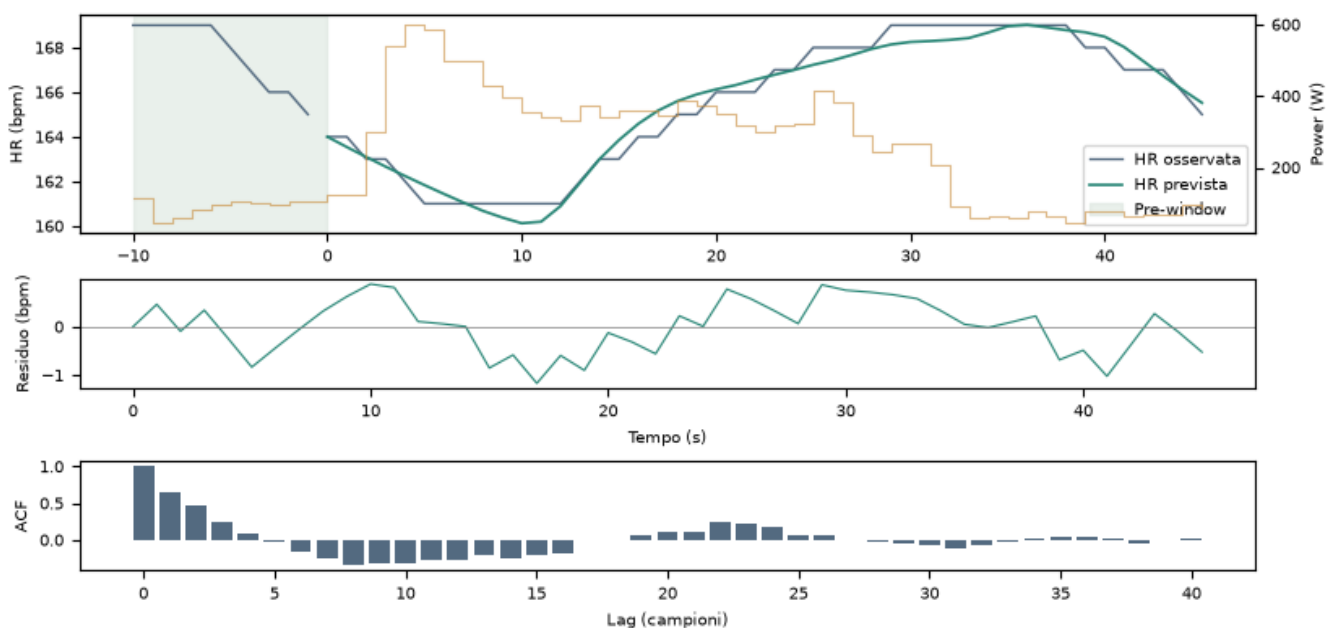
ID: ce19bcc7ca6ef72ce4d4118e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=91.6 W; HR0=164 bpm.

HR(0) osservata fissata; B=148.52 bpm stimato; x(0)=15.479 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.5374	0.43566	0.0029271	0.96624	1.112
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.0029271	0.54333	0.6406	-0.035007	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.097855	0.02225	0.054244 / 0.14147	22.738	no	no
L	8.3787	0.33969	7.7129 / 9.0444	4.0543	no	no
tau	33.141	8.1241	17.218 / 49.065	24.514	no	no
B	148.52	3.8021	141.07 / 155.97	2.56	no	no

rho=1.105; Jacobian rank=4/4; cond=11402; accordo multistart=no.

Correlazioni: {"K_L": -0.8663065072192196, "K_tau": 0.9848794293274817, "K_B": -0.9979437883581685, "L_tau": -0.8520658274339253, "L_B": 0.8742714756006398, "tau_B": -0.9754039080588461}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 13.285 / 216.69

Griglia: 42441 coppie L/tau; SSE multistart originale: 13.285; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/250729173622__laps_010-011.csv

ID: ce7e43fbfe9c120c9d8b3574 | Stato: fitted

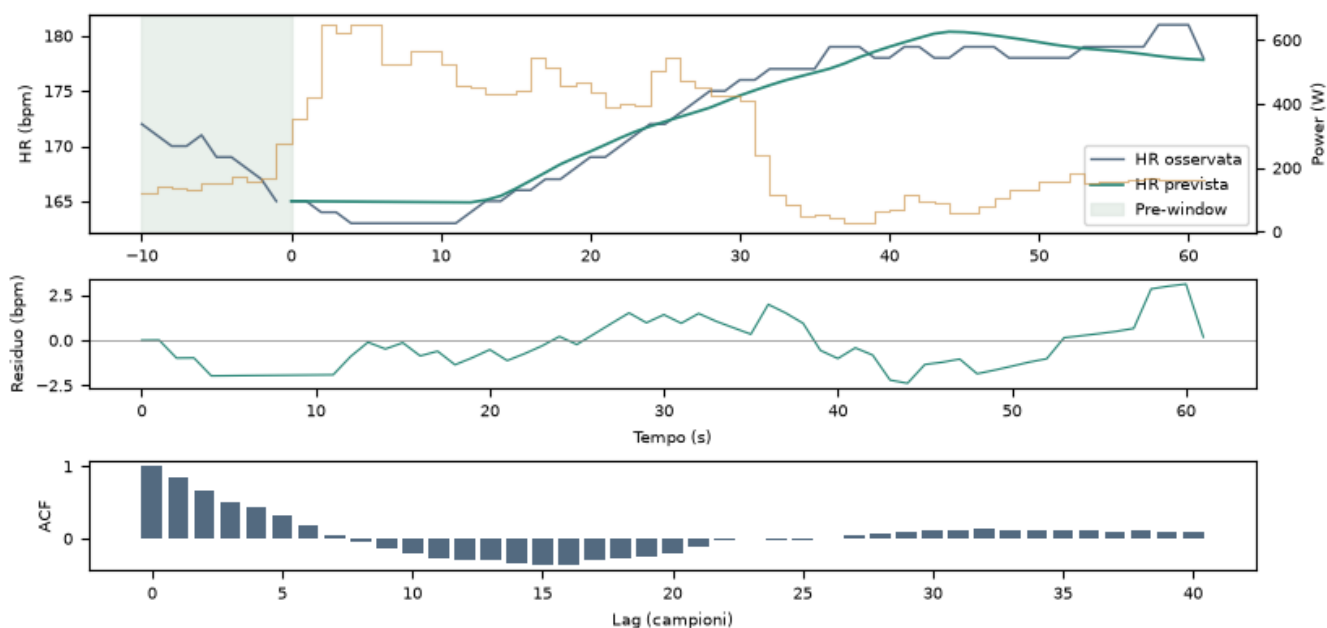
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=161.8 W; HR0=165 bpm.

HR(0) osservata fissata; B=149.25 bpm stimato; x(0)=15.748 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.3458	1.1161	-0.29824	0.95582	0.85993

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.29824	1.323	0.83994	0.32192



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	2.8176	25.015	-46.211 / 51.846	887.79	no	no
L	12.264	1.081	10.145 / 14.383	8.8145	no	no
tau	1800	15971	-29504 / 33104	887.3	si	si
B	149.25	207.54	-257.52 / 556.03	139.05	no	no

rho=0.027075; Jacobian rank=4/4; cond=3.3326e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.8833597045338741, "K_tau": 0.9999557052800788, "K_B": -0.9786671285384415, "L_tau": -0.8838412077784902, "L_B": 0.8614207854615916, "tau_B": -0.9767415803078806}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 112.29 / 416.22

Griglia: 42441 coppie L/tau; SSE multistart originale: 112.29; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_037-038.csv

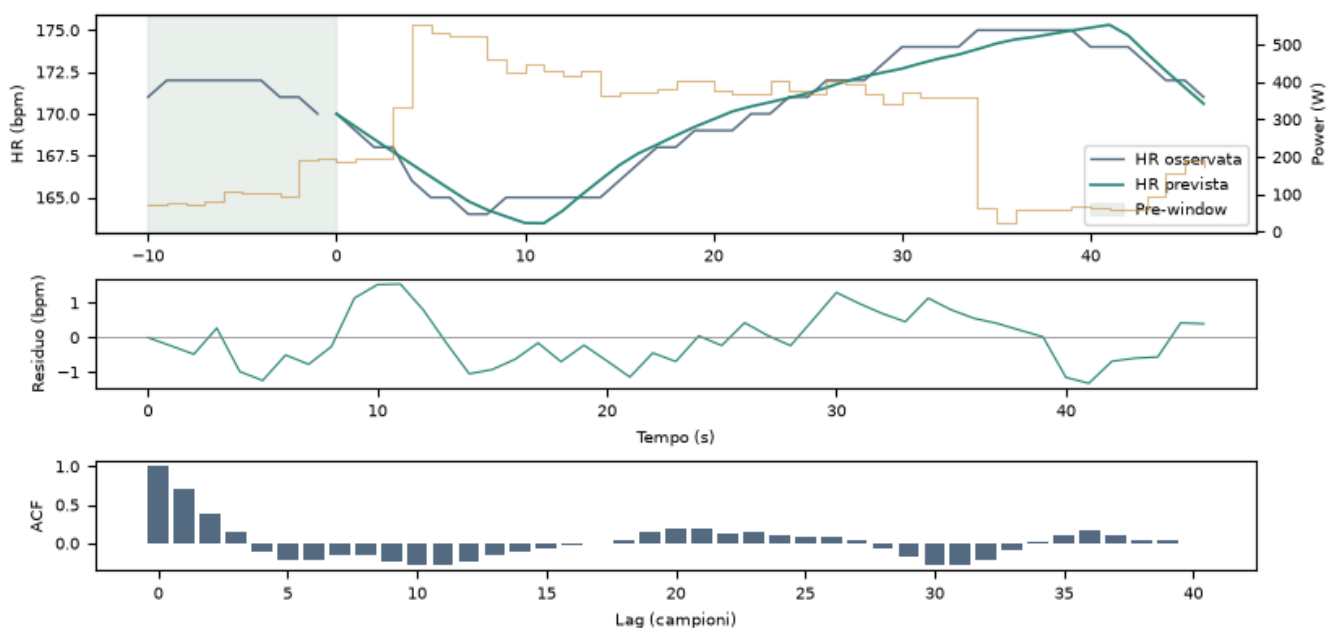
ID: cff71f333cde932d2c21be3f | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=110.7 W; HR0=170 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=110 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.75376	0.63303	-0.048802	0.95801	1.0786
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.048802	0.76031	0.70305		-0.22735



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.56483	0.67406	-0.75632 / 1.886	119.34	no	no
L	7.3063	0.37435	6.5726 / 8.04	5.1237	no	no
tau	143.76	173.68	-196.65 / 484.18	120.81	no	no
B	60	130.91	-196.59 / 316.59	218.19	no	si

rho=0.26915; Jacobian rank=4/4; cond=66838; accordo multistart=si.

Correlazioni: {"K_L": -0.8742091553742306, "K_tau": 0.9994758712791603, "K_B": -0.9999648047896595, "L_tau": -0.8680735100898163, "L_B": 0.8770020521354049, "tau_B": -0.9992857857673235}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 26.703 / 26.703

Griglia: 42441 coppie L/tau; SSE multistart originale: 26.703; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_062-063.csv

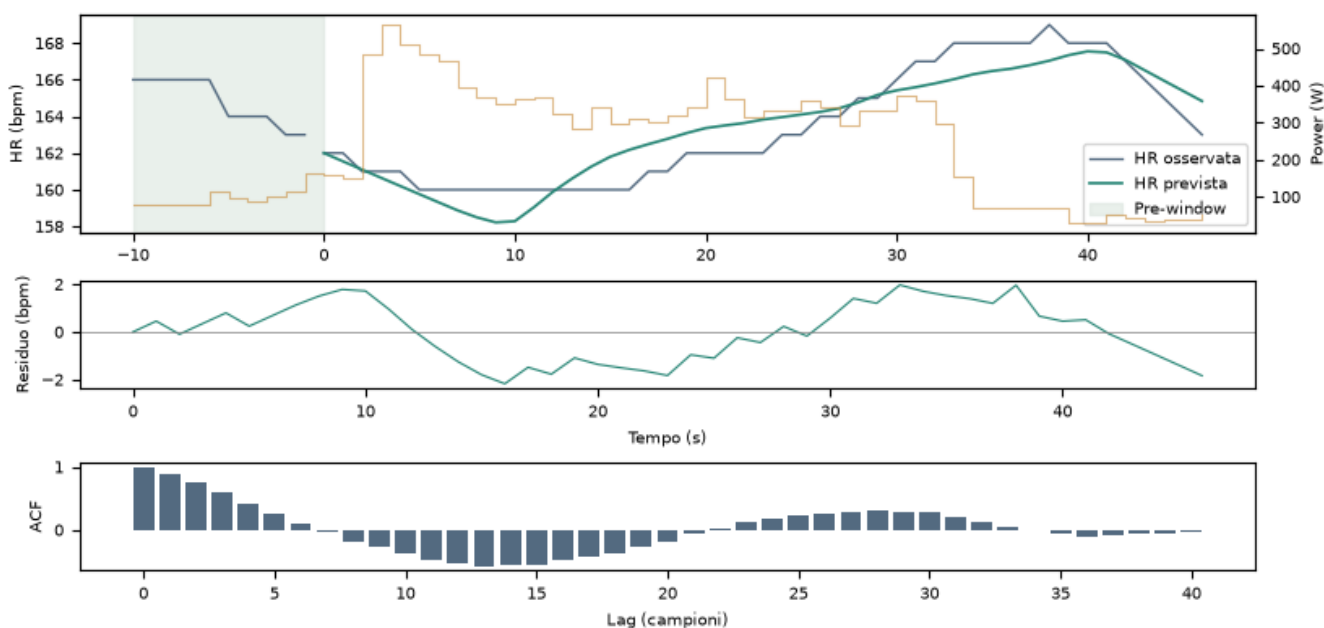
ID: d1fbc070b05f0795e23a7cf3 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=99.9 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.209	1.0395	0.0044706	0.8451	1.0368
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.0044706	1.222	0.87838		0.26345



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.6538	2.0606	-3.385 / 4.6925	315.17	no	no
L	7.6353	0.74578	6.1735 / 9.097	9.7676	no	no
tau	224.92	715.92	-1178.3 / 1628.1	318.3	no	no
B	60	324.27	-575.57 / 695.57	540.45	si	si

rho=0.17057; Jacobian rank=4/4; cond=90294; accordo multistart=no.

Correlazioni: {"K_L": -0.8214724336637975, "K_tau": 0.9994355613918338, "K_B": -0.9999356608409723, "L_tau": -0.8133790214997845, "L_B": 0.8257162591318267, "tau_B": -0.9990632851457231}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 68.697 / 245.87

Griglia: 42441 coppie L/tau; SSE multistart originale: 68.697; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_024-025.csv

ID: d717207fcc134f1506b1f372 | Stato: fitted

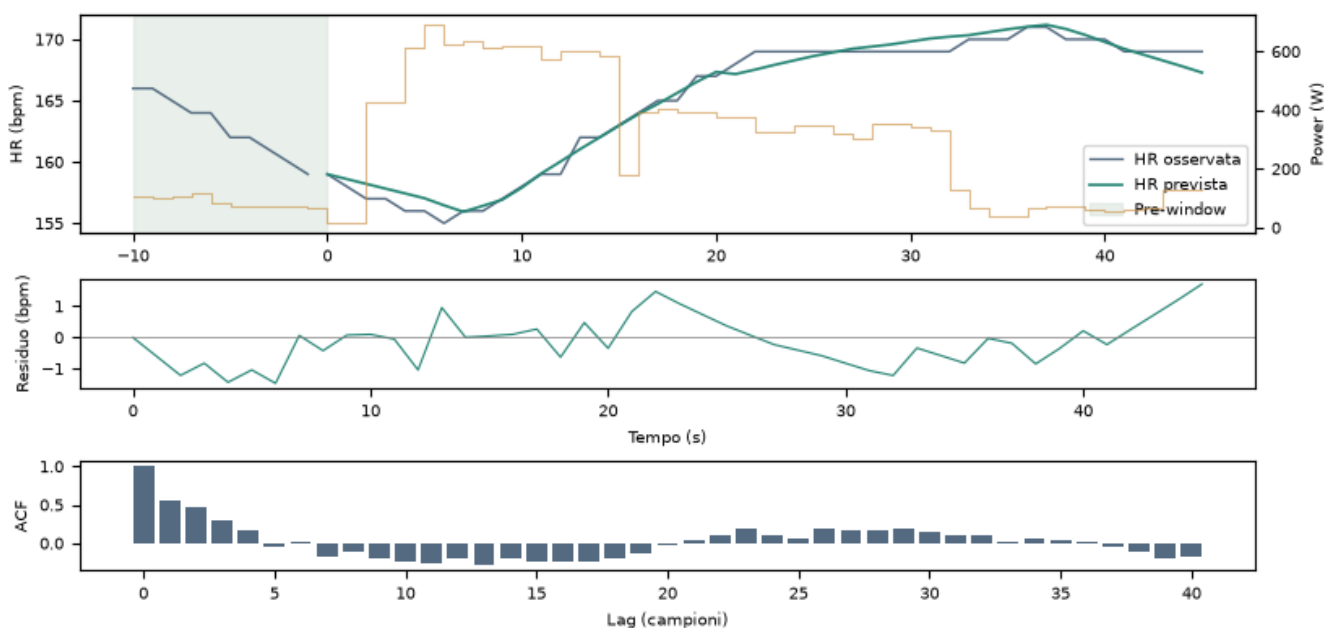
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=87.7 W; HR0=159 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=99 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.75832	0.59852	-0.13023	0.98026	0.95314

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.13023	0.75531	0.55807	-0.054275



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.64505	0.89671	-1.1125 / 2.4026	139.01	no	no
L	4.9624	0.40994	4.1589 / 5.7658	8.261	no	no
tau	250.77	345.06	-425.55 / 927.09	137.6	no	no
B	60	146.52	-227.18 / 347.18	244.2	si	si

rho=0.15966; Jacobian rank=4/4; cond=86861; accordo multistart=no.

Correlazioni: {"K_L": -0.9065661929003445, "K_tau": 0.9994345769077443, "K_B": -0.9996800092673425, "L_tau": -0.9056773977970667, "L_B": 0.9084199108406587, "tau_B": -0.9983242060402537}

Multistart: 16 tentativi, 13 convergenti; SSE min/max: 26.453 / 325

Griglia: 42441 coppie L/tau; SSE multistart originale: 26.462; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_076-077.csv

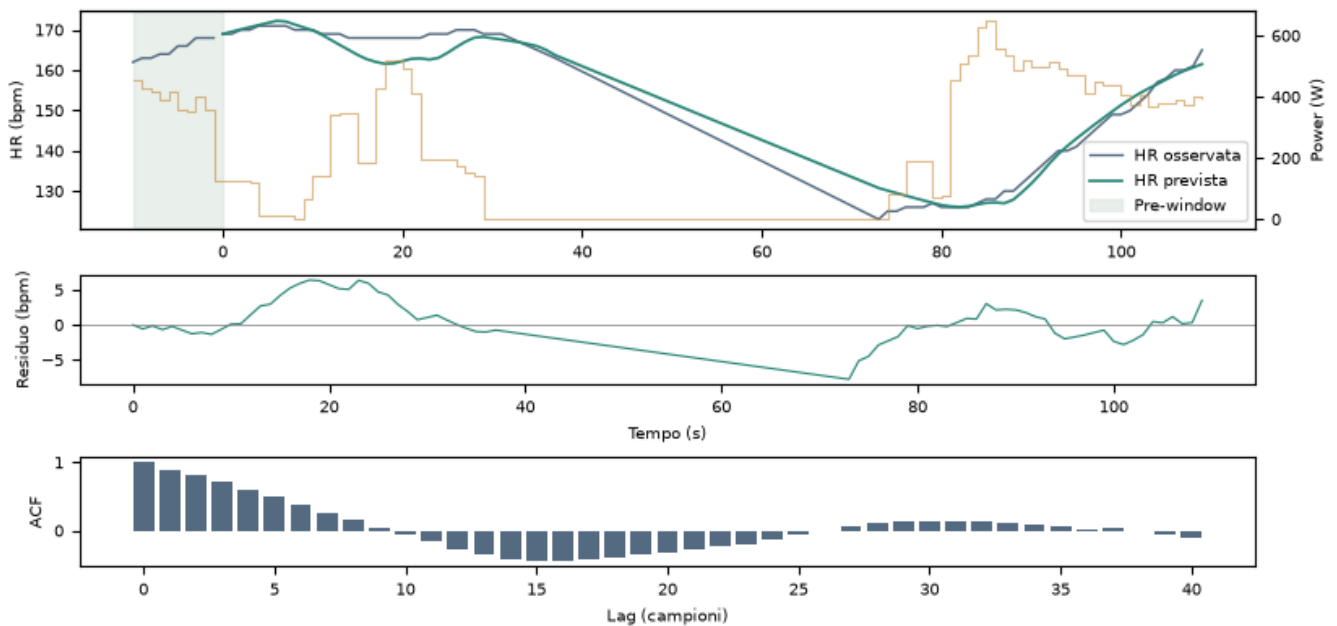
ID: dd5fce51dcbdefd113edc71b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=370 W; HR0=169 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-31 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.8418	2.0654	0.6783	0.97362	0.96258
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.6783	2.7782	0.89341	0.5096	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.28131	0.027288	0.22782 / 0.33479	9.7003	no	no
L	6.4552	0.44699	5.5791 / 7.3313	6.9244	no	no
tau	54.043	5.8659	42.546 / 65.54	10.854	no	no
B	200	4.8283	190.54 / 209.46	2.4141	si	si

rho=1.8975; Jacobian rank=4/4; cond=3164.8; accordo multistart=no.

Correlazioni: {"K_L": -0.6698256500797929, "K_tau": 0.9516425697230789, "K_B": 0.9883208418368663, "L_tau": -0.7118600964637901, "L_B": -0.6120356562925416, "tau_B": 0.9119490229734306}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 605.67 / 10764

Griglia: 42441 coppie L/tau; SSE multistart originale: 605.67; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, recording_gaps, irregular_sampling, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_021-022.csv

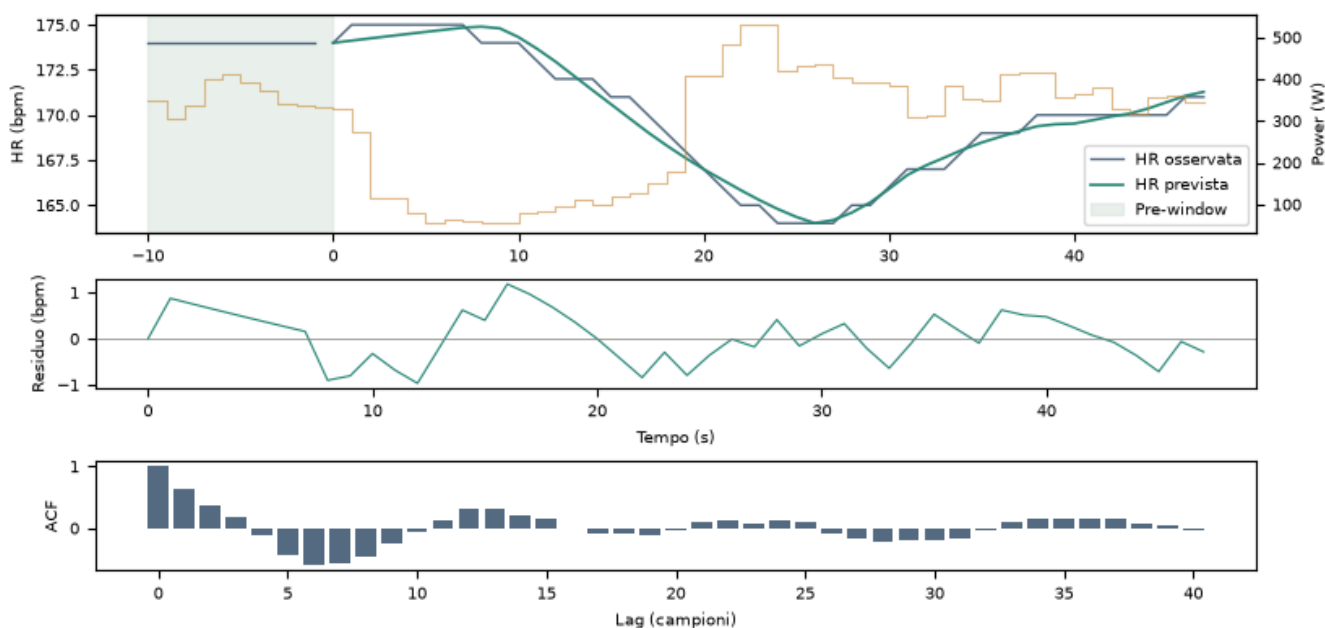
ID: dddcb726d819464fc34b5908 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=357.5 W; HR0=174 bpm.

HR(0) osservata fissata; B=183.52 bpm stimato; x(0)=-9.5248 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.52496	0.43237	0.041981	0.97745	0.98998
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.041981	0.52882	0.63943	-0.43436	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.24185	0.10444	0.037157 / 0.44655	43.182	no	no
L	7.3417	0.32825	6.6983 / 7.985	4.4711	no	no
tau	75.722	32.086	12.834 / 138.61	42.373	no	no
B	183.52	5.9943	171.78 / 195.27	3.2662	no	no

rho=0.52374; Jacobian rank=4/4; cond=13463; accordo multistart=no.

Correlazioni: {"K_L": -0.8589810690847152, "K_tau": 0.9977245808927158, "K_B": 0.997346879964718, "L_tau": -0.8685171306259774, "L_B": -0.8485511003538025, "tau_B": 0.991266192010217}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 13.228 / 318.3

Griglia: 42441 coppie L/tau; SSE multistart originale: 13.228; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_017-018.csv

ID: e33316f4acf5932c9fe28a6e | Stato: fitted

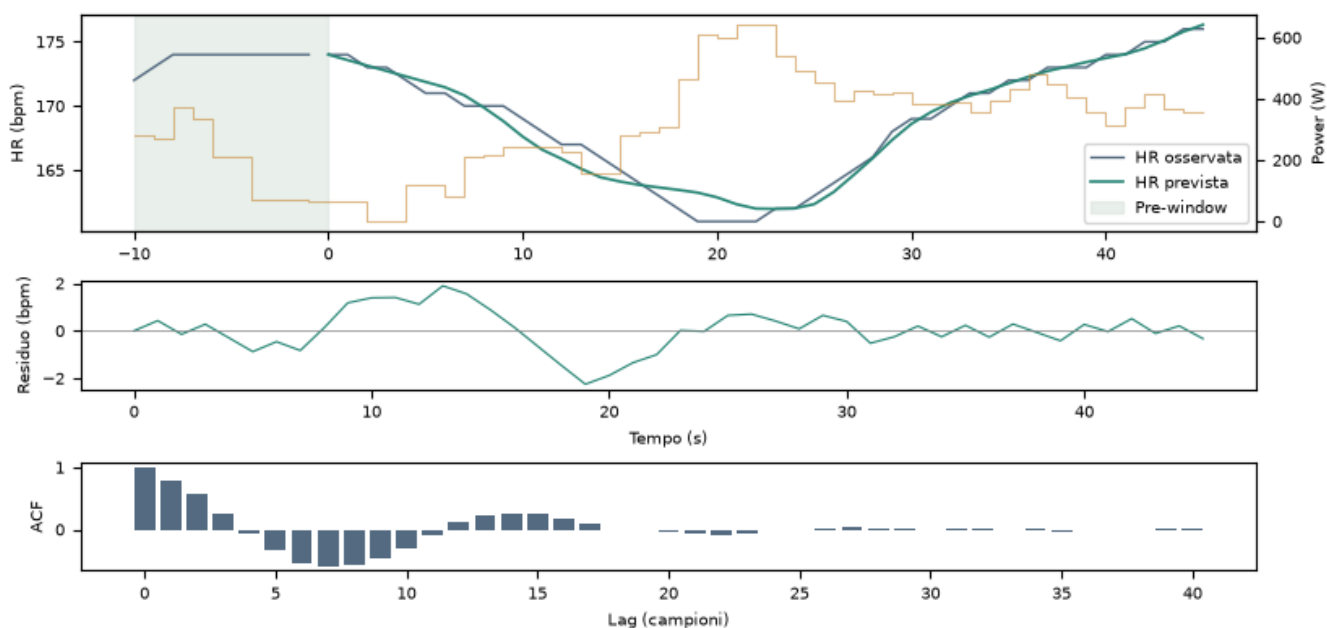
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=195.9 W; HR0=174 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=114 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.83908	0.61965	0.03374	0.96697	0.95643

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.03374	0.84767	0.7806	-0.32628



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.0921	2.3926	-3.5973 / 5.7816	219.07	no	no
L	6.6069	0.40634	5.8105 / 7.4033	6.1502	no	no
tau	265.99	583.18	-877.04 / 1409	219.25	no	no
B	60	238.12	-406.72 / 526.72	396.87	si	si

rho=0.14434; Jacobian rank=4/4; cond=38467; accordo multistart=no.

Correlazioni: {"K_L": -0.7701762340272507, "K_tau": 0.9999432847139735, "K_B": -0.9999443996888352, "L_tau": -0.7712596573463789, "L_B": 0.775225654527609, "tau_B": -0.9999392042517932}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 32.387 / 870.24

Griglia: 42441 coppie L/tau; SSE multistart originale: 32.387; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_016-017.csv

ID: e4e3cf9c9fc62b7256a3db10 | Stato: fitted

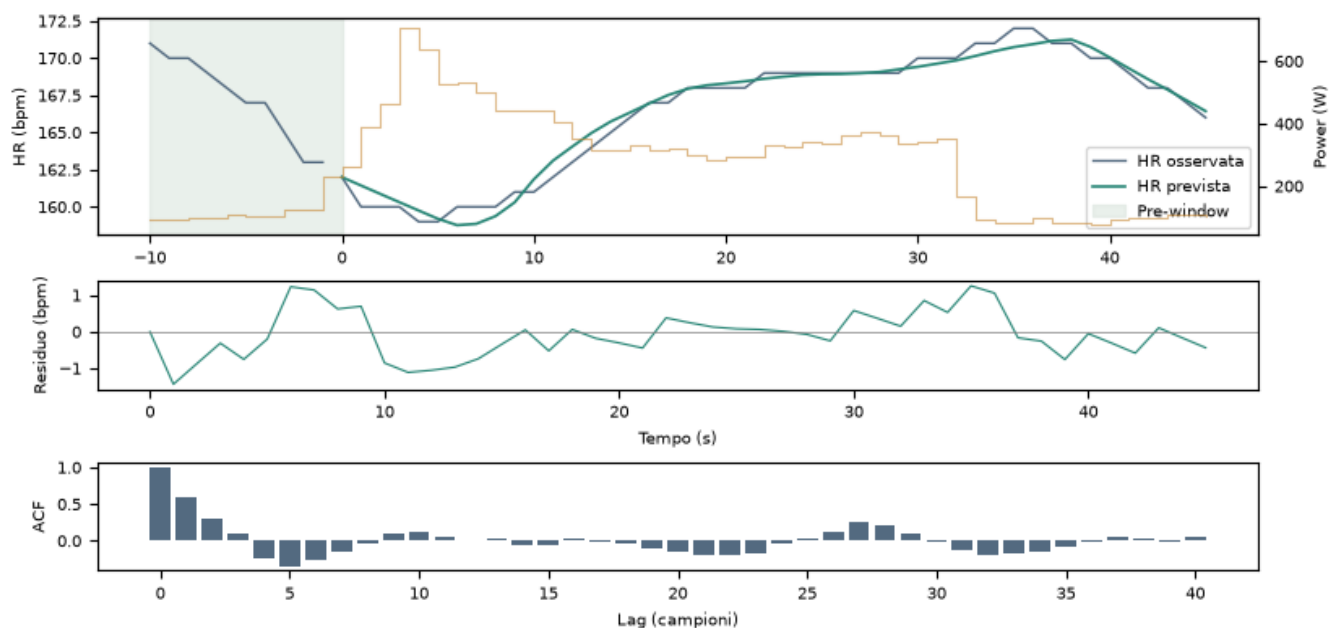
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=117.4 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.63103	0.49473	-0.074989	0.97594	0.96056

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.074989	0.63348	0.58516	-0.33932



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.66272	0.67801	-0.66618 / 1.9916	102.31	no	no
L	5.7763	0.28815	5.2115 / 6.3411	4.9884	no	no
tau	179.59	181.49	-176.13 / 535.32	101.06	no	no
B	60	109.2	-154.02 / 274.02	181.99	si	si

rho=0.2184; Jacobian rank=4/4; cond=63037; accordo multistart=no.

Correlazioni: {"K_L": -0.8836045219935736, "K_tau": 0.9990716453041917, "K_B": -0.9998568030981705, "L_tau": -0.884212395475271, "L_B": 0.8847192340976743, "tau_B": -0.9982791841607203}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 18.317 / 238.08

Griglia: 42441 coppie L/tau; SSE multistart originale: 18.317; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_080-081.csv

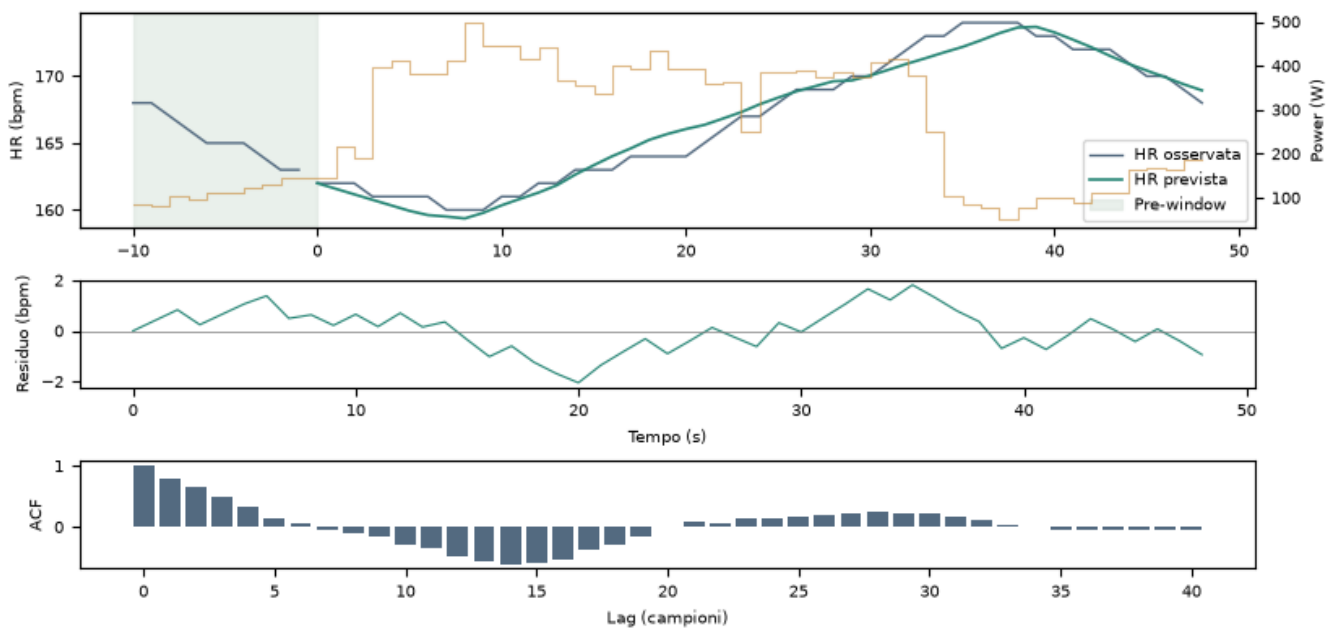
ID: ee8ebd9da0c05baf35d6cd83 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=113 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.83834	0.67455	0.052241	0.96875	1.0232
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.052241	0.84539	0.78384	0.14093	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.80176	1.3332	-1.8113 / 3.4148	166.28	no	no
L	5.164	0.57493	4.0372 / 6.2909	11.133	no	no
tau	244.36	409.16	-557.59 / 1046.3	167.44	no	no
B	60	176.34	-285.63 / 405.63	293.91	si	si

rho=0.1753; Jacobian rank=4/4; cond=76865; accordo multistart=no.

Correlazioni: {"K_L": -0.9219151670019607, "K_tau": 0.9994570498118681, "K_B": -0.999801632057462, "L_tau": -0.9200905635519071, "L_B": 0.9242812831255357, "tau_B": -0.9986605490417747}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 34.438 / 305.58

Griglia: 42441 coppie L/tau; SSE multistart originale: 34.438; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_070-071.csv

ID: eede334349b5661bffa82d | Stato: fitted

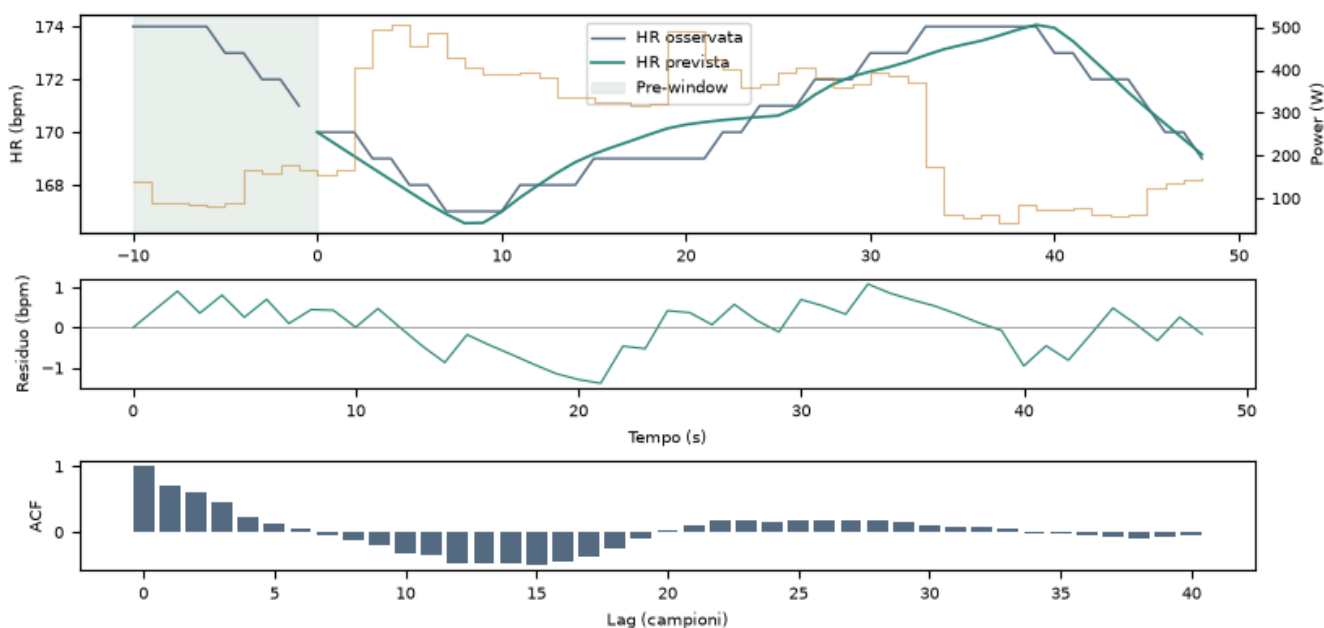
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=124.5 W; HR0=170 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=110 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.59672	0.48939	0.027176	0.92798	1.0741

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.027176	0.60228	0.70251	0.12877



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.62825	1.4871	-2.2865 / 3.543	236.71	no	no
L	6.4402	0.43183	5.5938 / 7.2866	6.7053	no	no
tau	241.02	573.87	-883.76 / 1365.8	238.1	no	no
B	60	261.03	-451.63 / 571.63	435.06	si	si

rho=0.17243; Jacobian rank=4/4; cond=1.4833e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.8922520018302582, "K_tau": 0.9997802394645364, "K_B": -0.9999896943607844, "L_tau": -0.8904519916984919, "L_B": 0.8933397394466426, "tau_B": -0.9997168599712012}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 17.448 / 182.03

Griglia: 42441 coppie L/tau; SSE multistart originale: 17.448; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_015-016.csv

ID: f0b1831cd2d81446639a6ee5 | Stato: fitted

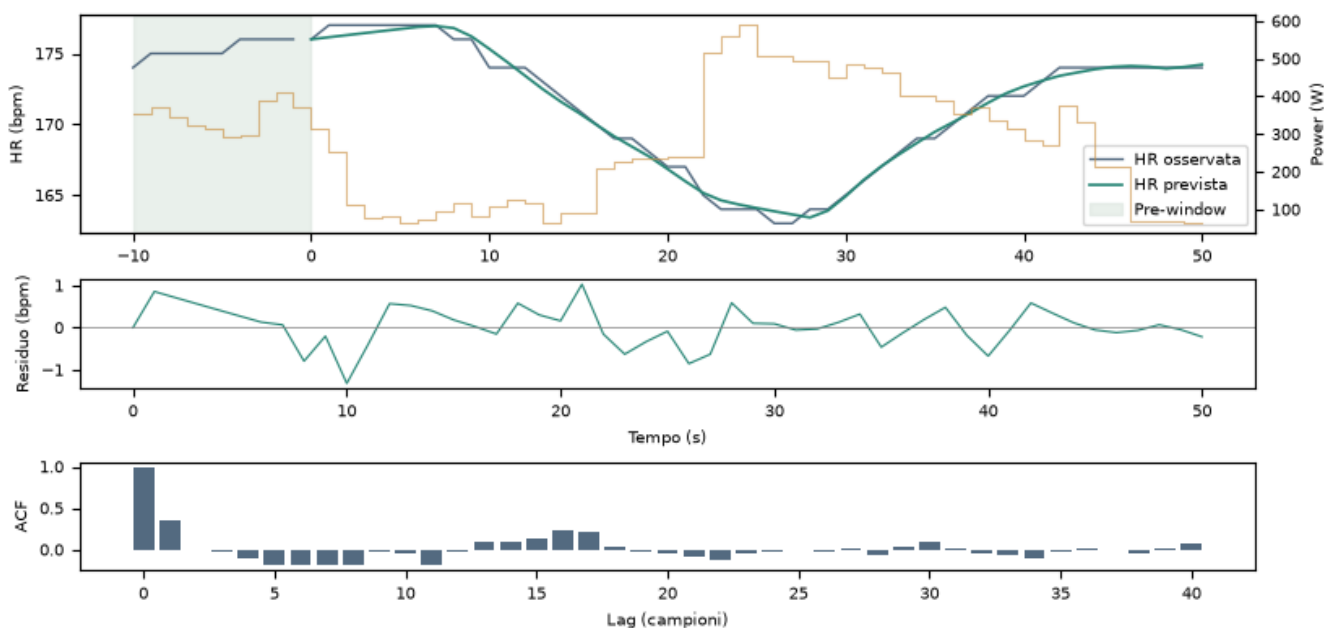
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=346.9 W; HR0=176 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-24 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.45058	0.3412	0.041071	0.98965	0.96575

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.041071	0.45317	0.35323	-0.19504



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.65589	0.48453	-0.29379 / 1.6056	73.874	no	no
L	6.3756	0.24213	5.901 / 6.8501	3.7977	no	no
tau	162.33	119	-70.919 / 395.57	73.311	no	no
B	200	21.673	157.52 / 242.48	10.836	si	si

rho=0.26874; Jacobian rank=4/4; cond=27406; accordo multistart=no.

Correlazioni: {"K_L": -0.896447665952719, "K_tau": 0.9997871955325649, "K_B": 0.9996628464990372, "L_tau": -0.8988152894608497, "L_B": -0.8939785645817253, "tau_B": 0.9990851504285067}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 10.354 / 713.86

Griglia: 42441 coppie L/tau; SSE multistart originale: 10.354; selezione: multistart.

Segnalazioni: at_bounds, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260427170343__laps_084-085.csv

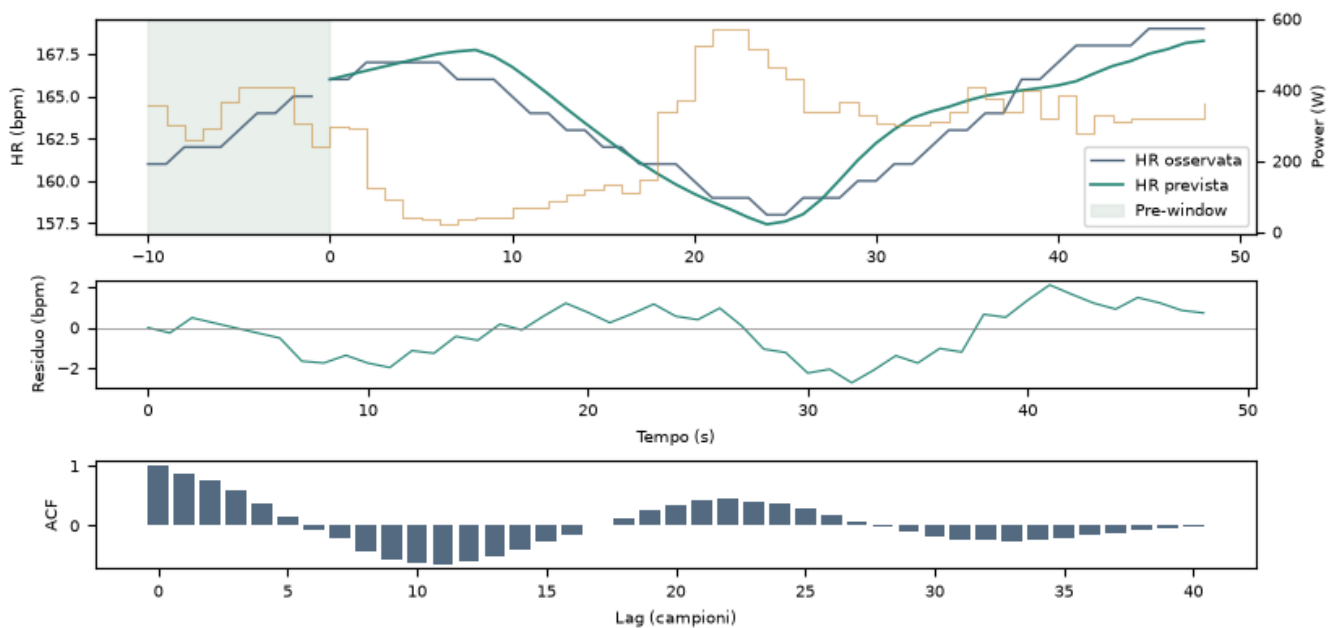
ID: f1f23b37c75cdc88c845c59d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=335.3 W; HR0=166 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-34 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.2141	1.0204	-0.19536	0.8746	0.98528
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.19536	1.2107	0.86534		0.15161



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.48474	1.059	-1.5909 / 2.5603	218.46	no	no
L	6.2457	0.63334	5.0044 / 7.487	10.14	no	no
tau	131.8	289.36	-435.35 / 698.95	219.54	no	no
B	200	79.712	43.764 / 356.24	39.856	si	si

rho=0.31679; Jacobian rank=4/4; cond=30014; accordo multistart=no.

Correlazioni: {"K_L": -0.8601971342560235, "K_tau": 0.9996945674622942, "K_B": 0.9999136838889522, "L_tau": -0.8635916402006022, "L_B": -0.859758005033006, "tau_B": 0.99949369810258}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 72.227 / 526.95

Griglia: 42441 coppie L/tau; SSE multistart originale: 72.227; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260407082440__laps_053-054.csv

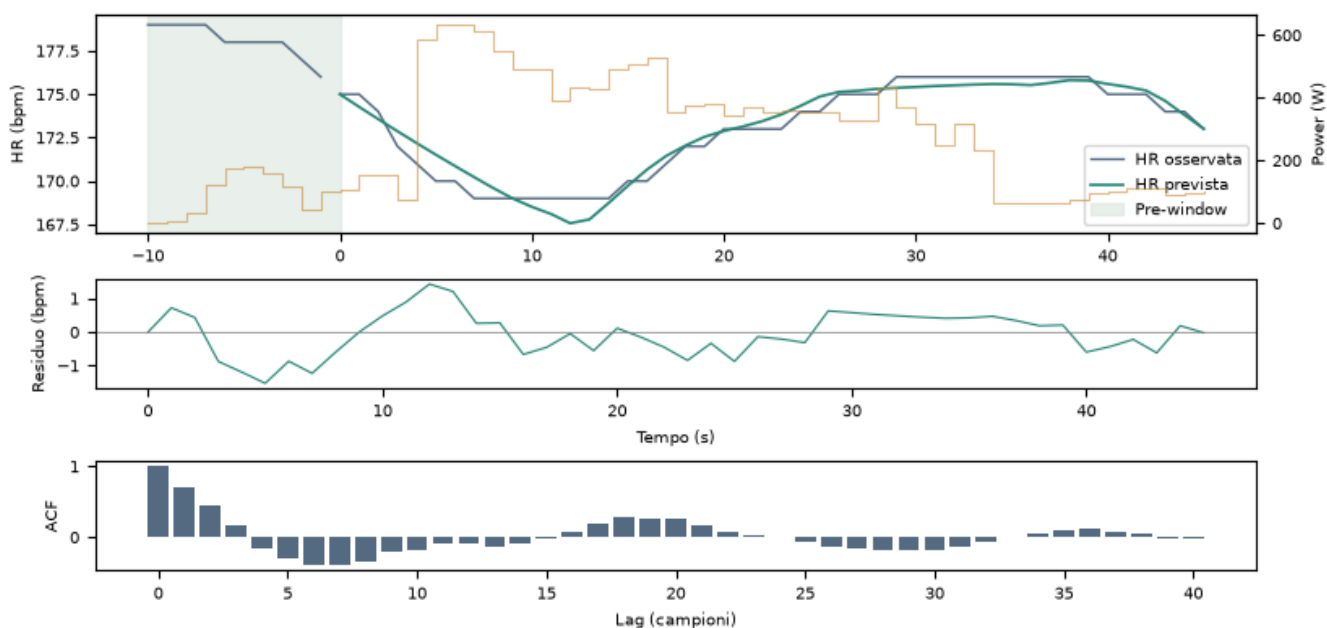
ID: f9e793bc7f4e768a378c4659 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=93.7 W; HR0=175 bpm.

HR(0) osservata fissata; B=138.55 bpm stimato; x(0)=36.452 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.63586	0.51881	-0.05166	0.94016	1.177
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.05166	0.64077	0.69735	-0.30605	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.14812	0.075027	0.0010663 / 0.29517	50.653	no	no
L	8.4391	0.34975	7.7536 / 9.1247	4.1444	no	no
tau	49.832	25.716	-0.57168 / 100.24	51.606	no	no
B	138.55	17.475	104.3 / 172.8	12.613	no	no

rho=0.73368; Jacobian rank=4/4; cond=28520; accordo multistart=no.

Correlazioni: {"K_L": -0.8454544645993518, "K_tau": 0.9975510901496623, "K_B": -0.999838267432923, "L_tau": -0.8317427739640659, "L_B": 0.8508697851968927, "tau_B": -0.9971600851497946}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 18.599 / 58.261

Griglia: 42441 coppie L/tau; SSE multistart originale: 18.599; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G1/260424160919__laps_020-021.csv

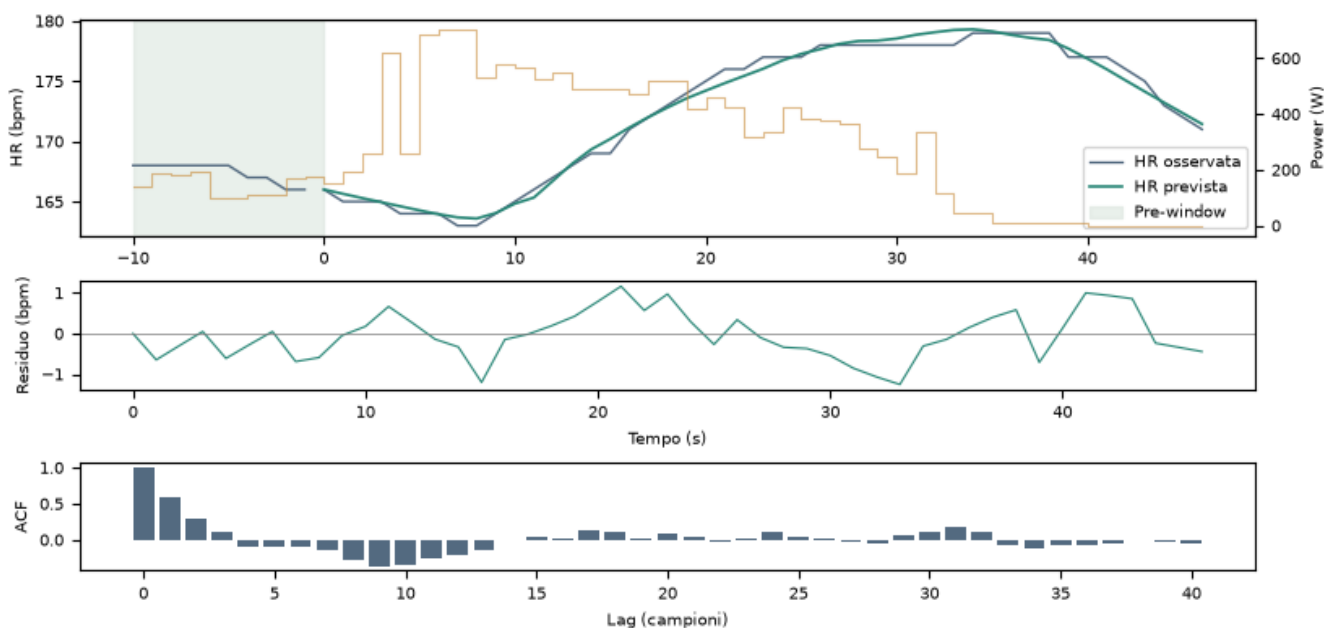
ID: fee0d8774af1c7bfb7b4a557 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=146.9 W; HR0=166 bpm.

HR(0) osservata fissata; B=141.25 bpm stimato; x(0)=24.749 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.57892	0.46679	-0.040682	0.98939	0.98235
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.040682	0.58373	0.59215	-0.089488	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.22221	0.066276	0.092311 / 0.35211	29.826	no	no
L	5.6441	0.30262	5.051 / 6.2372	5.3616	no	no
tau	69.199	20.832	28.369 / 110.03	30.104	no	no
B	141.25	9.177	123.26 / 159.24	6.497	no	no

rho=0.58319; Jacobian rank=4/4; cond=18404; accordo multistart=no.

Correlazioni: {"K_L": -0.9161637773091524, "K_tau": 0.9957666236392135, "K_B": -0.9976555239749798, "L_tau": -0.924242299749582, "L_B": 0.9088707558540704, "tau_B": -0.9878906716914366}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 15.752 / 501.47

Griglia: 42441 coppie L/tau; SSE multistart originale: 15.752; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Configurazione completa dell'esperimento

Configurazione, split, sorgenti, hash e ambiente della run. I CSV originali restano conservati nel blob store. Manifest integrale e risultati sono inclusi nell'export tabellare ZIP.

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